Unit 2. Multiple Regression I: Matrix Theory

1 Data, Correlation and the Model

FREES §2.1–2.2, §3.1, §3.2.1–3.2.3 ·ISLR §3.1, §3.2 ·CLM §1. SRM learning outcomes 2(a), 2(c).

1.1 Scatter plots and correlation

Example 2.1 (Wisconsin lottery sales; FREES §2.1)**.** Average weekly online lottery sales in dollars (`SALES`) over forty weeks of 1998–99 and the population (`POP`) of 50 Wisconsin ZIP-code areas chosen at random.

```
lottery <- read.csv("data/frees/csv/WiscLottery.csv")
descr <- function(v) c(mean = mean(v), median = median(v), sd = sd(v), min = min(v),
                       max = max(v))
round(rbind(POP = descr(lottery$POP), SALES = descr(lottery$SALES)), 1)
```

```
##        mean median    sd min   max
## POP    9311   4406 11098 280 39098
## SALES 6495   2426  8103 189 33181
```

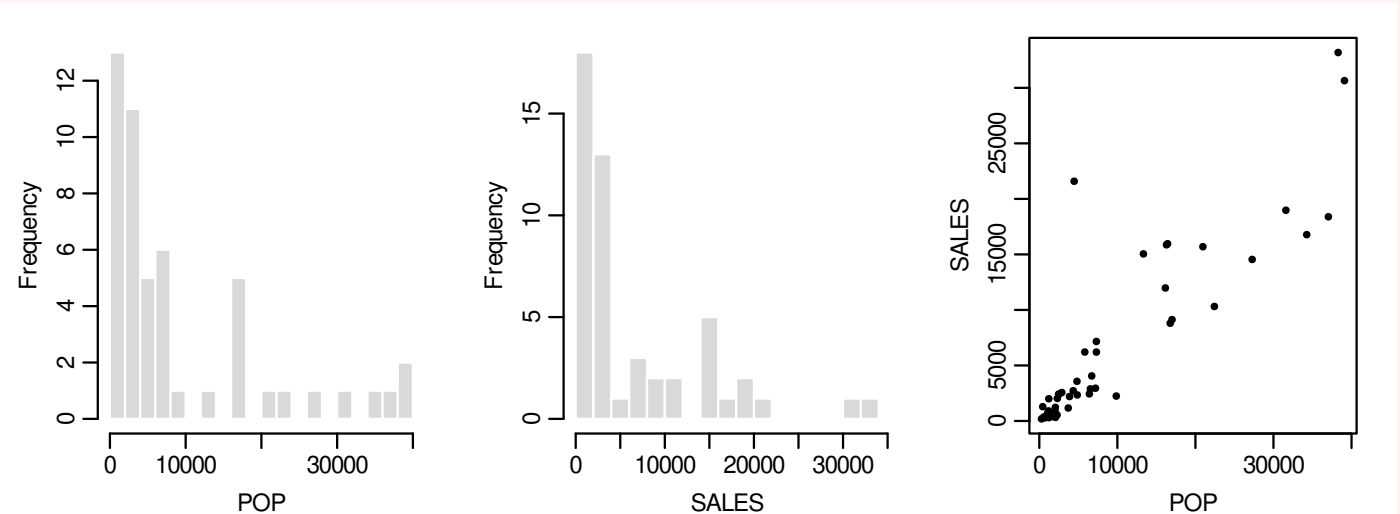


Figure 2.1: The lottery data: histograms of population and of sales, and the scatter plot of sales against population (FREES Figures 2.1–2.2).

Both variables are skewed to the right, each mean well above its median; the scatter plot shows sales rising with population.

Definition 2.1 (Pearson correlation; FREES §2.1, ISLR §3.1.3; CLM Definition 2.15)**.** For pairs (x_i, y_i), $i = 1, \dots, n$, of variables that are not constant, with $s_x^2 = (n-1)^{-1}\sum_i (x_i - \bar{x})^2$ and s_y^2 likewise,

$$r = \frac{1}{(n-1)s_x s_y}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}. \tag{1}$$

The variables are **positively correlated** if $r > 0$ and **negatively correlated** if $r < 0$.

Proposition 2.2 (Bounds and invariance; CLM Proposition 2.17)**.** For variables that are not constant, $-1 \leqslant r \leqslant 1$; $r = 1$ iff the points lie on a line of positive slope and $r = -1$ iff they lie on a line of negative slope. Replacing x_i by $a + bx_i$ and y_i by $c + dy_i$ leaves r unchanged when $bd > 0$ and changes its sign when $bd < 0$.

Proof. Step 1: the bound. Put $u_i = x_i - \bar{x}$ and $v_i = y_i - \bar{y}$. By the Cauchy–Schwarz inequality of the prerequisites, $(\sum_i u_i v_i)^2 \leqslant (\sum_i u_i^2)(\sum_i v_i^2)$, with equality iff the vectors (u_i) and (v_i) are proportional. When neither variable is constant the right side is positive, and dividing by it gives $r^2 \leqslant 1$.

Step 2: the cases of equality. Equality means $v_i = cu_i$ for every i and some $c \neq 0$, that is $y_i - \bar{y} = c(x_i - \bar{x})$: the points lie on the line through $(\bar{x}, \bar{y})$ of slope c, and then $r = c\sum_i u_i^2 \big/ \sqrt{\sum_i u_i^2 \cdot c^2 \sum_i u_i^2} = c/|c|$, which is 1 for $c > 0$ and -1 for $c < 0$. Conversely, points on a line $y = \alpha + \beta x$ with $\beta \neq 0$ have $v_i = \beta u_i$, so equality holds.

Step 3: invariance. The transformed deviations are bu_i and dv_i, so the numerator of (1) is multiplied by bd and the denominator by $|b|\,|d|$; the ratio is r when $bd > 0$ and $-r$ when $bd < 0$. □

Remark (What r measures; FREES §2.1)**.** r is dimensionless and measures **linear** association only. The five points $(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)$ lie on $y = x^2$ and have $r = 0$: with $\bar{x} = 0$ and $\bar{y} = 2$, $\sum_i x_i(y_i - 2) = (-2)(2) + (-1)(-1) + 0 + (1)(-1) + (2)(2) = 0$.

Example 2.2 (Lottery sales, continued; FREES §2.1)**.**

```
1 cor(lottery$POP, lottery$SALES)
```

```
## [1] 0.8863
```

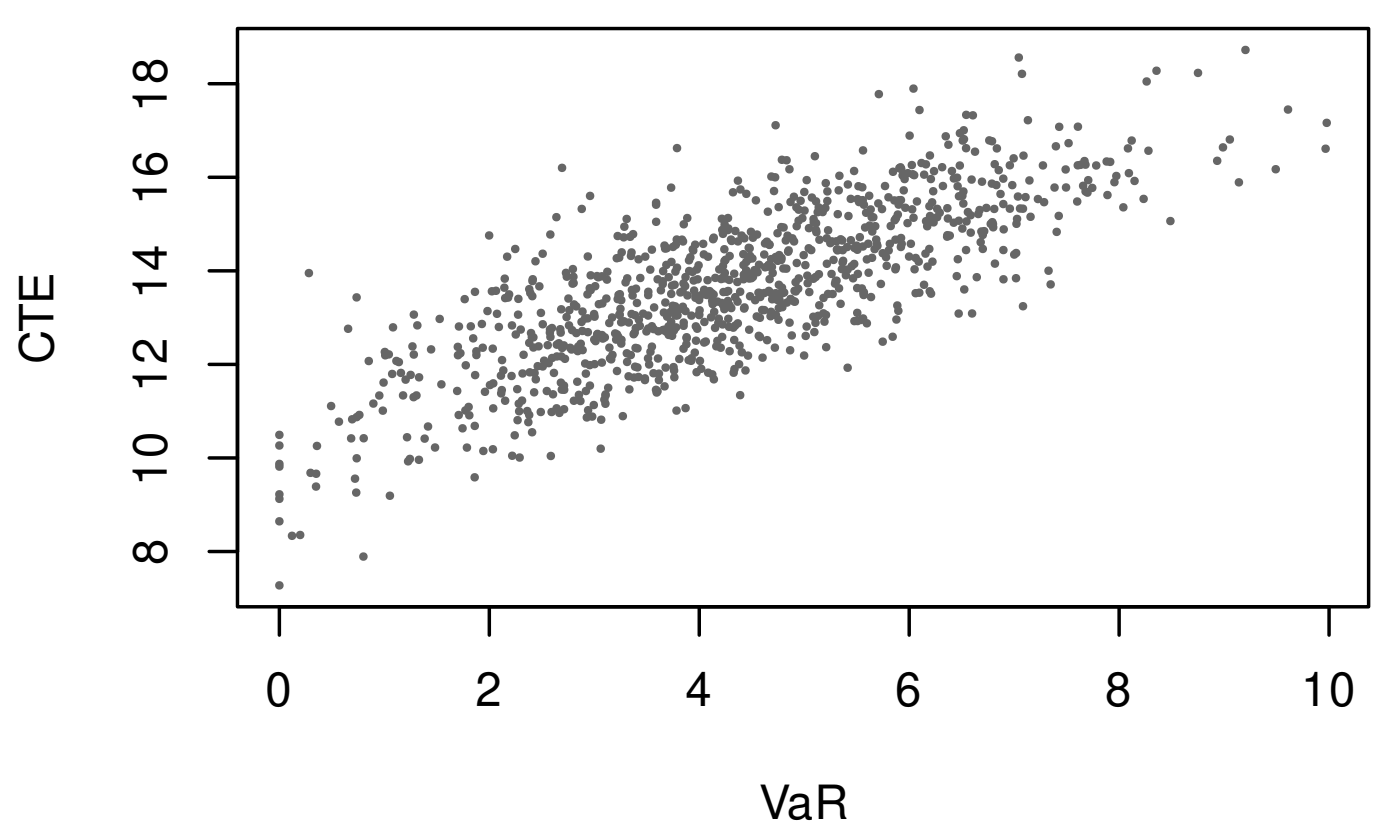
Sales and population are strongly and positively correlated.

Example 2.3 (Summarizing simulations; FREES §2.1, Figure 2.3)**.** Manistre and Hancock's put option: a stock worth \$100 now is worth $S(Z) = 100\exp(0.08 \cdot 10 + 0.15\sqrt{10}\,Z)$ in ten years, Z standard normal, and a put with strike \$110 has present value $C(Z) = e^{-0.06\cdot 10}\max\{0, 110 - S(Z)\}$. Each of 1000 replicates simulates 1000 values of C and records their 95th percentile, the value at risk VaR, and the mean of the largest 50, the conditional tail expectation CTE:

```
set.seed(1)
Cput <- function(z) exp(-0.6) * pmax(0, 110 - 100 * exp(0.8 + 0.15 * sqrt(10) * z))
vc <- t(replicate(1000, { v <- sort(Cput(rnorm(1000)))
  c(VaR = v[950], CTE = mean(v[951:1000])) }))
cor(vc[, "VaR"], vc[, "CTE"])
```

```
## [1] 0.7835
```

```
par(mar = c(4, 4.2, 0.5, 0.5), cex = 0.85)
plot(vc, pch = 16, cex = 0.4, col = "grey40")
```



The correlation is 0.78, FREES's 0.782 up to simulation noise: a strong but not perfect linear

relation between two risk measures computed from the same simulated values. The summaries of this unit apply to the output of a simulation study as they do to data.

Example 2.4 (Term life insurance; FREES §3.1)**.** The 2004 Survey of Consumer Finances: of 500 households with positive income, the 275 that bought term life insurance, with the policy face amount `FACE`, annual `INCOME`, years of `EDUCATION` of the respondent and the household size `NUMHH`. Face amount and income are analysed in logarithms.

```
term <- subset(read.csv("data/frees/csv/TermLife.csv"), FACE > 0)
term$LNFACE <- log(term$FACE); term$LNINCOME <- log(term$INCOME)
nrow(term)
```

```
## [1] 275
```

```
round(t(sapply(term[, c("FACE", "INCOME", "EDUCATION", "NUMHH", "LNFACE",
 ↪  "LNINCOME")],
                descr)), 3)
```

```
##                mean    median          sd     min         max
## FACE      747581.45 150000.00 1674362.433 800.000 14000000.00
## INCOME    208974.62  65000.00  824009.775 260.000 10000000.00
```

```
## EDUCATION     14.52     16.00       2.549   2.000       17.00
## NUMHH          2.96      3.00       1.493   1.000        9.00
## LNFACE        11.99     11.92       1.871   6.685       16.45
## LNINCOME      11.15     11.08       1.295   5.561       16.12
```

```
round(cor(term[, c("NUMHH", "EDUCATION", "LNINCOME", "LNFACE")]), 3)
```

```
##             NUMHH EDUCATION LNINCOME LNFACE
## NUMHH       1.000    -0.064    0.179  0.288
## EDUCATION -0.064     1.000    0.343  0.383
## LNINCOME    0.179     0.343    1.000  0.482
## LNFACE      0.288     0.383    0.482  1.000
```

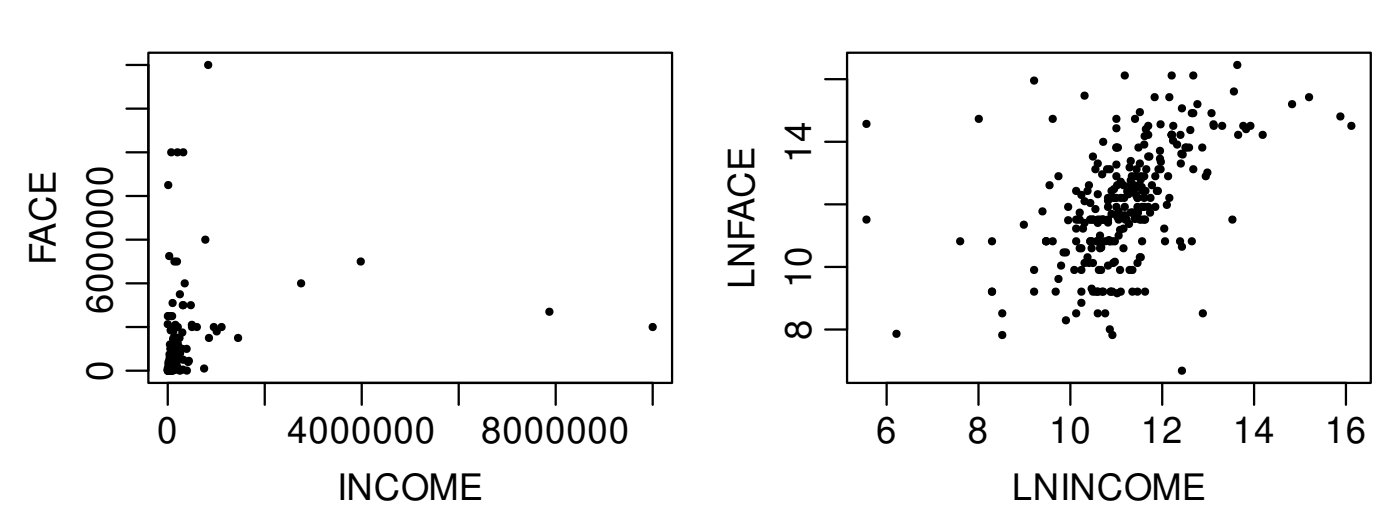


Figure 2.2: Face amount against income, in dollars and in logarithms (FREES Figure 3.1).

In dollars the points cluster in the lower left corner and the relation is far from linear; in logarithms a line is a reasonable summary. The correlation matrix summarises the pairwise scatter plots: each entry is a pairwise linear association and cannot show how the response depends on several variables jointly.

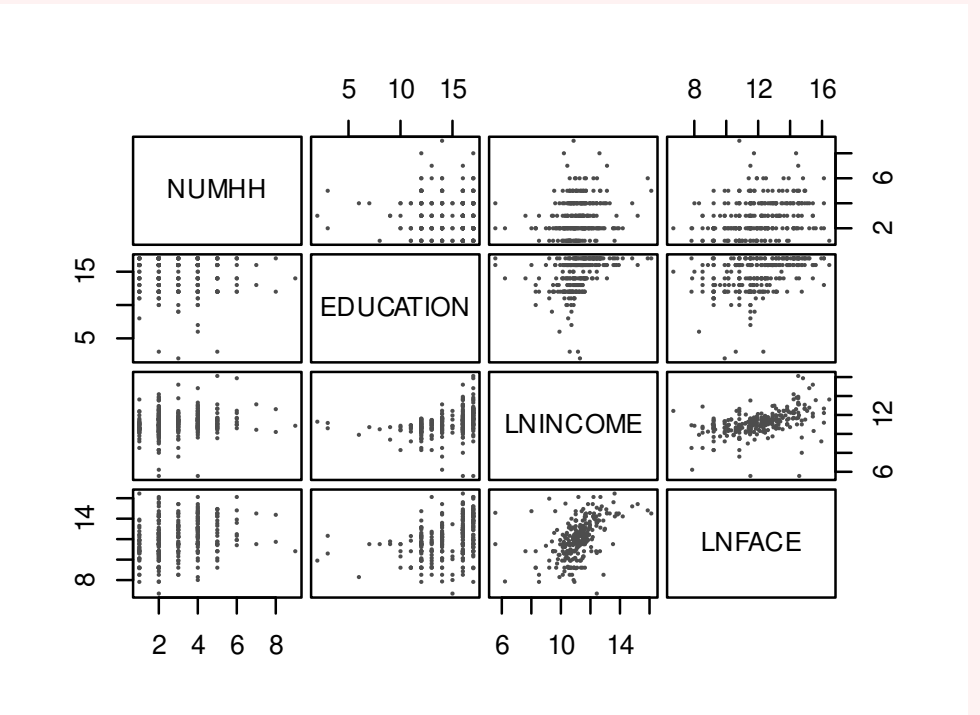


Figure 2.3: The scatterplot matrix of the four variables, whose correlations are printed above (FREES Figure 3.2): `LNFACE` rises with `LNINCOME` and `EDUCATION`; `NUMHH` takes few values.

Example 2.5 (Advertising; ISLR §2.1, §3.1–3.2)**.** Sales of a product in thousands of units (`sales`) in 200 markets, with the advertising budgets in thousands of dollars for `TV`, `radio` and `newspaper`. The file is distributed at the book's site and is not in `ISLR2`.

```
adv <- read.csv("data/islr/Advertising.csv")[, -1]
round(cor(adv), 4)
```

```
##                TV  radio newspaper  sales
## TV         1.0000 0.0548    0.0566 0.7822
## radio      0.0548 1.0000    0.3541 0.5762
## newspaper 0.0566 0.3541    1.0000 0.2283
## sales      0.7822 0.5762    0.2283 1.0000
```

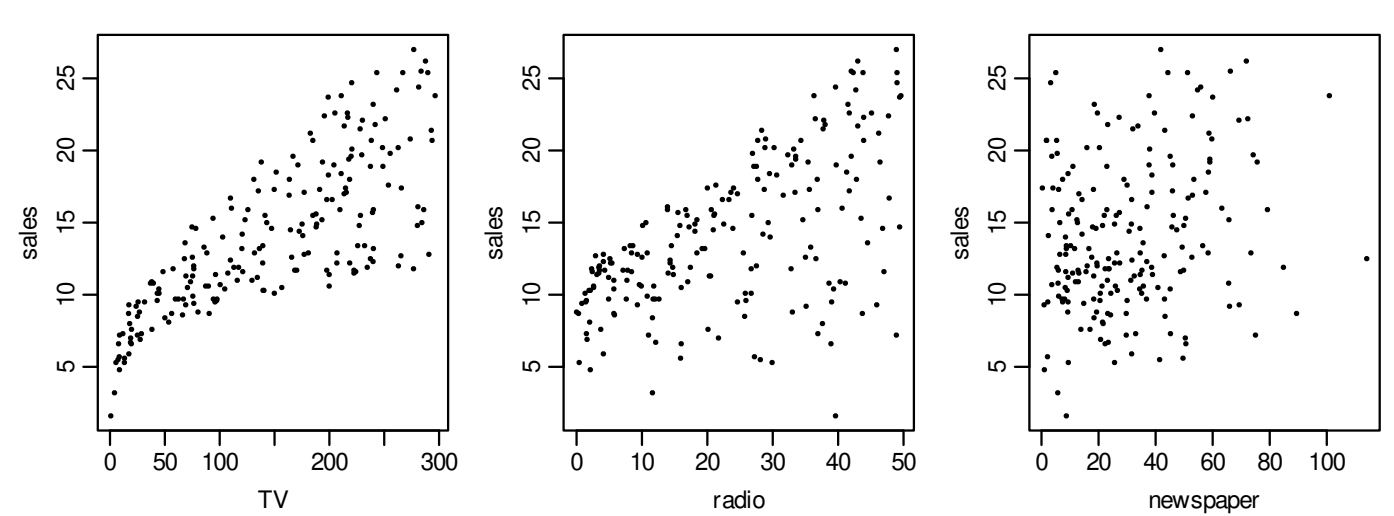


Figure 2.4: Sales against each advertising budget (ISLR Figure 2.1).

1.2 The model and its assumptions

Definition 2.3 (Multiple linear regression model; FREES §3.2, ISLR §3.2; CLM Definition 1.1, equation (2))**.** Given n observations, each consisting of a response Y_i and k explanatory variables $X_{i1}, \dots, X_{ik}$, the **multiple linear regression model** is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i, \qquad i = 1, \dots, n, \tag{2}$$

or in matrix form

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{3}$$

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{nk} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

$\mathbf{X}$ is the $n \times (k+1)$ **design matrix**, whose first column $\mathbf{1}$ carries the intercept; $\mathbf{x}_i'$ is its i-th row, so $Y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$. The **regression function** is $\mathsf{E}[Y \mid X_1, \dots, X_k] = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$, linear in the parameters.

Remark (Counts and notation; FREES §3.1, ISLR §3.2)**.** k is the number of explanatory variables, $k+1$ the number of parameters and of columns of $\mathbf{X}$, and $n-k-1$ the residual degrees of freedom. FREES writes k and $\mathbf{b}$ for the estimate of $\boldsymbol{\beta}$; ISLR writes p for k and $\hat{\beta}$; the SOA questions use both. The case $k=1$, $\mathbf{X}=[\mathbf{1}\ \mathbf{x}]$, is FREES's **basic** and ISLR's **simple** linear regression, and fitting (3) is called **regressing** Y **on** $X_1,\dots,X_k$.

Definition 2.4 (Classical assumptions; FREES §2.2, §3.2.3, ISLR §3.1.2; CLM Definition 1.2)**.** On model (3):

(A1) **Linearity**: $\mathsf{E}[\mathbf{Y}\mid\mathbf{X}]=\mathbf{X}\boldsymbol{\beta}$.

(A2) **Zero conditional mean**: $\mathsf{E}[\boldsymbol{\varepsilon}\mid\mathbf{X}]=\mathbf{0}$.

(A3) **Spherical errors**: $\operatorname{var}(\boldsymbol{\varepsilon}\mid\mathbf{X})=\sigma^2\mathbf{I}_n$ with $\sigma^2>0$, that is $\operatorname{var}(\varepsilon_i)=\sigma^2$ for every i (**homoscedasticity**) and $\operatorname{cov}(\varepsilon_i,\varepsilon_j)=0$ for $i\neq j$.

(A4) **Full rank**: $\operatorname{rank}(\mathbf{X})=k+1$, which requires $n\geqslant k+1$; inference requires $n>k+1$.

(A5) **Normality**: $\boldsymbol{\varepsilon}\mid\mathbf{X}\sim N(\mathbf{0},\sigma^2\mathbf{I}_n)$.

A1–A4 are the **Gauss–Markov assumptions**; A5 is used only for exact finite-sample distributions. $\mathbf{X}$ is treated as fixed, and every expectation, variance and distribution is conditional on

it, with the conditioning suppressed.

Remark (FREES's two representations; FREES §2.2, §3.2.3)**.** FREES states the model through **observables** assumptions F1–F5, namely $\mathsf{E}[y_i] = \mathbf{x}_i'\boldsymbol{\beta}$, nonstochastic x_{ij}, $\operatorname{var}(y_i) = \sigma^2$, independent y_i, normal y_i, or through **error** assumptions E1–E5 on $\varepsilon_i = y_i - \mathbf{x}_i'\boldsymbol{\beta}$. The two sets are equivalent, and A1–A3 and A5 are their matrix form with independence weakened to zero covariance, which is all the proofs use; A4 is FREES's requirement that the explanatory variables are not linear combinations of one another. Nonstochastic x's are motivated by stratified sampling: at each value of $\mathbf{x}_i$ a response is drawn from the population at that value, so $\mathbf{X}$ is fixed and the responses are independent. Since $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, A1 and A2 are the same statement.

Remark (Interpretation of the coefficients; FREES §3.2.1, ISLR §3.1.2, §3.2.1)**.** $\beta_0 = \mathsf{E}[Y \mid X_1 = \cdots = X_k = 0]$ sets the height of the plane and is rarely of interest. For $j \geqslant 1$, $\beta_j = \partial\, \mathsf{E}[Y \mid X_1, \ldots, X_k]/\partial X_j$ is the expected change in Y per unit change in X_j with the other explanatory variables held fixed; ISLR's phrase is the average effect on Y of a one-unit increase in X_j, holding all other predictors fixed. For $k = 1$ the equation $Y = \beta_0 + \beta_1 X + \varepsilon$ is the **population regression line**, and ε collects what the line misses: curvature, omitted variables and measurement error.

2 Least Squares

FREES §2.1, §3.1, §3.2.2 ·ISLR §3.1.1, §3.2.1 ·CLM §2. SRM learning outcomes 2(a), 2(c).

2.1 The normal equations

Definition 2.5 (Least squares; FREES §2.1, §3.1, ISLR §3.1.1; CLM Definition 2.1)**.** The **least squares estimator** $\hat{\boldsymbol{\beta}}$ minimises the **residual sum of squares**

$$Q(\boldsymbol{\beta}^*) = \sum_{i=1}^{n} (Y_i - \mathbf{x}_i'\boldsymbol{\beta}^*)^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}^*)'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}^*) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}^*\|^2 \tag{4}$$

over $\boldsymbol{\beta}^* \in \mathbb{R}^{k+1}$. The **fitted values** are $\hat{Y}_i = \mathbf{x}_i'\hat{\boldsymbol{\beta}}$, the **residuals** $e_i = Y_i - \hat{Y}_i$, and $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$ is the **fitted regression plane**, a line when $k = 1$.

Theorem 2.6 (The least squares estimator; CLM Theorem 2.4, equation (6))**.** Under A4 the

minimiser of (4) is unique and equals

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}. \tag{5}$$

Proof. Step 1: expand. The scalar $\mathbf{Y}'\mathbf{X}\boldsymbol{\beta}$ equals its transpose $\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y}$, so

$$Q(\boldsymbol{\beta}) = \mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{Y}'\mathbf{Y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \tag{6}$$

Step 2: differentiate. By the two rules of matrix differentiation of the prerequisites, $\partial(\mathbf{a}'\boldsymbol{\beta})/\partial\boldsymbol{\beta}' = \mathbf{a}'$ and $\partial(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta})/\partial\boldsymbol{\beta}' = 2\boldsymbol{\beta}'\mathbf{A}$ for symmetric $\mathbf{A}$, applied with $\mathbf{a} = \mathbf{X}'\mathbf{Y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$,

$$\frac{\partial Q}{\partial \boldsymbol{\beta}'} = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}.$$

Setting this to $\mathbf{0}'$ and transposing gives the **normal equations**

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}. \tag{7}$$

Row j of (7) is $\sum_i x_{ij}(Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}) = 0$, which is $\partial Q/\partial\beta_j = 0$; for $k = 1$ the two rows are $\sum_i(Y_i - b_0 - b_1 x_i) = 0$ and $\sum_i x_i(Y_i - b_0 - b_1 x_i) = 0$.

Step 3: $\mathbf{X}'\mathbf{X}$ is invertible. $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}$ have the same null space: $\mathbf{X}\mathbf{v} = \mathbf{0}$ gives $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$, and $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 = 0$, hence $\mathbf{X}\mathbf{v} = \mathbf{0}$. Both matrices have $k+1$ columns, so by the rank–nullity theorem of the prerequisites they have the same rank, which is $k+1$ under A4. The square matrix $\mathbf{X}'\mathbf{X}$ is therefore invertible and (7) has the unique solution (5).

Step 4: it is the minimum. For any $\boldsymbol{\beta}$ put $\mathbf{d} = \boldsymbol{\beta} - \hat{\boldsymbol{\beta}}$. Expanding $Q(\boldsymbol{\beta}) = \|(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) - \mathbf{X}\mathbf{d}\|^2$ and using $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$, which is (7) rearranged,

$$Q(\boldsymbol{\beta}) = Q(\hat{\boldsymbol{\beta}}) - 2\mathbf{d}'\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \mathbf{d}'\mathbf{X}'\mathbf{X}\mathbf{d} = Q(\hat{\boldsymbol{\beta}}) + \|\mathbf{X}\mathbf{d}\|^2 \geqslant Q(\hat{\boldsymbol{\beta}}),$$

with equality iff $\mathbf{X}\mathbf{d} = \mathbf{0}$, that is iff $\mathbf{d} = \mathbf{0}$ by A4. So $\hat{\boldsymbol{\beta}}$ is the unique global minimiser. □

Remark (Without full rank; FREES §3.1)**.** Only Step 3 and the equality case of Step 4 use A4. If one column of $\mathbf{X}$ is a linear combination of the others, $\mathbf{X}'\mathbf{X}$ is singular, the normal equations have infinitely many solutions (one exists because the prerequisites' theorem **orthogonal projection** gives a closest point $\mathbf{X}\boldsymbol{\beta}$ of $\mathcal{C}(\mathbf{X})$ with $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} \perp \mathcal{C}(\mathbf{X})$, which is (7); adding any nonzero vector of the null space of $\mathbf{X}$ gives others), all with the same fitted vector by Step 4, and R reports `NA` for the redundant coefficients.

2.2 Orthogonal projection

Theorem 2.7 (Least squares as projection; CLM Theorem 2.5)**.** Under A4, $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{Y}$ onto the column space $\mathcal{C}(\mathbf{X})$: it is the unique point of $\mathcal{C}(\mathbf{X})$ closest to $\mathbf{Y}$, and $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

Proof. Minimising $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta}$ is finding the point of $\mathcal{C}(\mathbf{X})$ closest to $\mathbf{Y}$.

Step 1: closest iff orthogonal residual. A point $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimises $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2$ over $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ iff $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$. ($\Leftarrow$) For $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$, $\hat{\mathbf{Y}} - \tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$, so $(\mathbf{Y} - \hat{\mathbf{Y}}) \perp (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$ and Pythagoras gives

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{Y}} - \tilde{\mathbf{Y}}\|^2 \geqslant \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2,$$

with equality iff $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}}$. ($\Rightarrow$) If some unit vector $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ had $\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$, then $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}} + t\mathbf{v} \in \mathcal{C}(\mathbf{X})$ gives $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - 2t\,\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) + t^2$, strictly smaller at $t = \mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}})$, contradicting optimality.

Step 2: the orthogonality condition is the normal equations. $\mathbf{Y} - \hat{\mathbf{Y}}$ is orthogonal to $\mathcal{C}(\mathbf{X})$ iff it is orthogonal to each column of $\mathbf{X}$, that is $\mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) = \mathbf{0}$; with $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ this is (7),

whose solution Theorem 2.6 gives. Uniqueness of the closest point is the equality case of Step 1. □

Definition 2.8 (Hat matrix and residual maker; CLM Definition 2.6)**.** Under A4,

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \tag{8}$$

so that $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ and $\mathbf{e} = \mathbf{M}\mathbf{Y}$.

Theorem 2.9 (Properties of $\mathbf{H}$ and $\mathbf{M}$; CLM Theorem 2.9)**.**

(i) $\mathbf{H}$ and $\mathbf{M}$ are symmetric and idempotent.
(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $\mathbf{M}\mathbf{X} = \mathbf{0}$.
(iii) $\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}$.
(iv) $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Proof. **(i)** $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric, being the inverse of a symmetric matrix, so

$$\mathbf{H}' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{H}, \qquad \mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{\mathbf{I}_{k+1}}\mathbf{X}' = \mathbf{H}.$$

Then $\mathbf{M}' = \mathbf{I} - \mathbf{H}' = \mathbf{M}$ and $\mathbf{M}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - \mathbf{H} = \mathbf{M}$.
(ii) $\mathbf{HX} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}$, hence $\mathbf{MX} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.
(iii) $\mathbf{HM} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$, and $\mathbf{MH} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$.
(iv) By the prerequisites' lemma **trace equals rank** for idempotent matrices, $\operatorname{tr}(\mathbf{H}) = \operatorname{rank}(\mathbf{H})$. Now $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$: every column of $\mathbf{H}$ is $\mathbf{X}$ times a vector, and $\mathbf{HX} = \mathbf{X}$ from (ii) puts every column of $\mathbf{X}$ in $\mathcal{C}(\mathbf{H})$. By A4, $\dim \mathcal{C}(\mathbf{X}) = k+1$, so $\operatorname{tr}(\mathbf{H}) = k+1$ and $\operatorname{tr}(\mathbf{M}) = \operatorname{tr}(\mathbf{I}_n) - \operatorname{tr}(\mathbf{H}) = n - k - 1$. □

Remark. $\mathbf{H}$ is symmetric, idempotent, with $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$; by the prerequisites' proposition **projections are the symmetric idempotents**, $\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, so Theorem 2.7 and Definition 2.8 describe one object, and $\mathbf{M}$ is the projection onto $\mathcal{C}(\mathbf{X})^{\perp}$, of dimension $n - k - 1$.

Lemma 2.10 (The intercept projection; CLM Remark 6.2, Lemma 7.2)**.** Let $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}'$ and suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Then $\mathbf{H}\mathbf{H}_0 = \mathbf{H}_0\mathbf{H} = \mathbf{H}_0$; $\mathbf{H} - \mathbf{H}_0$ is symmetric idempotent with $\operatorname{tr}(\mathbf{H} - \mathbf{H}_0) = k$, and is the orthogonal projection onto $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$; and $(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{0}$.

Proof. $\mathbf{H}_0$ is symmetric, and $\mathbf{H}_0^2 = \frac{1}{n^2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{H}_0$ since $\mathbf{1}'\mathbf{1} = n$; so $\mathbf{H}_0$ is the projection onto $\mathcal{C}(\mathbf{1})$. Since $\mathbf{H}$ projects onto $\mathcal{C}(\mathbf{X}) \ni \mathbf{1}$, $\mathbf{H}\mathbf{1} = \mathbf{1}$ and $\mathbf{H}\mathbf{H}_0 = \frac{1}{n}(\mathbf{H}\mathbf{1})\mathbf{1}' = \mathbf{H}_0$; transposing, $\mathbf{H}_0\mathbf{H} = \mathbf{H}_0$. Hence $\mathbf{H} - \mathbf{H}_0$ is symmetric and

$$(\mathbf{H} - \mathbf{H}_0)^2 = \mathbf{H} - \mathbf{H}\mathbf{H}_0 - \mathbf{H}_0\mathbf{H} + \mathbf{H}_0^2 = \mathbf{H} - \mathbf{H}_0 - \mathbf{H}_0 + \mathbf{H}_0 = \mathbf{H} - \mathbf{H}_0,$$

with $\operatorname{tr}(\mathbf{H}-\mathbf{H}_0) = \operatorname{tr}(\mathbf{H})-\operatorname{tr}(\mathbf{H}_0) = (k+1)-1 = k$ by Theorem 2.9(iv) and $\operatorname{tr}(\mathbf{H}_0) = \frac{1}{n}\operatorname{tr}(\mathbf{1}\mathbf{1}') = 1$. Its column space: $(\mathbf{H} - \mathbf{H}_0)\mathbf{v} = \mathbf{H}\mathbf{v} - \mathbf{H}_0\mathbf{v}$ lies in $\mathcal{C}(\mathbf{X})$, and $\mathbf{1}'(\mathbf{H} - \mathbf{H}_0)\mathbf{v} = \mathbf{1}'\mathbf{H}\mathbf{v} - \mathbf{1}'\mathbf{H}_0\mathbf{v} = \mathbf{1}'\mathbf{v} - \mathbf{1}'\mathbf{v} = 0$ using $\mathbf{1}'\mathbf{H} = (\mathbf{H}\mathbf{1})' = \mathbf{1}'$ and $\mathbf{1}'\mathbf{H}_0 = \mathbf{1}'$; conversely a vector $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ with $\mathbf{1}'\mathbf{u} = 0$ has $(\mathbf{H}-\mathbf{H}_0)\mathbf{u} = \mathbf{u}-\mathbf{0} = \mathbf{u}$. So $\mathbf{H}-\mathbf{H}_0$ is the symmetric idempotent with column space $\mathcal{C}(\mathbf{X})\cap\mathbf{1}^\perp$. Finally $(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{H}\mathbf{M} - \mathbf{H}_0\mathbf{H}\mathbf{M} = \mathbf{0}$ by Theorem 2.9(iii) and $\mathbf{H}_0 = \mathbf{H}_0\mathbf{H}$. □

2.3 Residuals and leverage

Theorem 2.11 (Residual identities; CLM Theorem 2.12)**.** The least squares residuals satisfy

(i) $\mathbf{X}'\mathbf{e} = \mathbf{0}$;

(ii) $\sum_i e_i = 0$ and $\overline{\hat{Y}} = \bar{Y}$, provided the model contains an intercept;

(iii) $\hat{\mathbf{Y}}'\mathbf{e} = 0$.

Proof. **(i)** Rearranging (7), $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$. **(ii)** The first column of $\mathbf{X}$ is $\mathbf{1}$, so the first component of $\mathbf{X}'\mathbf{e} = \mathbf{0}$ is $\mathbf{1}'\mathbf{e} = \sum_i e_i = 0$; then $\mathbf{1}'\hat{\mathbf{Y}} = \mathbf{1}'(\mathbf{Y} - \mathbf{e}) = \mathbf{1}'\mathbf{Y}$, and dividing by n gives $\overline{\hat{Y}} = \bar{Y}$. **(iii)** $\hat{\mathbf{Y}}'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = 0$ by (i). □

Remark (Errors versus residuals; FREES §2.3.2, §3.3)**.** The **error** $\varepsilon_i = Y_i - \mathbf{x}_i'\boldsymbol{\beta}$ is unobservable; the **residual** $e_i = Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}$ is observable. Since $\mathbf{MX} = \mathbf{0}$,

$$\mathbf{e} = \mathbf{MY} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}, \tag{9}$$

a linear map of rank $n - k - 1$ of the errors: the residuals satisfy the $k+1$ constraints (i) whatever the data, and FREES's phrase is that only $n - (k+1)$ of them are free. The identities hold for

every fit, so a residual mean of zero is not evidence that the model is right.

Definition 2.12 (Leverage; ISLR §3.3.3; CLM Definition 2.13)**.** The **leverage** of observation i is h_{ii}, the i-th diagonal entry of $\mathbf{H}$, so that $\hat{Y}_i = \sum_j h_{ij} Y_j$ and $h_{ii} = \partial \hat{Y}_i / \partial Y_i$.

Theorem 2.13 (Properties of leverage; CLM Theorem 2.14, Exercise D3)**.**

(i) $0 \leqslant h_{ii} \leqslant 1$, and $h_{ii} = 1$ forces $h_{ij} = 0$ for $j \neq i$ and $e_i = 0$;

(ii) $\sum_{i=1}^{n} h_{ii} = k + 1$, so the mean leverage is $(k+1)/n$;

(iii) with an intercept, $h_{ii} \geqslant 1/n$, and for $k = 1$

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}. \tag{10}$$

Proof. **(i)** $\mathbf{H} = \mathbf{H}'\mathbf{H}$ by Theorem 2.9(i); comparing (i, i) entries with $h_{ji} = h_{ij}$,

$$h_{ii} = \sum_{j=1}^{n} h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2.$$

The left equality gives $h_{ii} \geqslant 0$; dropping the sum gives $h_{ii} \geqslant h_{ii}^2$, hence $h_{ii} \leqslant 1$. At $h_{ii} = 1$ the sum $\sum_{j \neq i} h_{ij}^2$ must vanish, so $\hat{Y}_i = \sum_j h_{ij} Y_j = Y_i$ and $e_i = 0$.
(ii) $\sum_i h_{ii} = \operatorname{tr}(\mathbf{H}) = k + 1$ by Theorem 2.9(iv).
(iii) With $\mathbf{u}_i$ the i-th coordinate vector and $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}'$, whose diagonal entries are $1/n$, Lemma 2.10 gives $h_{ii} - 1/n = \mathbf{u}_i'(\mathbf{H} - \mathbf{H}_0)\mathbf{u}_i = \|(\mathbf{H} - \mathbf{H}_0)\mathbf{u}_i\|^2 \geqslant 0$. For $k = 1$, $\mathbf{X} = [\mathbf{1}\ \mathbf{x}]$ and, with $S_{xx} = \sum_j (x_j - \bar{x})^2$,

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_j x_j^2 \end{pmatrix}, \qquad (\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_j x_j^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix},$$

the determinant being $n \sum_j x_j^2 - n^2\bar{x}^2 = nS_{xx}$. Then with $\mathbf{x}_i' = (1, x_i)$,

$$h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i = \frac{\sum_j x_j^2 - 2n\bar{x}x_i + nx_i^2}{nS_{xx}} = \frac{S_{xx} + n(x_i - \bar{x})^2}{nS_{xx}} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}},$$

using $\sum_j x_j^2 = S_{xx} + n\bar{x}^2$. □

Remark (ISLR §3.3.3)**.** h_{ii} depends on $\mathbf{X}$ alone and, with one explanatory variable, measures how far x_i lies from $\bar{x}$: by (10) it is a quadratic in $x_i - \bar{x}$ with minimum $1/n$. ISLR flags a leverage that greatly exceeds the mean $(k+1)/n$; FREES §5.4.1 flags three times the mean, and twice the mean, $2(k+1)/n$, is also in common use.

2.4 The case $k = 1$

Corollary 2.14 (The fitted line; FREES §2.1, ISLR §3.1.1; CLM Corollary 2.18)**.** With one explanatory variable and an intercept, and $x_1, \ldots, x_n$ not all equal (and, for the last form of b_1, $y_1, \ldots, y_n$ not all equal),

$$b_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2} = r\,\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1\bar{x}, \tag{11}$$

so the fitted line $\hat{y} = b_0 + b_1 x$ passes through $(\bar{x}, \bar{y})$ and $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$.

Proof. With $\mathbf{X} = [\mathbf{1}\ \mathbf{x}]$ the normal equations (7) are

$$\begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \end{pmatrix} = \begin{pmatrix} \sum_i y_i \\ \sum_i x_i y_i \end{pmatrix}.$$

The first row is $nb_0 + b_1 \sum_i x_i = \sum_i y_i$, that is $b_0 = \bar{y} - b_1 \bar{x}$. Substituting into the second row, $\sum_i x_i y_i = (\bar{y} - b_1\bar{x})n\bar{x} + b_1 \sum_i x_i^2$, so $b_1(\sum_i x_i^2 - n\bar{x}^2) = \sum_i x_i y_i - n\bar{x}\bar{y}$, the middle form of (11). The identities $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - 2\bar{x}\sum_i x_i + n\bar{x}^2 = \sum_i x_i^2 - n\bar{x}^2$ and $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - \bar{y}\sum_i x_i - \bar{x}\sum_i y_i + n\bar{x}\bar{y} = \sum_i x_i y_i - n\bar{x}\bar{y}$ give the first form, and dividing numerator and denominator by $n-1$ gives $b_1 = \{(n-1)s_x s_y r\}/\{(n-1)s_x^2\} = r\, s_y/s_x$. Finally $\hat{y}_i - \bar{y} = b_0 + b_1 x_i - \bar{y} = b_1(x_i - \bar{x})$. □

Corollary 2.15 (Regression through the origin; FREES exercise 2.7, ISLR §3.7; CLM Corollary 2.20)**.** For the model $y_i = \beta x_i + \varepsilon_i$ without an intercept, and some $x_i \neq 0$, the least squares estimate is $\hat{\beta} = \sum_i x_i y_i / \sum_i x_i^2$ and $\hat{y}_i = x_i \hat{\beta}$.

Proof. Here $\mathbf{X} = \mathbf{x}$, so $\mathbf{X}'\mathbf{X} = \sum_i x_i^2$, a positive scalar when some $x_i \neq 0$, and $\mathbf{X}'\mathbf{Y} = \sum_i x_i y_i$; Theorem 2.6 gives the ratio. □

Example 2.6 (Lottery sales, continued; FREES §2.1)**.** From Example 2.2 and the summary statistics, $b_1 = r\, s_y/s_x = 0.8863(8103)/11{,}098 = 0.6471$ and $b_0 = \bar{y} - b_1\bar{x} = 6495 - 0.6471(9311) = 470$: the fitted line is $\widehat{\texttt{SALES}} = 470 + 0.647\,\texttt{POP}$, and at a population of 10,000 it gives $470 + 6471 = 6941$ dollars of sales.

```
r <- cor(lottery$POP, lottery$SALES)
b1 <- r * sd(lottery$SALES) / sd(lottery$POP)
b0 <- mean(lottery$SALES) - b1 * mean(lottery$POP)
fit_lot <- lm(SALES ~ POP, data = lottery)
rbind(formula = c(b0 = b0, b1 = b1), lm = coef(fit_lot))
```

```
##             b0     b1
## formula 469.7 0.6471
## lm      469.7 0.6471
```

```
c(at_10000 = b0 + b1 * 10000, predict = unname(predict(fit_lot, data.frame(POP =
↪  10000))))
```

```
## at_10000  predict
##     6941     6941
```

Example 2.7 (Advertising, continued; ISLR §3.1.1)**.** Regressing `sales` on `TV` gives $\hat{\beta}_0 = 7.03$ and $\hat{\beta}_1 = 0.0475$: an additional \$1,000 of television advertising is associated with about 47.5 additional units sold.

```
fit_tv <- lm(sales ~ TV, data = adv)
coef(fit_tv)
```

```
## (Intercept)          TV
##     7.03259     0.04754
```

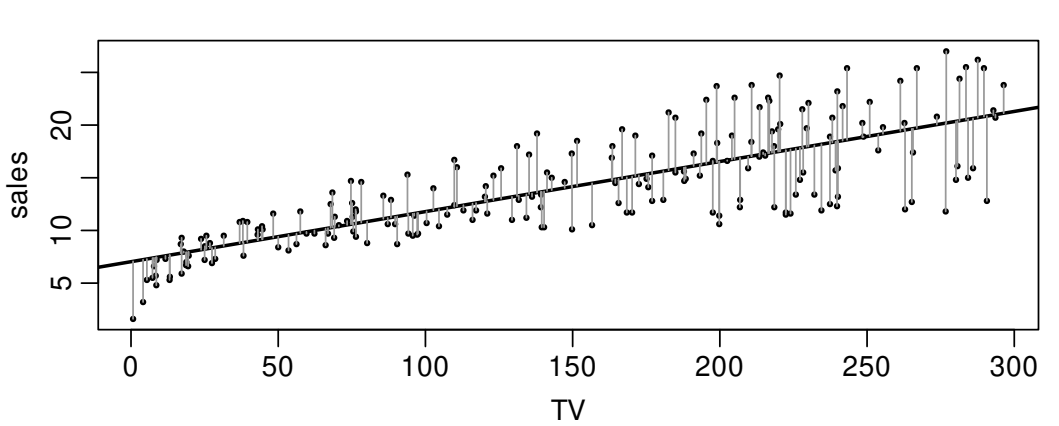


Figure 2.5: The least squares line for sales on television advertising; each grey segment is a residual (ISLR Figure 3.1).

Example 2.8 (Term life insurance, continued; FREES §3.1–3.2)**.** The regression of `LNFACE` on `EDUCATION`, `NUMHH` and `LNINCOME` has $n = 275$ and $k = 3$; the first rows of $\mathbf{X}$ and $\mathbf{Y}$ and the fitted plane $\hat{y} = 2.584 + 0.206x_1 + 0.306x_2 + 0.494x_3$:

```r
fit_term <- lm(LNFACE ~ EDUCATION + NUMHH + LNINCOME, data = term)
X_term <- model.matrix(fit_term)
round(cbind(X_term, LNFACE = term$LNFACE)[c(1, 2, 275), ], 3)
```

```
##     (Intercept) EDUCATION NUMHH LNINCOME LNFACE
## 1             1        16     3   10.669  9.903
## 2             1         9     3    9.393 11.775
## 500           1        12     1   10.545  9.210
```

```r
round(coef(fit_term), 3)
```

```
## (Intercept)   EDUCATION       NUMHH    LNINCOME
##       2.584       0.206       0.306       0.494
```

The three slopes are positive: with income and education fixed, a larger household demands more insurance; with household size and education fixed, a household with more income buys more; more education, other things equal, goes with more insurance.

Remark (Reading a coefficient on a logarithmic scale; FREES §3.2.2)**.** With y in logarithms, $\partial \ln y/\partial x = (\partial y/\partial x)/y$: a small change in $\ln y$ is a small proportional change in y, and a change of δ in x_j changes y by a factor $e^{\beta_j \delta} \approx 1 + \beta_j \delta$. With x_j also in logarithms, $\beta_j = \partial \ln y/\partial \ln x_j =$

$(\partial y/y)/(\partial x_j/x_j)$ is the ratio of the percentage changes, an **elasticity**. Natural logarithms are used because these identities need them.

Example 2.9 (Term life insurance, continued; FREES §3.2.2)**.** Holding `NUMHH` and `LNINCOME` at their sample means and moving `EDUCATION` in steps of 0.1 year raises $\widehat{\texttt{LNFACE}}$ by a constant 0.0206 per step, and the fitted face amount by a constant 2.09% per step; the continuous version is $0.206 \times 0.1 = 2.06\%$. Moving `LNINCOME` in steps of 0.1 raises income by 10.52% per step and the fitted face amount by 5.06%, a ratio of 0.481 against the coefficient 0.494.

```
b <- coef(fit_term); mN <- mean(term$NUMHH); mL <- mean(term$LNINCOME)
ed <- c(14, 14.1, 14.2, 14.3)
lnface_ed <- b[1] + b[2] * ed + b[3] * mN + b[4] * mL
round(rbind(EDUCATION = ed, LNFACE = lnface_ed, FACE = exp(lnface_ed),
            pct_change = c(NA, 100 * diff(exp(lnface_ed)) / exp(lnface_ed)[-4])), 3)
```

```
##                [,1]       [,2]       [,3]       [,4]
## EDUCATION     14.00     14.100     14.200     14.300
## LNFACE        11.88     11.903     11.923     11.944
## FACE      144669.50 147686.713 150766.852 153911.229
## pct_change       NA      2.086      2.086      2.086
```

```r
li <- c(11, 11.1, 11.2, 11.3)
lnface_li <- b[1] + b[2] * mean(term$EDUCATION) + b[3] * mN + b[4] * li
inc_pct <- 100 * diff(exp(li)) / exp(li)[-4]
face_pct <- 100 * diff(exp(lnface_li)) / exp(lnface_li)[-4]
round(rbind(LNINCOME = li, INCOME_pct = c(NA, inc_pct), FACE_pct = c(NA, face_pct),
            ratio = c(NA, face_pct / inc_pct)), 3)
```

```
##             [,1]   [,2]   [,3]   [,4]
## LNINCOME      11 11.100 11.200 11.300
## INCOME_pct    NA 10.517 10.517 10.517
## FACE_pct      NA  5.059  5.059  5.059
## ratio         NA  0.481  0.481  0.481
```

So, holding household size and education fixed, a 1% increase in income is associated with about a 0.48% increase in the face amount.

Example 2.10 (Simple against multiple regression; ISLR §3.2.1)**.** Regressed alone on `newspaper`, sales rise by 0.055 thousand units per thousand dollars; with `TV` and `radio` in the same model the coefficient is -0.001.

```r
fit_all <- lm(sales ~ TV + radio + newspaper, data = adv)
rbind(alone    = coef(lm(sales ~ newspaper, data = adv))["newspaper"],
      together = coef(fit_all)["newspaper"])
```

```
##          newspaper
## alone     0.054693
## together -0.001037
```

```r
round(coef(fit_all), 4)
```

```
## (Intercept)          TV       radio   newspaper
##      2.9389      0.0458      0.1885     -0.0010
```

The simple coefficient measures the association ignoring the other media; the multiple coefficient measures it with `TV` and `radio` held fixed. The two differ because `newspaper` and `radio` spending are correlated (0.35 in Example 2.5): markets that spend on radio, which does raise sales, also spend on newspapers, so newspaper spending stands in for radio spending when radio is left out. ISLR's illustration is a regression of shark attacks on ice cream sales, which is positive until temperature is added.

3 Sampling Properties of Least Squares

FREES §2.3.2, §2.4, §3.2.4, §3.3 ·ISLR §3.1.2 ·CLM §3. SRM learning outcome 2(a).

Definition 2.16 (Sums of squares, mean squares and the residual standard deviation; FREES §2.3, §3.3, ISLR §3.1.3; CLM Definition 3.1)**.** For the fitted model $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$,

$$\text{SST} = \sum_i (Y_i - \bar{Y})^2, \qquad \text{SSR} = \sum_i (\hat{Y}_i - \bar{Y})^2, \qquad \text{SSE} = \sum_i (Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e},$$

$$\text{MSE} = s^2 = \frac{\text{SSE}}{n-k-1} \quad (n > k+1), \qquad \text{MSR} = \frac{\text{SSR}}{k} \quad (k \geqslant 1).$$

s is the **residual standard deviation** (FREES) or **residual standard error**, RSE (ISLR). FREES writes **Total SS**, **Regression SS** and **Error SS**; ISLR writes TSS and RSS for SST and SSE; the SOA questions use both.

Proposition 2.17 (Coefficients are weighted sums of the responses; FREES §2.4, §3.2.4; CLM Proposition 2.19)**.** $\hat{\boldsymbol{\beta}} = \mathbf{WY}$ with $\mathbf{W} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ fixed by the design, so each $\hat{\beta}_j = \sum_i w_{ji} Y_i$. For $k = 1$,

$$b_1 = \sum_{i=1}^{n} w_i Y_i, \qquad w_i = \frac{x_i - \bar{x}}{(n-1)s_x^2}, \qquad \sum_i w_i = 0, \quad \sum_i w_i x_i = 1, \quad \sum_i w_i^2 = \frac{1}{(n-1)s_x^2}. \tag{12}$$

Proof. The first statement is (5). For $k = 1$, Corollary 2.14 gives $b_1 = \sum_i (x_i - \bar{x})(Y_i - \bar{Y})/\{(n-1)s_x^2\} = \sum_i w_i (Y_i - \bar{Y}) = \sum_i w_i Y_i - \bar{Y} \sum_i w_i$, and $\sum_i w_i = \sum_i (x_i - \bar{x})/\{(n-1)s_x^2\} = 0$. Next $\sum_i w_i x_i = \sum_i (x_i - \bar{x}) x_i / \{(n-1)s_x^2\}$, and $\sum_i (x_i - \bar{x}) x_i = \sum_i (x_i - \bar{x})^2 + \bar{x} \sum_i (x_i - \bar{x}) = (n-1)s_x^2$, so the sum is 1. Finally $\sum_i w_i^2 = \sum_i (x_i - \bar{x})^2 / \{(n-1)s_x^2\}^2 = 1/\{(n-1)s_x^2\}$. □

3.1 Unbiasedness and variance

Theorem 2.18 (Unbiasedness; CLM Theorem 3.3, equation (12))**.** Under A1–A2 and A4, $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$.

Proof. Substituting $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into (5),

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}, \tag{13}$$

and $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ by A2, so $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{0} = \boldsymbol{\beta}$. □

Theorem 2.19 (Variance–covariance matrix; CLM Theorem 3.4)**.** Under A1–A4, $\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$; in particular $\operatorname{var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ and $\operatorname{cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{ij}$.

Proof. By A3, $\operatorname{var}(\mathbf{Y}) = \operatorname{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$. By (13) and the prerequisites' proposition **variance under a linear map**, $\operatorname{var}(\mathbf{A}\mathbf{z}) = \mathbf{A}\operatorname{var}(\mathbf{z})\mathbf{A}'$, with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$,

$$\operatorname{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I}_n)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$ □

Corollary 2.20 (The case $k = 1$; FREES §2.4, ISLR §3.1.2; CLM Corollary 3.13)**.** Under A1–A4, with $x_1, \ldots, x_n$ not all equal and $S_{xx} = \sum_i (x_i - \bar{x})^2 = (n-1)s_x^2$,

$$\operatorname{var}(b_1) = \frac{\sigma^2}{S_{xx}}, \qquad \operatorname{var}(b_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right), \qquad \operatorname{cov}(b_0, b_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}}. \tag{14}$$

Proof. Reading $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ off the explicit inverse in the proof of Theorem 2.13(iii): the $(2,2)$ entry is $\sigma^2 n/(nS_{xx})$, the $(1,1)$ entry $\sigma^2 \sum_i x_i^2/(nS_{xx}) = \sigma^2(S_{xx}+n\bar{x}^2)/(nS_{xx})$, and the off-diagonal entry $-\sigma^2 n\bar{x}/(nS_{xx})$. FREES's route is Proposition 2.17: the Y_i are uncorrelated with common variance, so $\operatorname{var}(b_1) = \sum_i w_i^2 \sigma^2 = \sigma^2/\{(n-1)s_x^2\}$. □

Remark (The three determinants of precision; FREES §2.4, ISLR §3.1.2)**.** $\operatorname{var}(b_1) = \sigma^2/\{(n-1)s_x^2\}$ falls with more observations, with less scatter about the line, and with more spread in the explanatory variable: doubling s_x at fixed n and σ halves the standard deviation of the slope, which is what ISLR means by more **leverage** to estimate a slope. The spread of x is a design choice. $\operatorname{var}(b_0)$ reduces to σ^2/n, the variance of a sample mean, when $\bar{x} = 0$, in which case $b_0 = \bar{Y}$.

3.2 The Gauss–Markov theorem

Theorem 2.21 (Gauss–Markov; FREES §3.2.4; CLM Theorem 3.5)**.** Under A1–A4, $\hat{\boldsymbol{\beta}}$ is the **best linear unbiased estimator** of $\boldsymbol{\beta}$: for any estimator $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{Y}$ with $\mathbf{C}$ a constant matrix and $\mathsf{E}[\tilde{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ for every $\boldsymbol{\beta}$, the matrix $\mathrm{var}(\tilde{\boldsymbol{\beta}}) - \mathrm{var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite, so $\mathrm{var}(\tilde{\beta}_j) \geqslant \mathrm{var}(\hat{\beta}_j)$ for every j, and equality of the two variance matrices forces $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$.

Proof. Step 1: unbiasedness constrains C. $\mathsf{E}[\tilde{\boldsymbol{\beta}}] = \mathbf{C}\,\mathsf{E}[\mathbf{Y}] = \mathbf{C}\mathbf{X}\boldsymbol{\beta}$, and this equals $\boldsymbol{\beta}$ for every $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$ iff $\mathbf{C}\mathbf{X} = \mathbf{I}_{k+1}$.
Step 2: decompose C. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$. Then $\mathbf{D}\mathbf{X} = \mathbf{C}\mathbf{X} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_{k+1} - \mathbf{I}_{k+1} = \mathbf{0}$.
Step 3: expand the variance. By the prerequisites' **variance under a linear map** and A3, $\mathrm{var}(\tilde{\boldsymbol{\beta}}) = \mathbf{C}(\sigma^2\mathbf{I})\mathbf{C}' = \sigma^2\mathbf{C}\mathbf{C}'$, and

$$\begin{aligned}\mathbf{C}\mathbf{C}' &= \left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}\right]\left[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'\right] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' + \mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}\mathbf{D}'.\end{aligned}$$

The first term is $(\mathbf{X}'\mathbf{X})^{-1}$. The third vanishes because $\mathbf{DX} = \mathbf{0}$, and the second is its transpose, $(\mathbf{DX}(\mathbf{X}'\mathbf{X})^{-1})' = \mathbf{0}$. Hence

$$\operatorname{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{DD}' = \operatorname{var}(\hat{\boldsymbol{\beta}}) + \sigma^2\mathbf{DD}'.$$

Step 4: the difference is positive semi-definite. For any $\mathbf{v}$, $\mathbf{v}'\mathbf{DD}'\mathbf{v} = \|\mathbf{D}'\mathbf{v}\|^2 \geqslant 0$. Taking $\mathbf{v}$ the j-th coordinate vector gives $\operatorname{var}(\tilde{\beta}_j) \geqslant \operatorname{var}(\hat{\beta}_j)$. If $\mathbf{DD}' = \mathbf{0}$ then every row of $\mathbf{D}$ has zero length, so $\mathbf{D} = \mathbf{0}$ and $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. □

Remark. The theorem ranks least squares within the estimators that are linear in $\mathbf{Y}$ and unbiased; it uses no normality and says nothing about prediction error on new data. Biased estimators, such as those that shrink $\hat{\boldsymbol{\beta}}$ toward zero, can have smaller mean squared error.

3.3 Estimating the error variance

Theorem 2.22 (Unbiasedness of the mean square error; FREES §2.3.2, §3.3; CLM Theorem 3.7)**.** Under A1–A4, $\mathsf{E}[\text{SSE}] = (n-k-1)\sigma^2$, so $\mathsf{E}[s^2] = \sigma^2$.

Proof. By (9) and the symmetry and idempotency of $\mathbf{M}$, $\text{SSE} = \mathbf{e}'\mathbf{e} = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$. The prerequisites' lemma **expectation of a quadratic form**, $\mathsf{E}[\mathbf{z}'\mathbf{A}\mathbf{z}] = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$ for $\mathsf{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\operatorname{var}(\mathbf{z}) = \boldsymbol{\Sigma}$, applied with $\mathbf{z} = \boldsymbol{\varepsilon}$, $\boldsymbol{\mu} = \mathbf{0}$ and $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}_n$, gives $\mathsf{E}[\text{SSE}] = \operatorname{tr}(\mathbf{M}\sigma^2\mathbf{I}_n) = \sigma^2 \operatorname{tr}(\mathbf{M}) = \sigma^2(n-k-1)$ by Theorem 2.9(iv). Dividing by $n-k-1$ gives $\mathsf{E}[s^2] = \sigma^2$. □

Remark (The divisor; FREES §2.3.2, §3.3)**.** The divisor is $n-k-1 = \operatorname{tr}(\mathbf{M})$ because the residual vector lies in the $(n-k-1)$-dimensional space $\mathcal{C}(\mathbf{X})^{\perp}$: $k+1$ observations determine the plane exactly and leave no residual, and a further observation is needed before there is any variation about it. Dividing SSE by n gives an estimator biased downward by the factor $(n-k-1)/n$.

Theorem 2.23 (Variance of the residuals; CLM Theorem 3.8)**.** Under A1–A4, $\operatorname{var}(\mathbf{e}) = \sigma^2\mathbf{M}$: $\operatorname{var}(e_i) = \sigma^2(1-h_{ii})$ and $\operatorname{cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

Proof. By (9) and **variance under a linear map**, $\operatorname{var}(\mathbf{e}) = \mathbf{M}(\sigma^2\mathbf{I}_n)\mathbf{M}' = \sigma^2\mathbf{M}^2 = \sigma^2\mathbf{M}$; the (i,i) entry of $\mathbf{I}-\mathbf{H}$ is $1-h_{ii}$ and the (i,j) entry is $-h_{ij}$. □

Definition 2.24 (Standard errors; FREES §2.4, §3.4, ISLR §3.1.2)**.** The **standard error** of $\hat{\beta}_j$ is its estimated standard deviation,

$$\operatorname{SE}(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}, \qquad \operatorname{SE}(b_1) = \frac{s}{s_x\sqrt{n-1}}, \qquad \operatorname{SE}(b_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2}}, \tag{15}$$

the last two for $k=1$, by Theorem 2.19 and Corollary 2.20 with σ replaced by s. ISLR writes $\operatorname{SE}(\hat{\beta}_j)$ for the same quantity and notes that, strictly, it is an estimate.

Example 2.11 (Lottery sales, continued; FREES §2.4)**.** With $s = 3792$, $s_x = 11{,}098$ and $n = 50$, $\operatorname{SE}(b_1) = 3792/(11{,}098\sqrt{49}) = 0.0488$.

```
s_lot <- summary(fit_lot)$sigma
c(s = s_lot, se_b1 = s_lot / (sd(lottery$POP) * sqrt(49)),
  se_b1_lm = coef(summary(fit_lot))["POP", "Std. Error"])
```

```
##          s      se_b1   se_b1_lm
## 3791.75848    0.04881    0.04881
```

Example 2.12 (Advertising, continued; ISLR §3.1.2)**.** The standard errors of ISLR Table 3.1, from the formulas (15) and from `lm`:

```
s_tv <- summary(fit_tv)$sigma; xt <- adv$TV; n_adv <- nrow(adv)
rbind(formula = c(intercept = s_tv * sqrt(1/n_adv + mean(xt)^2 / sum((xt -
 ↪  mean(xt))^2)),
                  TV = s_tv / sqrt(sum((xt - mean(xt))^2))),
      lm = coef(summary(fit_tv))[, "Std. Error"])
```

```
##         intercept       TV
## formula    0.4578 0.002691
## lm         0.4578 0.002691
```

Example 2.13 (The population line and the least squares line; ISLR §3.1.2)**.** ISLR Figure 3.3 draws 100 observations from $Y = 2 + 3X + \varepsilon$ and fits a line; repeated on ten independent samples, the fitted lines scatter about the population line, and averaged over many samples they reproduce it, which is Theorem 2.18.

```r
set.seed(2)
x_pop <- runif(100, -2, 2)
fits <- replicate(2000, coef(lm(I(2 + 3 * x_pop + rnorm(100, sd = 2)) ~ x_pop)))
Sxx_pop <- sum((x_pop - mean(x_pop))^2)
rbind(true = c(2, 3), mean_of_fits = rowMeans(fits), sd_of_fits = apply(fits, 1, sd),
      sd_by_formula = 2 * sqrt(c(1 / 100 + mean(x_pop)^2 / Sxx_pop, 1 / Sxx_pop)))
```

```
##                (Intercept)  x_pop
## true                2.0000 3.0000
## mean_of_fits        2.0043 2.9930
## sd_of_fits          0.2094 0.1646
## sd_by_formula       0.2001 0.1683
```

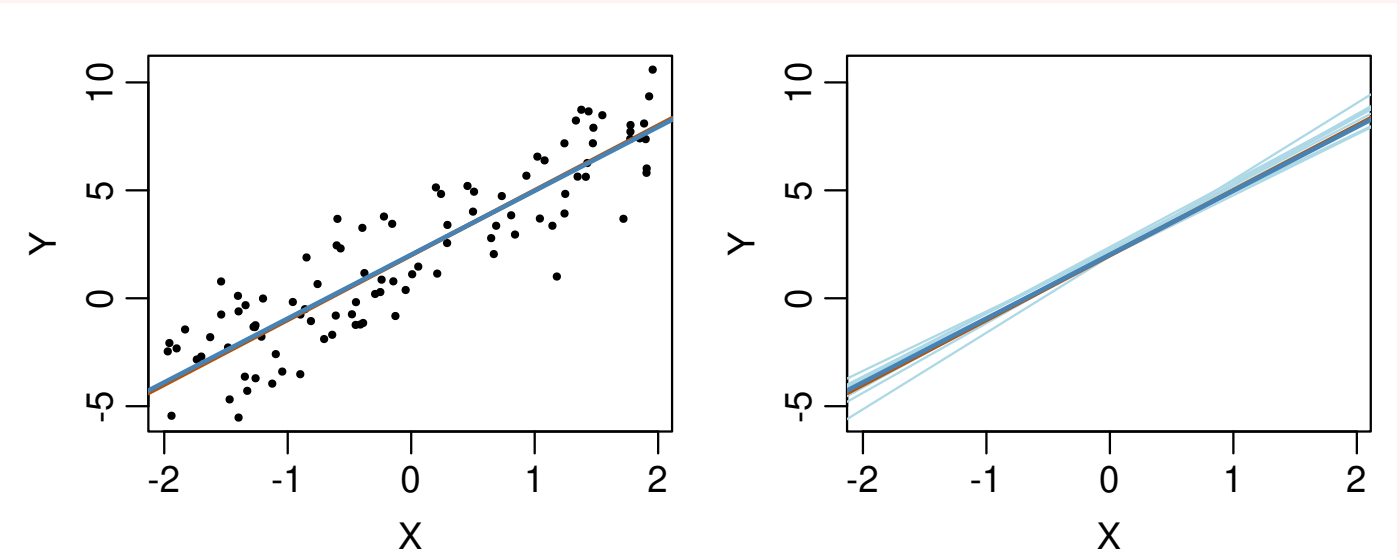


Figure 2.6: Left: the population line $2 + 3X$ (orange) and the least squares line (blue) for one sample. Right: ten least squares lines from ten samples (ISLR Figure 3.3).

The mean of 2000 fitted intercepts and slopes agrees with $(2, 3)$ within 0.01, and the spread of the fits is close to the standard deviation (14) (row `sd_by_formula`).

4 Goodness of Fit

FREES §2.3, §3.3 ·ISLR §3.1.3, §3.2.2 ·CLM §6. SRM learning outcome 2(e).

4.1 The analysis of variance

Definition 2.25 (Centering matrix; CLM Definition 6.1)**.** $\mathbf{M}_0 = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}' = \mathbf{I}_n - \mathbf{H}_0$, so that $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$. $\mathbf{M}_0$ is the residual maker of the intercept-only model: symmetric idempotent, with $\text{tr}(\mathbf{M}_0) = n - 1$, projecting onto $\mathbf{1}^\perp$.

Theorem 2.26 (Variance decomposition; FREES §2.3.1, §3.3, ISLR §3.1.3; CLM Theorem 6.4, equation (24))**.** In a model with an intercept,

$$\text{SST} = \text{SSR} + \text{SSE}, \tag{16}$$

with $\text{SST} = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$, $\text{SSR} = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}} = \|(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}\|^2$ and $\text{SSE} = \mathbf{e}'\mathbf{e} = \mathbf{Y}'\mathbf{M}\mathbf{Y}$.

Proof. Step 1: the matrix forms. For any $\mathbf{v}$, since $\mathbf{M}_0'\mathbf{M}_0 = \mathbf{M}_0$, $\mathbf{v}'\mathbf{M}_0\mathbf{v} = \|\mathbf{M}_0\mathbf{v}\|^2 = \|\mathbf{v} - \bar{v}\mathbf{1}\|^2 = \sum_i (v_i - \bar{v})^2$. With $\mathbf{v} = \mathbf{Y}$ this is SST; with $\mathbf{v} = \hat{\mathbf{Y}}$ it is SSR, because $\bar{\hat{Y}} = \bar{Y}$ by Theorem 2.11(ii). Also $\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{M}_0\mathbf{H}\mathbf{Y} = (\mathbf{H} - \mathbf{H}_0)\mathbf{Y}$, since $\mathbf{H}_0\mathbf{H} = \mathbf{H}_0$ by Lemma 2.10; and $\mathbf{Y}'\mathbf{M}\mathbf{Y} = \|\mathbf{M}\mathbf{Y}\|^2 = \|\mathbf{e}\|^2$.
Step 2: decompose. Since $\bar{e} = 0$, $\mathbf{M}_0\mathbf{e} = \mathbf{e}$ and

$$\mathbf{M}_0\mathbf{Y} = \mathbf{M}_0(\hat{\mathbf{Y}} + \mathbf{e}) = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}. \tag{17}$$

Step 3: expand. $\text{SST} = \|\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}\|^2 = \text{SSR} + 2\,\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} + \text{SSE}$.
Step 4: the cross term vanishes. $\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{e}'\hat{\mathbf{Y}} - \bar{Y}\,\mathbf{e}'\mathbf{1} = 0 - 0$ by Theorem 2.11(iii) and (ii). □

Remark (FREES §2.3.1)**.** Equation (16) is FREES's "total deviation equals unexplained deviation plus explained deviation", $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$, squared and summed; the cross products sum to zero for the least squares fit because of the normal equations (Step 4); for an arbitrary line they need not vanish. Without an intercept $\bar{e}$ is in general nonzero (unless $\mathbf{1} \in \mathcal{C}(\mathbf{X})$), the cross term $-2n\bar{Y}\bar{e}$ survives, and (16) in general fails.

Source	Sum of squares	df	Mean square
Regression	SSR	k	$\text{MSR} = \text{SSR}/k$
Error	SSE	$n-k-1$	$\text{MSE} = \text{SSE}/(n-k-1) = s^2$
Total	SST	$n-1$	

Remark (The ANOVA table; FREES §2.3.2, §3.3)**.** The degrees of freedom are the dimensions of the three subspaces of Lemma 2.10: $\mathbf{M}_0\mathbf{Y}$ lies in $\mathbf{1}^\perp$ $(n-1)$, and splits into $(\mathbf{H}-\mathbf{H}_0)\mathbf{Y} \in \mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ (k) and $\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp$ $(n-k-1)$. For $k=1$ the regression sum of squares equals its mean square.

Example 2.14 (Lottery sales, continued; FREES §2.3)**.**

```
1 anova(fit_lot)
```

```
## Analysis of Variance Table
##
## Response: SALES
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## POP         1 2527165015 2527165015     176 <2e-16 ***
## Residuals 48  690116755   14377432
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
c(R2 = summary(fit_lot)$r.squared, s = s_lot, s_y = sd(lottery$SALES))
```

```
##        R2         s       s_y
##    0.7855 3791.7585 8103.0125
```

The regression explains 78.5% of the variation and the typical deviation from the line is $s = 3792$ against $s_y = 8103$ without the explanatory variable.

4.2 The coefficient of determination

Definition 2.27 (R^2 and adjusted R^2; FREES §2.3.1, §3.3, ISLR §3.1.3; CLM Definition 6.6)**.**

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}, \qquad R^2_a = 1 - \frac{\text{SSE}/(n-k-1)}{\text{SST}/(n-1)} = 1 - \frac{s^2}{s^2_y} = 1 - \frac{n-1}{n-k-1}(1-R^2), \quad (18)$$

the proportion of the variation in Y explained by the regression and its version adjusted for degrees of freedom; R^2_a can be negative. $R = \sqrt{R^2}$ is the **multiple correlation coefficient**.

Proposition 2.28 (Bounds and monotonicity; FREES §2.3, §3.3, ISLR §3.2.2; CLM Proposition 6.7, Theorem 6.8)**.** In a model with an intercept and SST > 0, $0 \leqslant R^2 \leqslant 1$; $R^2 = 1$ iff every residual is zero and $R^2 = 0$ iff $\hat{\mathbf{Y}} = \bar{Y}\mathbf{1}$. Adding an explanatory variable cannot decrease R^2, and can decrease R^2_a.

Proof. SSR $\geqslant 0$ and SST > 0 give $R^2 \geqslant 0$, with equality iff $\hat{Y}_i = \bar{Y}$ for every i; SST = SSR+SSE with SSE $\geqslant 0$ gives $R^2 \leqslant 1$, with equality iff SSE $= 0$. For monotonicity let $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ be the column spaces of the smaller and the enlarged design. By Theorem 2.7, $\hat{\mathbf{Y}}_{k+1}$ is the point of $\mathcal{C}_{k+1}$ closest to $\mathbf{Y}$, and $\hat{\mathbf{Y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so $\text{SSE}_{k+1} = \|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\|^2 \leqslant \|\mathbf{Y} - \hat{\mathbf{Y}}_k\|^2 = \text{SSE}_k$, while SST does not involve the design; hence $R^2_{k+1} \geqslant R^2_k$. For R^2_a, SST and $n-1$ are fixed and $s^2 = \text{SSE}/(n-k-1)$ rises whenever SSE falls by less than the factor $(n-k-2)/(n-k-1)$. □

Proposition 2.29 (R^2 as a squared correlation; FREES §3.3, ISLR §3.2.2; CLM Proposition 6.12)**.** In a model with an intercept, SST > 0 and SSR > 0, $R^2 = \text{cor}(Y, \hat{Y})^2$, the squared sample correlation between the responses and the fitted values; and among all vectors $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ that are not constant, $\text{cor}(Y, v)^2$ is largest at $\mathbf{v} = \hat{\mathbf{Y}}$.

Proof. Step 1: the correlation with $\hat{\mathbf{Y}}$. By Theorem 2.11(ii), $\overline{\hat{Y}} = \bar{Y}$, so the numerator of $\operatorname{cor}(Y, \hat{Y})$ is

$$\sum_i (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y}) = \hat{\mathbf{Y}}'\mathbf{M}_0\mathbf{Y} = \hat{\mathbf{Y}}'\mathbf{M}_0(\hat{\mathbf{Y}} + \mathbf{e}) = \mathrm{SSR} + \hat{\mathbf{Y}}'\mathbf{e} = \mathrm{SSR},$$

using $\mathbf{M}_0\mathbf{e} = \mathbf{e}$ and Theorem 2.11(iii). The two sums of squares in the denominator are SST and SSR, so $\operatorname{cor}(Y, \hat{Y})^2 = \mathrm{SSR}^2/(\mathrm{SST} \cdot \mathrm{SSR}) = R^2$.

Step 2: the maximum. Let $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and put $\tilde{\mathbf{Y}} = \mathbf{M}_0\mathbf{Y}$, $\tilde{\mathbf{v}} = \mathbf{M}_0\mathbf{v}$. Since $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{H}_0\mathbf{v} \in \mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$, $\tilde{\mathbf{v}} = \mathbf{v} - \mathbf{H}_0\mathbf{v}$ lies in $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^{\perp}$, the column space of $\mathbf{P} = \mathbf{H} - \mathbf{H}_0$ (Lemma 2.10), so $\mathbf{P}\tilde{\mathbf{v}} = \tilde{\mathbf{v}}$ and $\tilde{\mathbf{Y}}'\tilde{\mathbf{v}} = \tilde{\mathbf{Y}}'\mathbf{P}\tilde{\mathbf{v}} = (\mathbf{P}\tilde{\mathbf{Y}})'\tilde{\mathbf{v}}$. By the Cauchy–Schwarz inequality of the prerequisites,

$$\operatorname{cor}(Y, v)^2 = \frac{(\tilde{\mathbf{Y}}'\tilde{\mathbf{v}})^2}{\|\tilde{\mathbf{Y}}\|^2\|\tilde{\mathbf{v}}\|^2} \leqslant \frac{\|\mathbf{P}\tilde{\mathbf{Y}}\|^2\|\tilde{\mathbf{v}}\|^2}{\|\tilde{\mathbf{Y}}\|^2\|\tilde{\mathbf{v}}\|^2} = \frac{\|\mathbf{P}\mathbf{M}_0\mathbf{Y}\|^2}{\mathrm{SST}} = \frac{\|(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}\|^2}{\mathrm{SST}} = R^2,$$

using $\mathbf{P}\mathbf{M}_0 = \mathbf{P} - \mathbf{P}\mathbf{H}_0 = \mathbf{P}$ (as $\mathbf{H}\mathbf{H}_0 = \mathbf{H}_0$ and $\mathbf{H}_0^2 = \mathbf{H}_0$) and Theorem 2.26. Equality holds when $\tilde{\mathbf{v}}$ is proportional to $\mathbf{P}\tilde{\mathbf{Y}} = \mathbf{M}_0\hat{\mathbf{Y}}$, which is the case $\mathbf{v} = \hat{\mathbf{Y}}$ by Step 1. □

Lemma 2.30 (Adding one explanatory variable; CLM Lemma 6.9, equation (28))**.** Let $\mathbf{X}$ satisfy A4, let $\mathbf{X}_+ = [\mathbf{X} \;\; \mathbf{z}]$ with $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$, so that $\mathbf{z}'\mathbf{M}\mathbf{z} > 0$, and let $\hat{\gamma}$ be the coefficient of $\mathbf{z}$ in the enlarged fit, with residual sum of squares SSE_+. Then

$$\hat{\gamma} = \frac{\mathbf{z}'\mathbf{M}\mathbf{Y}}{\mathbf{z}'\mathbf{M}\mathbf{z}}, \qquad \mathrm{SSE} - \mathrm{SSE}_+ = \hat{\gamma}^2\,\mathbf{z}'\mathbf{M}\mathbf{z} = \frac{(\mathbf{z}'\mathbf{M}\mathbf{Y})^2}{\mathbf{z}'\mathbf{M}\mathbf{z}}, \qquad \mathrm{var}(\hat{\gamma}) = \frac{\sigma^2}{\mathbf{z}'\mathbf{M}\mathbf{z}}, \tag{19}$$

the last under A1–A4; and for $n > k+2$ the standard error of $\hat{\gamma}$ in the enlarged model is $\mathrm{SE}(\hat{\gamma}) = s_+/\sqrt{\mathbf{z}'\mathbf{M}\mathbf{z}}$ with $s_+^2 = \mathrm{SSE}_+/(n-k-2)$.

Proof. Step 1: the partitioned normal equations. Write the enlarged coefficient vector as $(\hat{\boldsymbol{\beta}}_+', \hat{\gamma})'$. The normal equations $\mathbf{X}_+'\mathbf{X}_+(\hat{\boldsymbol{\beta}}_+', \hat{\gamma})' = \mathbf{X}_+'\mathbf{Y}$ split into

$$\text{(a)}\;\; \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_+ + \mathbf{X}'\mathbf{z}\hat{\gamma} = \mathbf{X}'\mathbf{Y}, \qquad \text{(b)}\;\; \mathbf{z}'\mathbf{X}\hat{\boldsymbol{\beta}}_+ + \mathbf{z}'\mathbf{z}\hat{\gamma} = \mathbf{z}'\mathbf{Y}.$$

From (a), $\hat{\boldsymbol{\beta}}_+ = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{Y} - \mathbf{z}\hat{\gamma}) = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{z}\hat{\gamma}$. Substituting into (b), with $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$,

$$\mathbf{z}'\mathbf{H}\mathbf{Y} - \mathbf{z}'\mathbf{H}\mathbf{z}\hat{\gamma} + \mathbf{z}'\mathbf{z}\hat{\gamma} = \mathbf{z}'\mathbf{Y} \iff \mathbf{z}'(\mathbf{I}-\mathbf{H})\mathbf{z}\,\hat{\gamma} = \mathbf{z}'(\mathbf{I}-\mathbf{H})\mathbf{Y},$$

which is the first formula.

Step 2: the residuals. Multiplying the expression for $\hat{\boldsymbol{\beta}}_+$ by $\mathbf{X}$, $\mathbf{X}\hat{\boldsymbol{\beta}}_+ = \hat{\mathbf{Y}} - \mathbf{H}\mathbf{z}\hat{\gamma}$, so $\hat{\mathbf{Y}}_+ = \mathbf{X}\hat{\boldsymbol{\beta}}_+ + \mathbf{z}\hat{\gamma} = \hat{\mathbf{Y}} + \mathbf{M}\mathbf{z}\hat{\gamma}$ and $\mathbf{e}_+ = \mathbf{e} - \hat{\gamma}\mathbf{M}\mathbf{z}$. Taking squared lengths, with $\mathbf{e} = \mathbf{M}\mathbf{Y}$ and $\mathbf{M}^2 = \mathbf{M}$,

$$\mathrm{SSE}_+ = \mathrm{SSE} - 2\hat{\gamma}\,\mathbf{Y}'\mathbf{M}\mathbf{z} + \hat{\gamma}^2\mathbf{z}'\mathbf{M}\mathbf{z} = \mathrm{SSE} - 2\hat{\gamma}^2\mathbf{z}'\mathbf{M}\mathbf{z} + \hat{\gamma}^2\mathbf{z}'\mathbf{M}\mathbf{z} = \mathrm{SSE} - \hat{\gamma}^2\mathbf{z}'\mathbf{M}\mathbf{z},$$

substituting $\mathbf{Y}'\mathbf{M}\mathbf{z} = \mathbf{z}'\mathbf{M}\mathbf{Y} = \hat{\gamma}\,\mathbf{z}'\mathbf{M}\mathbf{z}$.

Step 3: the variance. $\hat{\gamma} = \mathbf{z}'\mathbf{M}\mathbf{Y}/(\mathbf{z}'\mathbf{M}\mathbf{z})$ is a linear map of $\mathbf{Y}$, so by **variance under a linear map** and A3, $\mathrm{var}(\hat{\gamma}) = \sigma^2\mathbf{z}'\mathbf{M}\mathbf{M}'\mathbf{z}/(\mathbf{z}'\mathbf{M}\mathbf{z})^2 = \sigma^2/(\mathbf{z}'\mathbf{M}\mathbf{z})$; the standard error replaces σ^2 by the enlarged model's s_+^2. □

Theorem 2.31 (Adjusted R^2 rises iff $|t| > 1$; CLM Theorem 6.10)**.** Let $n > k+2$ and $\mathrm{SST} > 0$, and add one explanatory variable $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$ with enlarged residual sum of squares $\mathrm{SSE}_+ > 0$. The addition raises R_a^2 iff the t statistic $t = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$ of the new variable in the enlarged model satisfies $|t| > 1$, and lowers it iff $|t| < 1$.

Proof. By (18), $R_a^2 = 1 - s^2/s_y^2$ with s_y^2 fixed, so R_a^2 rises iff $s_+^2 < s^2$, that is

$$\frac{\text{SSE}_+}{n-k-2} < \frac{\text{SSE}}{n-k-1} \iff (n-k-1)\text{SSE}_+ < (n-k-2)\text{SSE}$$
$$\iff \text{SSE}_+ < (n-k-2)(\text{SSE} - \text{SSE}_+),$$

subtracting $(n-k-2)\text{SSE}_+$ from both sides. Dividing by $\text{SSE}_+ = (n-k-2)s_+^2 > 0$ gives $1 < (\text{SSE} - \text{SSE}_+)/s_+^2$, and by Lemma 2.30

$$\frac{\text{SSE} - \text{SSE}_+}{s_+^2} = \frac{\hat{\gamma}^2\, \mathbf{z}'\mathbf{M}\mathbf{z}}{s_+^2} = \frac{\hat{\gamma}^2}{\text{SE}(\hat{\gamma})^2} = t^2.$$

The same equivalences hold with $<$ replaced by $>$ or $=$, which gives the second statement. □

Corollary 2.32 (The case $k = 1$; FREES §2.5.1, ISLR §3.1.3; CLM Corollary 6.13)**.** With one explanatory variable whose values are not all equal, an intercept and SST > 0, $R^2 = r^2$,

SSR $= b_1^2(n-1)s_x^2$, and, for $n > 2$,

$$s_y^2 = b_1^2 s_x^2 + s^2 \frac{n-2}{n-1}. \tag{20}$$

Proof. By Corollary 2.14, $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$, so SSR $= b_1^2 \sum_i (x_i - \bar{x})^2 = b_1^2(n-1)s_x^2$; substituting $b_1 = r\, s_y/s_x$ gives SSR $= r^2(n-1)s_y^2 = r^2$SST, that is $R^2 = r^2$. Dividing (16) by $n-1$ and using SSE $= (n-2)s^2$ gives (20). □

Remark (ISLR §3.1.3, §3.2.2)**.** For $k = 1$ the sign of the association is carried by r and its strength by $R^2 = r^2$, so ISLR's $R^2 = 0.61$ for sales on television advertising is $r = 0.78$ from Example 2.5. For $k > 1$, R^2 is the squared correlation of Y with the fitted combination of all the explanatory variables (Proposition 2.29), not with any one of them. What counts as a good R^2 depends on the field: near 1 in a physical law with small measurement error, well below 0.1 in a social or marketing setting where the linear model is a rough approximation.

Example 2.15 (Term life insurance, continued; FREES §3.3).

```
a <- anova(fit_term)
SSR_t <- sum(a[["Sum Sq"]][1:3]); SSE_t <- a[["Sum Sq"]][4]; SST_t <- SSR_t + SSE_t
rbind(Regression = c(SS = SSR_t, df = 3, MS = SSR_t / 3),
      Error      = c(SS = SSE_t, df = 271, MS = SSE_t / 271),
      Total      = c(SS = SST_t, df = 274, MS = NA))
```

```
##                SS  df      MS
## Regression 328.5   3 109.489
## Error      630.4 271   2.326
## Total      958.9 274      NA
```

```
c(R2 = SSR_t / SST_t, s = sqrt(SSE_t / 271), s_y = sqrt(SST_t / 274),
  R2_adj = 1 - (SSE_t / 271) / (SST_t / 274))
```

```
##     R2      s    s_y R2_adj
## 0.3425 1.5252 1.8707 0.3353
```

`anova` prints one row per variable, sequential sums of squares whose total is SSR. The three explanatory variables explain 34.3% of the variation in the logarithmic face amount; the typical error falls from $s_y = 1.871$ to $s = 1.525$, and $R_a^2 = 1 - s^2/s_y^2 = 0.335$.

Remark (Why do females live longer than males?; FREES §3.3, Table 3.4)**.** Lemaire's regression of the female advantage in life expectancy, 4.51 years on average over $n = 169$ countries, chose $k = 4$ of 45 candidate measures and reached $R^2 = 61\%$: the logarithm of persons per physician (-0.473), fertility (-0.444), the percentage of Hindus and Buddhists (-0.018) and an indicator of the former Soviet Union (4.922), each with $|t| > 3$. The four stand for wider groups of correlated variables, economic modernisation, education, religion and geography; with so much collinearity among the candidates another analyst could pick another four. The data are not published, so the table is quoted, not reproduced; its point is that a parsimonious model can explain most of the variation.

Example 2.16 (Advertising, continued; ISLR §3.1.3, §3.2.2)**.**

```
fit_two <- lm(sales ~ TV + radio, data = adv)
fit_stats <- function(m) c(RSE = summary(m)$sigma, R2 = summary(m)$r.squared,
                           adj_R2 = summary(m)$adj.r.squared)
round(rbind(TV_only = fit_stats(fit_tv), TV_radio = fit_stats(fit_two),
            all_three = fit_stats(fit_all)), 5)
```

```
##              RSE     R2 adj_R2
## TV_only    3.259 0.6119 0.6099
## TV_radio   1.681 0.8972 0.8962
## all_three 1.686 0.8972 0.8956
```

```
sprintf("R2: TV+radio %.5f, all three %.5f", summary(fit_two)$r.squared,
  ↪  summary(fit_all)$r.squared)
```

```
## [1] "R2: TV+radio 0.89719, all three 0.89721"
```

```
c(mean_sales = mean(adv$sales))
```

```
## mean_sales
##      14.02
```

Television alone explains 61% of the variation with a residual standard error of 3.26, so a market's sales deviate from the line by about 3,260 units, 23% of the mean sales of about 14,000. Adding radio raises R^2 to 0.897 and lowers the residual standard error to 1.68. Adding newspaper raises R^2 from 0.89719 to 0.89721, the increase Proposition 2.28 guarantees, while the residual standard error **rises** from 1.681 to 1.686: the decrease in SSE is smaller than the loss of a degree of freedom, and by Theorem 2.31 the newspaper t statistic must be below 1 in absolute value, which Example 2.10's

coefficient of -0.001 against its standard error confirms.

5 Exact Inference Under Normality

FREES §2.5, §2.7, §3.2.4 ·ISLR §3.1.2, §3.2.2 ·CLM §4–5, §7, §9. SRM learning outcomes 2(e), 2(f). A5 is in force throughout.

5.1 Sampling distributions

Theorem 2.33 (Normality of $\hat{\boldsymbol{\beta}}$; FREES §3.2.4; CLM Theorem 5.1)**.** Under A1–A5, $\hat{\boldsymbol{\beta}} \sim N\big(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big)$, and for any constant $\mathbf{c}$, $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta}, \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$.

Proof. By (13), $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \mathbf{A}\boldsymbol{\varepsilon}$ with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$. By the prerequisites' theorem **closure under affine maps**, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2\mathbf{A}\mathbf{A}')$, and $\mathbf{A}\mathbf{A}' = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}$. Applying the same theorem to the map $\hat{\boldsymbol{\beta}} \mapsto \mathbf{c}'\hat{\boldsymbol{\beta}}$ gives the second statement. □

Theorem 2.34 (Distribution of SSE and independence from $\hat{\boldsymbol{\beta}}$; CLM Theorem 5.2)**.** Under A1–A5, $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$, and SSE is independent of $\hat{\boldsymbol{\beta}}$.

Proof. Step 1. With $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$ and (9), $\mathrm{SSE}/\sigma^2 = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$.
Step 2. $\mathbf{M}$ is symmetric idempotent with $\mathrm{tr}(\mathbf{M}) = n-k-1$ (Theorem 2.9), so by the prerequisites' theorem **chi-square quadratic forms**, $\mathbf{z}'\mathbf{M}\mathbf{z} \sim \chi^2_{n-k-1}$.
Step 3. The stacked vector

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \mathbf{M} \end{pmatrix} \boldsymbol{\varepsilon}$$

is one affine map of one normal vector, hence jointly normal by **closure under affine maps**, with cross-covariance $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I}_n)\mathbf{M}' = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0}$, since $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ by Theorem 2.9(ii). By the prerequisites' theorem **uncorrelated blocks are independent**, $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are independent, and $\mathrm{SSE} = \mathbf{e}'\mathbf{e}$ is a function of $\mathbf{e}$ alone. □

Remark. A t statistic is a standard normal divided by $\sqrt{\chi^2_\nu/\nu}$ with numerator and denominator **independent**. Here the numerator comes from $\hat{\boldsymbol{\beta}}$ and the denominator from SSE; Step 3 supplies the independence, and without A5 zero covariance would not give it.

5.2 Tests and confidence intervals for a coefficient

Theorem 2.35 (The t statistic; FREES §2.4–2.5, §3.4, ISLR §3.1.2; CLM Theorem 7.1)**.** Under A1–A5, for any constant $\mathbf{c} \neq \mathbf{0}$,

$$\frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-k-1}; \tag{21}$$

in particular $(\hat{\beta}_j - \beta_j)/\mathrm{SE}(\hat{\beta}_j) \sim t_{n-k-1}$, and for $k = 1$, $(b_1 - \beta_1)/\mathrm{SE}(b_1) \sim t_{n-2}$.

Proof. Step 1: a standard normal. By Theorem 2.33, $Z = (\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta})/\{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}\} \sim N(0,1)$, so Z is standard normal by the prerequisites' corollary **standardising a normal variable**.
Step 2: an independent chi-square. By Theorem 2.34, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ is independent of $\hat{\boldsymbol{\beta}}$, hence of Z, which is a function of $\hat{\boldsymbol{\beta}}$ alone.
Step 3: the ratio. By the prerequisites' definition of Student's t, $Z/\sqrt{V/(n-k-1)} \sim t_{n-k-1}$,

and

$$\frac{Z}{\sqrt{V/(n-k-1)}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \cdot \frac{\sigma}{\sqrt{\mathrm{SSE}/(n-k-1)}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}},$$

since $s = \sqrt{\mathrm{SSE}/(n-k-1)}$. Taking $\mathbf{c}$ the j-th coordinate vector gives the statement for $\hat{\beta}_j$, and $k = 1$ gives $n-2$ degrees of freedom. □

Definition 2.36 (t tests, p-values and confidence intervals; FREES §2.5.1–2.5.2, Tables 2.3–2.4 and 3.5–3.6, ISLR §3.1.2; CLM Definition 7.11, Theorem 7.9)**.** To test $H_0\colon \beta_j = d$ at significance level α, compute the t **ratio** $t(\hat{\beta}_j) = (\hat{\beta}_j - d)/\mathrm{SE}(\hat{\beta}_j)$ and, with $t_{\nu,q}$ the q-th quantile of t_ν and $\nu = n-k-1$:

Alternative H_a	Reject H_0 if	p-value
$\beta_j > d$	$t(\hat{\beta}_j) > t_{\nu,1-\alpha}$	$\mathsf{P}(t_\nu > t(\hat{\beta}_j))$
$\beta_j < d$	$t(\hat{\beta}_j) < -t_{\nu,1-\alpha}$	$\mathsf{P}(t_\nu < t(\hat{\beta}_j))$
$\beta_j \neq d$	$\lvert t(\hat{\beta}_j)\rvert > t_{\nu,1-\alpha/2}$	$\mathsf{P}(\lvert t_\nu\rvert > \lvert t(\hat{\beta}_j)\rvert)$

The p-value is the probability, under H_0, of a t ratio at least as extreme as the one observed, and

H_0 is rejected at level α iff $p < \alpha$. The usual test is $H_0\colon \beta_j = 0$ against $\beta_j \neq 0$, whose t ratio $\hat\beta_j/\mathrm{SE}(\hat\beta_j)$ and p-value software prints for every coefficient. A $100(1-\alpha)\%$ **confidence interval** for β_j is

$$\hat\beta_j \pm t_{n-k-1,1-\alpha/2}\,\mathrm{SE}(\hat\beta_j), \tag{22}$$

and for $\mathbf{c}'\boldsymbol\beta$ it is $\mathbf{c}'\hat{\boldsymbol\beta} \pm t_{n-k-1,1-\alpha/2}\, s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}$.

Proposition 2.37 (Validity of the procedures; CLM Proposition 7.12)**.** Under A1–A5 each test of Definition 2.36 rejects a true H_0 with probability exactly α, and the interval (22) contains β_j with probability exactly $1-\alpha$.

Proof. Under $H_0\colon \beta_j = d$ the ratio $t(\hat\beta_j)$ is (21) with $\mathbf{c}$ the j-th coordinate vector, hence t_{n-k-1} by Theorem 2.35. For the two-sided test, $\mathsf{P}(|t_\nu| > t_{\nu,1-\alpha/2}) = 2\,\mathsf{P}(t_\nu > t_{\nu,1-\alpha/2}) = \alpha$ by the symmetry of t_ν, which follows from the symmetry of Z in its definition; the one-sided cases are $\mathsf{P}(t_\nu > t_{\nu,1-\alpha}) = \alpha$ and its mirror image. For the interval, with $T = (\hat\beta_j - \beta_j)/\mathrm{SE}(\hat\beta_j) \sim t_\nu$,

$$1-\alpha = \mathsf{P}\big(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}\big) = \mathsf{P}\big(\hat\beta_j - t_{\nu,1-\alpha/2}\mathrm{SE}(\hat\beta_j) \leqslant \beta_j \leqslant \hat\beta_j + t_{\nu,1-\alpha/2}\mathrm{SE}(\hat\beta_j)\big),$$

rearranging the inequalities inside the probability. The same argument with (21) covers $\mathbf{c}'\boldsymbol{\beta}$. □

Remark (ISLR's factor of two; ISLR §3.1.2)**.** ISLR writes the 95% interval as $\hat{\beta}_j \pm 2\,\mathrm{SE}(\hat{\beta}_j)$; the exact multiplier is $t_{n-k-1,0.975}$, which is 2.01 at 48 degrees of freedom and 1.97 at 198, and the t law itself rests on A5. Rejecting $H_0\colon \beta_1 = 0$ is ISLR's "there is a relationship between X and Y"; the p-value cutoffs 5% and 1% correspond, at $n = 30$, to $|t|$ of about 2 and 2.75.

```
round(c(t48_975 = qt(0.975, 48), t198_975 = qt(0.975, 198),
        t28_975 = qt(0.975, 28), t28_995 = qt(0.995, 28)), 3)
```

```
##  t48_975 t198_975  t28_975  t28_995
##    2.011    1.972    2.048    2.763
```

Example 2.17 (Lottery sales, continued; FREES §2.5)**.** With $b_1 = 0.647$ and $\mathrm{SE}(b_1) = 0.0488$, $t(b_1) = 13.3$ on $n - 2 = 48$ degrees of freedom: the slope is 13 standard errors above zero, far beyond $t_{48,0.975} = 2.011$, so $H_0\colon \beta_1 = 0$ is rejected against $\beta_1 \neq 0$ at 5%, with p-value 10^{-17}. The 95% confidence interval $0.647 \pm 2.011(0.0488) = (0.549, 0.745)$ says that ZIP codes differing by 1,000 in population are expected to differ in sales by between \$549 and \$745; the 90% interval is narrower.

```r
coef(summary(fit_lot))
```

```
##             Estimate Std. Error t value  Pr(>|t|)
## (Intercept) 469.7036  702.90619  0.6682 5.072e-01
## POP           0.6471    0.04881 13.2579 1.158e-17
```

```r
c(t_crit = qt(0.975, 48))
```

```
## t_crit
##  2.011
```

```r
confint(fit_lot, level = 0.95)
```

```
##                2.5 %    97.5 %
## (Intercept) -943.584 1882.9912
## POP            0.549    0.7452
```

```r
confint(fit_lot, level = 0.90)
```

```
##                  5 %     95 %
## (Intercept) -709.2277 1648.635
## POP            0.5652    0.729
```

Example 2.18 (Advertising, continued; ISLR §3.1.2, §3.2.1)**.** For sales on television advertising both t ratios are large and both p-values are below 10^{-4}; the 95% intervals are $[6.13, 7.94]$ for the intercept, so sales without advertising fall between 6,130 and 7,935 units on average, and $[0.042, 0.053]$ for the slope. In the three-medium model the newspaper p-value is 0.86: with television and radio held fixed there is no evidence that newspaper advertising is associated with sales.

```
round(coef(summary(fit_tv)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.0326     0.4578   15.36        0
## TV            0.0475     0.0027   17.67        0
```

```
round(confint(fit_tv), 4)
```

```
##               2.5 % 97.5 %
## (Intercept) 6.1297 7.9355
## TV          0.0422 0.0528
```

```
round(coef(summary(fit_all)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.9389     0.3119  9.4223   0.0000
## TV             0.0458     0.0014 32.8086   0.0000
## radio          0.1885     0.0086 21.8935   0.0000
## newspaper     -0.0010     0.0059 -0.1767   0.8599
```

5.3 The overall F test

Theorem 2.38 (Overall F test; ISLR §3.2.2; CLM Theorem 7.3, Proposition 7.14, equation (34))**.** Under A1–A5, for H_0: $\beta_1 = \cdots = \beta_k = 0$,

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/k}{\text{SSE}/(n-k-1)} = \frac{(\text{SST}-\text{SSE})/k}{\text{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \tag{23}$$

is distributed as $F_{k,n-k-1}$ under H_0; it is defined with probability one, since $\text{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ (Theorem 2.34) is positive with probability one and SST $\geqslant$ SSE; and $\mathsf{E}[\text{SSR}/k] = \sigma^2 + \boldsymbol{\beta}_1'\mathbf{X}_1'\mathbf{M}_0\mathbf{X}_1\boldsymbol{\beta}_1/k \geqslant \sigma^2$ in general, with equality iff H_0 holds, where $\mathbf{X}_1$ holds the non-intercept columns and $\boldsymbol{\beta}_1$ the slopes.

Proof. Step 1: SSR as a quadratic form in the errors. By Theorem 2.26, $\text{SSR} = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}$ with $\mathbf{P} = \mathbf{H} - \mathbf{H}_0$ symmetric idempotent (Lemma 2.10). Since $\mathbf{P1} = \mathbf{H1} - \mathbf{H}_0\mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0}$, $\mathbf{PX}\boldsymbol{\beta} = \mathbf{P}(\beta_0\mathbf{1} + \mathbf{X}_1\boldsymbol{\beta}_1) = \mathbf{PX}_1\boldsymbol{\beta}_1$, and under H_0 this is $\mathbf{0}$, so $\mathbf{PY} = \mathbf{P}\boldsymbol{\varepsilon}$ and $\text{SSR} = \boldsymbol{\varepsilon}'\mathbf{P}\boldsymbol{\varepsilon}$.
Step 2: the two chi-squares. With $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I}_n)$, under H_0, $\text{SSR}/\sigma^2 = \mathbf{z}'\mathbf{Pz}$ and $\text{SSE}/\sigma^2 = \mathbf{z}'\mathbf{Mz}$. Both $\mathbf{P}$ and $\mathbf{M}$ are symmetric idempotent, with traces k and $n-k-1$, so by **chi-square quadratic forms** they are χ^2_k and χ^2_{n-k-1}.
Step 3: independence. $\mathbf{PM} = \mathbf{0}$ by Lemma 2.10, so by the prerequisites' **Cochran theorem** the two quadratic forms are independent.
Step 4: the ratio. By the prerequisites' definition of the F law, the ratio of independent chi-squares each divided by its degrees of freedom is $F_{k,n-k-1}$. The other forms of (23) substitute $\text{SSR} = \text{SST} - \text{SSE}$ and then divide numerator and denominator by SST.
Step 5: the expectation. Without H_0, $\mathbf{PY} = \mathbf{PX}_1\boldsymbol{\beta}_1 + \mathbf{P}\boldsymbol{\varepsilon}$, so by **expectation of a quadratic form** with mean $\mathbf{PX}_1\boldsymbol{\beta}_1$ and variance $\sigma^2\mathbf{I}_n$, $\mathsf{E}[\text{SSR}] = \mathsf{E}[\mathbf{Y}'\mathbf{PY}] = \sigma^2 \operatorname{tr}(\mathbf{P}) + \boldsymbol{\beta}_1'\mathbf{X}_1'\mathbf{PX}_1\boldsymbol{\beta}_1 = k\sigma^2 + \boldsymbol{\beta}_1'\mathbf{X}_1'\mathbf{M}_0\mathbf{X}_1\boldsymbol{\beta}_1$, using $\mathbf{PX}_1 = \mathbf{M}_0\mathbf{X}_1 - \mathbf{MX}_1 = \mathbf{M}_0\mathbf{X}_1$, which holds since $\mathbf{MX}_1 = \mathbf{0}$ and $\mathbf{P} = \mathbf{H} - \mathbf{H}_0 = (\mathbf{I} - \mathbf{H}_0) - (\mathbf{I} - \mathbf{H}) = \mathbf{M}_0 - \mathbf{M}$. The quadratic form $\|\mathbf{M}_0\mathbf{X}_1\boldsymbol{\beta}_1\|^2$ is zero iff $\mathbf{X}_1\boldsymbol{\beta}_1$ is constant, which by A4 happens iff $\boldsymbol{\beta}_1 = \mathbf{0}$. □

Corollary 2.39 (R^2 from F; CLM Corollary 7.15)**.** If SST $>$ 0 and SSE $>$ 0, then $R^2 = \dfrac{kF}{kF + n - k - 1}$, so F determines R^2 without SSE and SST.

Proof. Solving the last form of (23), $F(1 - R^2)(n - k - 1) = R^2 k$, for R^2. □

Remark (Why F and not the k separate t tests; ISLR §3.2.2)**.** $\mathsf{E}[\text{MSE}] = \sigma^2$ always and $\mathsf{E}[\text{MSR}] = \sigma^2$ under H_0, so F near 1 is consistent with no relation and F well above 1 is evidence of one; how far above depends on n and k through the $F_{k,n-k-1}$ quantile, and a value barely above 1 can be decisive when n is large. With $k = 100$ true null slopes, about five of the individual p-values fall below 0.05 by chance and the smallest very probably does (in 98% of the 2000 simulated data sets with $n = 250$ printed here), so declaring a relation whenever some t test rejects has a false-alarm rate far above 5%; the F test rejects a true H_0 with probability exactly 5% whatever k. When $k + 1 \geqslant n$ no residual degrees of freedom are left (and for $k \geqslant n$ the least squares coefficients are not unique), so neither statistic exists.

```
set.seed(5); n_s <- 250; k_s <- 100
X_s <- cbind(1, matrix(rnorm(n_s * k_s), n_s)); A_s <- solve(crossprod(X_s))
```

```r
sims <- replicate(2000, {
  y_s <- rnorm(n_s); b_s <- drop(A_s %*% crossprod(X_s, y_s))
  s_s <- sqrt(sum((y_s - X_s %*% b_s)^2) / (n_s - k_s - 1))
  p_s <- 2 * pt(-abs(b_s[-1] / (s_s * sqrt(diag(A_s)[-1]))), n_s - k_s - 1)
  c(n_below = sum(p_s < 0.05), some_below = any(p_s < 0.05))
})
round(rowMeans(sims), 3)
```

```
##    n_below some_below
##      5.032      0.980
```

Proposition 2.40 ($t^2 = F$ for one explanatory variable; ISLR §3.2.2; CLM Corollary 7.17)**.** Let $\mathbf{z}$ be one of the explanatory variables in a fitted model, t its t ratio for $H_0\colon \gamma = 0$, SSE_0 the residual sum of squares of the model without $\mathbf{z}$, and SSE that of the full model, with $n - k - 1$ residual degrees of freedom. Then, under A4 and $\mathrm{SSE} > 0$,

$$\frac{(\mathrm{SSE}_0 - \mathrm{SSE})/1}{\mathrm{SSE}/(n-k-1)} = t^2, \tag{24}$$

and under A1–A5 and H_0 this statistic is $F_{1,n-k-1}$. For $k = 1$ the overall F of (23) is $t(b_1)^2$.

Proof. Let $\mathbf{X}_{(\mathbf{z})}$ be the design without the column $\mathbf{z}$ and $\mathbf{M}_{(\mathbf{z})}$ its residual maker. By A4 the columns of $\mathbf{X}_{(\mathbf{z})}$ are linearly independent and $\mathbf{z} \notin \mathcal{C}(\mathbf{X}_{(\mathbf{z})})$, so Lemma 2.30 with $\mathbf{X}_{(\mathbf{z})}$ as the base gives $\mathrm{SSE}_0 - \mathrm{SSE} = \hat{\gamma}^2 \mathbf{z}'\mathbf{M}_{(\mathbf{z})}\mathbf{z}$ and $\mathrm{SE}(\hat{\gamma})^2 = s^2/(\mathbf{z}'\mathbf{M}_{(\mathbf{z})}\mathbf{z})$, and the left side of (24) is $\hat{\gamma}^2 \mathbf{z}'\mathbf{M}_{(\mathbf{z})}\mathbf{z}/s^2 = \hat{\gamma}^2/\mathrm{SE}(\hat{\gamma})^2 = t^2$. Under H_0, $t \sim t_{n-k-1}$ by Theorem 2.35, and by the prerequisites' corollary **the square of a t is an F**, $t^2 \sim F_{1,n-k-1}$. For $k = 1$, $\mathbf{X}_{(\mathbf{z})} = \mathbf{1}$, the intercept alone, whose residual sum of squares is SST, and (24) is (23). □

Example 2.19 (Advertising, continued; ISLR §3.2.2)**.** For the three-medium model $F = 570$ on $(3, 196)$ degrees of freedom, with a p-value of order 10^{-96}: at least one medium is associated with sales.

```
sm <- summary(fit_all); Fst <- sm$fstatistic
SSE_a <- sum(residuals(fit_all)^2); SST_a <- sum((adv$sales - mean(adv$sales))^2)
c(F_lm = Fst[["value"]], F_formula = ((SST_a - SSE_a) / 3) / (SSE_a / 196),
  F_from_R2 = (sm$r.squared / 3) / ((1 - sm$r.squared) / 196),
  p = pf(Fst[["value"]], 3, 196, lower.tail = FALSE))
```

```
##      F_lm F_formula F_from_R2         p
## 5.703e+02 5.703e+02 5.703e+02 1.575e-96
```

```r
c(t2_newspaper = coef(sm)["newspaper", "t value"]^2,
  F_drop_newspaper = anova(fit_two, fit_all)[["F"]][2])
```

```
##      t2_newspaper F_drop_newspaper
##           0.03123           0.03123
```

Dropping newspaper from the full model gives an F equal to the square of its t ratio, which is Proposition 2.40.

5.4 Prediction

Theorem 2.41 (Confidence and prediction intervals; FREES §2.5.3, ISLR §3.2.2; CLM Theorem 9.1, Theorem 9.2, Corollary 9.4, Corollary 3.10)**.** Let $\mathbf{x}_0$ be a new vector of explanatory variables, $\hat{Y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$, and $h_0 = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$. Under A1–A5:

(i) $\operatorname{var}(\hat{Y}_0) = \sigma^2 h_0$, and a $100(1-\alpha)\%$ confidence interval for the mean response $\mathsf{E}[Y_0] = \mathbf{x}_0'\boldsymbol{\beta}$ is $\hat{Y}_0 \pm t_{n-k-1,1-\alpha/2}\, s\sqrt{h_0}$;

(ii) for a new response $Y_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0$ with $\varepsilon_0 \sim N(0,\sigma^2)$ independent of $\boldsymbol{\varepsilon}$, $\operatorname{var}(Y_0 - \hat{Y}_0) = \sigma^2(1+h_0)$, and a $100(1-\alpha)\%$ **prediction interval** for Y_0 is $\hat{Y}_0 \pm t_{n-k-1,1-\alpha/2}\, s\sqrt{1+h_0}$;

(iii) for $k=1$, $\mathrm{SE(pred)} = s\sqrt{1+h_0} = s\sqrt{1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{(n-1)s_x^2}}$;

(iv) at an observed point $\mathbf{x}_0 = \mathbf{x}_i$, $\mathrm{var}(\hat{Y}_i) = \sigma^2 h_{ii}$ and $\mathrm{var}(\hat{Y}_i) + \mathrm{var}(e_i) = \sigma^2$.

Proof. **(i)** $\hat{Y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$ is the case $\mathbf{c} = \mathbf{x}_0$ of Theorem 2.33: $\hat{Y}_0 \sim N(\mathbf{x}_0'\boldsymbol{\beta}, \sigma^2 h_0)$, and the interval is Definition 2.36's interval for $\mathbf{c}'\boldsymbol{\beta}$ with Proposition 2.37.
(ii) $Y_0 - \hat{Y}_0 = \varepsilon_0 - \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, and $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ is a function of $\boldsymbol{\varepsilon}$, independent of ε_0. Hence the variance is the sum $\sigma^2 + \mathbf{x}_0'\,\mathrm{var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 = \sigma^2 + \sigma^2 h_0$, and the difference of two independent normals is normal: $Y_0 - \hat{Y}_0 \sim N(0, \sigma^2(1+h_0))$. It is independent of SSE, which is a function of $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$: $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}, \mathbf{e})$ are independent by Theorem 2.34, and ε_0 is independent of the pair $(\hat{\boldsymbol{\beta}}, \mathbf{e})$, both being functions of $\boldsymbol{\varepsilon}$. So $(Y_0 - \hat{Y}_0)/\{s\sqrt{1+h_0}\}$ is $(Y_0 - \hat{Y}_0)/\{\sigma\sqrt{1+h_0}\}$, standard normal by the prerequisites' corollary **standardising a normal variable**, over an independent $\sqrt{\chi^2_{n-k-1}/(n-k-1)}$, hence t_{n-k-1}, and the interval follows as in Proposition 2.37.
(iii) Equation (10) with x_0 in place of x_i and $\sum_j (x_j - \bar{x})^2 = (n-1)s_x^2$.
(iv) $h_0 = h_{ii}$ by Definition 2.12, and $\mathrm{var}(e_i) = \sigma^2(1-h_{ii})$ by Theorem 2.23. □

Remark (ISLR §3.2.2, FREES §2.5.3)**.** The confidence interval measures the reducible error, the uncertainty in the estimated plane, and shrinks to nothing as n grows whenever $h_0 \to 0$, as for $k = 1$ when $\bar{x}$ stays bounded and s_x stays away from 0 (by (10)); the prediction interval adds the irreducible error ε_0, is always the wider, and cannot be narrower than $2t\,s$. Both are centred at $\hat{Y}_0$. Leverage is the share of σ^2 that a point's own fitted value carries, and $1 - h_{ii}$ the share left in its residual. Model bias, the error from a linear model being only an approximation, is excluded from both intervals.

Example 2.20 (Lottery sales, continued; FREES §2.5.3)**.** At $x_0 = 10{,}000$ the point prediction is 6,941 and

$$\mathrm{SE(pred)} = 3792\sqrt{1 + \frac{1}{50} + \frac{(10{,}000 - 9{,}311)^2}{49(11{,}098)^2}} = 3830,$$

so the 95% prediction interval is $6941 \pm 2.011(3830)$, which `predict` gives as $(-759,\ 14{,}641)$: of many ZIP codes with population 10,000, about 19 in 20 would have sales between 0 and \$14,641, the negative lower limit being truncated at zero.

```r
x0 <- 10000; xp <- lottery$POP
h0 <- 1/50 + (x0 - mean(xp))^2 / sum((xp - mean(xp))^2)
c(se_pred = s_lot * sqrt(1 + h0), se_mean = s_lot * sqrt(h0))
```

```
## se_pred se_mean
##  3829.6   537.3
```

```r
predict(fit_lot, data.frame(POP = x0), interval = "prediction")
```

```
##    fit    lwr   upr
## 1 6941 -759.3 14641
```

```r
predict(fit_lot, data.frame(POP = x0), interval = "confidence")
```

```
##    fit  lwr  upr
## 1 6941 5860 8021
```

Example 2.21 (Advertising, continued; ISLR §3.2.2)**.** With \$100,000 on television and \$20,000 on radio, the fitted sales are 11,256 units; the 95% confidence interval for the **average** sales of such markets is [10,985, 11,528] and the prediction interval for one such market is [7,930, 14,583].

```r
nd <- data.frame(TV = 100, radio = 20)
round(1000 * rbind(confidence = predict(fit_two, nd, interval = "confidence"),
                   prediction = predict(fit_two, nd, interval = "prediction")))
```

```
##     fit   lwr   upr
## 1 11256 10985 11528
## 1 11256  7930 14583
```

Example 2.22 (The capital asset pricing model; FREES §2.7)**.** Monthly returns of the Lincoln National Insurance Corporation (y) and of the equally weighted S&P 500 index (x) over 1986–1990, $n = 60$. The CAPM posits $\mathsf{E}[r] = \beta_0 + \beta_1 r_m$; β_1 measures the return attributed to the market, and securities with $\beta_1 > 1$ are called **aggressive**, with $\beta_1 < 1$ **defensive**.

```r
capm <- read.csv("data/frees/csv/CAPM.csv")
round(rbind(LINCOLN = descr(capm$LINCOLN), MARKET = descr(capm$MARKET)), 4)
```

```
##           mean median     sd     min    max
## LINCOLN 0.0051 0.0075 0.0859 -0.2803 0.3147
## MARKET  0.0074 0.0142 0.0525 -0.2205 0.1275
```

```r
fit_capm <- lm(LINCOLN ~ MARKET, data = capm)
round(coef(summary(fit_capm)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.0021     0.0091 -0.2354   0.8147
## MARKET        0.9732     0.1725  5.6412   0.0000
```

```r
c(s = summary(fit_capm)$sigma, R2 = summary(fit_capm)$r.squared,
  crash_month = which.min(capm$MARKET), z_crash = (min(capm$MARKET) -
   ↪  mean(capm$MARKET)) / sd(capm$MARKET))
```

```
##           s          R2 crash_month     z_crash
##     0.06962     0.35429    22.00000    -4.33860
```

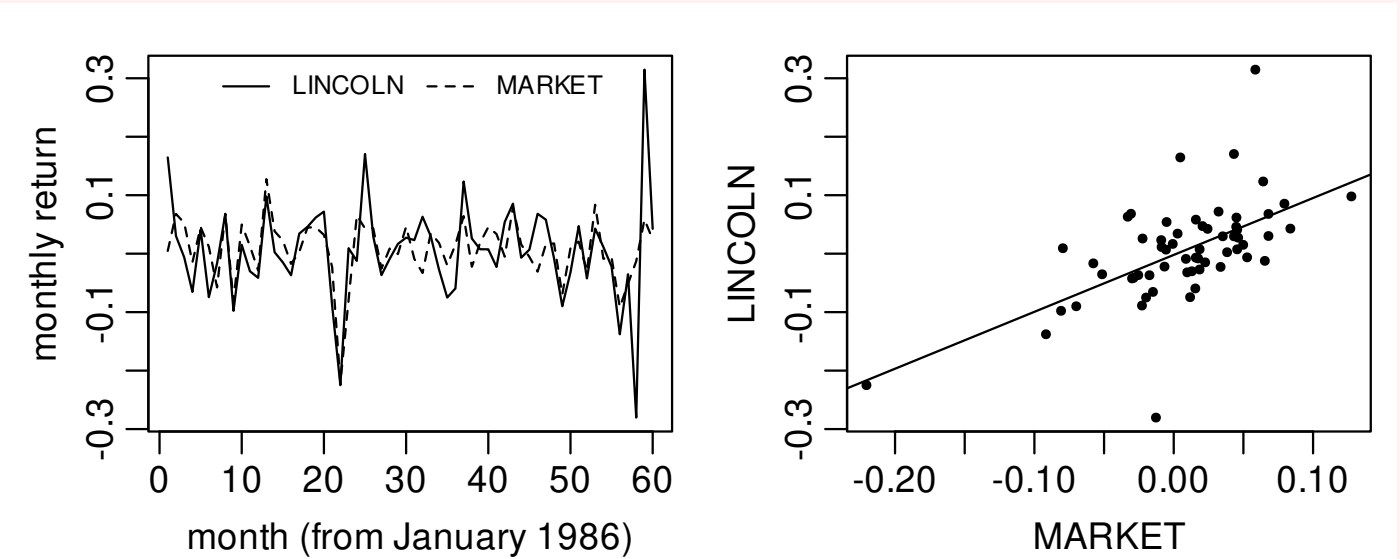


Figure 2.7: Time series plots of the two returns, and Lincoln's return against the market's with the fitted line (FREES Figures 2.10–2.11).

The fitted line is $\widehat{\texttt{LINCOLN}} = -0.0021 + 0.973\,\texttt{MARKET}$ with $t(b_1) = 5.64$: the market is clearly a determinant, yet $R^2 = 35.4\%$, so it explains only a third of the variation in Lincoln's return.

Lincoln is riskier than the market, $s_y = 0.0859 > 0.0525 = s_x$, and its five-year mean return, 0.51% a month, is below the market's 0.74%. The smallest market return, -0.221, is 4.3 standard deviations below its mean; it is October 1987, the month of the crash, a high leverage point. Corollary 2.32 splits the risk of the security into a market component and a diversifiable one, $s_y^2 = b_1^2 s_x^2 + s^2(n-2)/(n-1)$:

```
b1c <- coef(fit_capm)[["MARKET"]]; sc <- summary(fit_capm)$sigma
c(s_y2 = var(capm$LINCOLN), market = b1c^2 * var(capm$MARKET),
  diversifiable = sc^2 * 58 / 59, sum = b1c^2 * var(capm$MARKET) + sc^2 * 58 / 59)
```

```
##          s_y2        market diversifiable           sum
##      0.007379      0.002614      0.004765      0.007379
```

The CAPM question is whether $\beta_1 = 1$, the market's own sensitivity. By Definition 2.36 with $d = 1$, $t = (0.973 - 1)/\mathrm{SE}(b_1)$, and the hypothesis is not rejected:

```
se_b1c <- coef(summary(fit_capm))["MARKET", "Std. Error"]
c(t_beta1_equals_1 = (b1c - 1) / se_b1c, p = 2 * pt(-abs((b1c - 1) / se_b1c), 58))
```

```
## t_beta1_equals_1                p
##          -0.1552           0.8772
```

Refitting without the crash month gives $\widehat{\texttt{LINCOLN}} = -0.00181 + 0.956\,\texttt{MARKET}$ with $R^2 = 26.4\%$: removing the high leverage point **lowers** R^2, because that month showed the relation over a range the other months do not cover.

```
fit_capm1 <- lm(LINCOLN ~ MARKET, data = capm[-22, ])
c(coef(fit_capm1), R2 = summary(fit_capm1)$r.squared, s = summary(fit_capm1)$sigma,
  t = coef(summary(fit_capm1))["MARKET", "t value"])
```

```
## (Intercept)      MARKET          R2           s           t
##   -0.001806    0.956088    0.263536    0.070215    4.516283
```

5.5 Maximum likelihood (beyond the syllabus)

CLM §5. Not in the SRM or PA reading lists.

Proposition 2.42 (Least squares is maximum likelihood; CLM Theorem 5.3)**.** Under A1–A5, $\hat{\boldsymbol{\beta}}$ is the maximum likelihood estimator of $\boldsymbol{\beta}$, and $\hat{\sigma}^2_{\text{MLE}} = \text{SSE}/n$, which is biased downward by the factor $(n-k-1)/n$.

Proof. $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n)$ has density $(2\pi\sigma^2)^{-n/2}\exp\{-(\mathbf{Y}-\mathbf{X}\boldsymbol{\beta})'(\mathbf{Y}-\mathbf{X}\boldsymbol{\beta})/(2\sigma^2)\}$, so

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{Q(\boldsymbol{\beta})}{2\sigma^2}. \tag{25}$$

For fixed σ^2 only the last term involves $\boldsymbol{\beta}$, with the negative coefficient $-1/(2\sigma^2)$, so ℓ is maximised over $\boldsymbol{\beta}$ where Q is minimised, at $\hat{\boldsymbol{\beta}}$ by Theorem 2.6, whatever σ^2. Substituting $Q(\hat{\boldsymbol{\beta}}) = \mathrm{SSE}$,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\mathrm{SSE}}{2\sigma^4} = 0 \iff \sigma^2 = \frac{\mathrm{SSE}}{n},$$

and $\partial^2\ell/\partial(\sigma^2)^2 = n/(2\sigma^4) - \mathrm{SSE}/\sigma^6$ is negative there, equal to $-n^3/(2\mathrm{SSE}^2)$, and $\partial\ell/\partial\sigma^2 = (\mathrm{SSE} - n\sigma^2)/(2\sigma^4)$ changes sign from $+$ to $-$ there, so it is the global maximum ($\mathrm{SSE} > 0$ with probability one when $n > k+1$). Its expectation is $(n-k-1)\sigma^2/n$ by Theorem 2.22. □

6 Residual Analysis and Unusual Points

FREES §2.6 ·ISLR §3.3.3 (outliers, high leverage points) ·CLM §2–3. SRM learning outcome 2(d).

Definition 2.43 (Standardised residuals, outliers, high leverage points; FREES §2.6; CLM Definition 3.11)**.** The **standardised residual** of observation i is, for $s > 0$ and $h_{ii} < 1$, $e_i/\{s\sqrt{1-h_{ii}}\}$, the residual divided by its estimated standard deviation (Theorem 2.23); FREES's approximation e_i/s ignores the leverage. An observation is an **outlier** if it is unusual in the vertical direction, its standardised residual exceeding 2 in absolute value, and a **high leverage point** if it is unusual in the horizontal direction, flagged here by $h_{ii} > 2(k+1)/n$. A point can be both.

Remark (Model discrepancies; FREES §2.6)**.** Residual analysis checks the assumptions of Definition 2.4 against the residuals, which stand in for the unobservable errors, and uses any pattern found to respecify the model. FREES lists five discrepancies: lack of independence among the errors; heteroscedasticity, a variance that changes with the observation; a relation between the errors and an explanatory variable, which means the variable could predict better; non-normal errors, which invalidate the exact inference of §5; and unusual points, single observations that

move the fit. Standardised residuals have approximately a standard normal distribution under A5 when $n-k-1$ is large ($e_i/(\sigma\sqrt{1-h_{ii}})$ is exactly $N(0,1)$, e_i being a linear function of the normal errors with variance $\sigma^2(1-h_{ii})$ by Theorem 2.23), and are dimensionless, so experience of their size transfers between data sets.

Example 2.23 (Outliers and high leverage points; FREES §2.6)**.** FREES's fictitious data: 19 base points and three unusual points A, B and C.

```
ol <- read.csv("data/frees/csv/OutlierExample.csv")
one_fit <- function(rows) {
  m <- lm(Y ~ X, data = ol[rows, ])
  c(b0 = coef(m)[[1]], b1 = coef(m)[[2]], s = summary(m)$sigma,
    R2_pct = 100 * summary(m)$r.squared, t_b1 = coef(summary(m))["X", "t value"])
}
round(rbind(base = one_fit(1:19), base_A = one_fit(1:20),
            base_B = one_fit(c(1:19, 21)), base_C = one_fit(c(1:19, 22))), 3)
```

```
##            b0    b1     s R2_pct   t_b1
## base    1.869 0.611 0.288  88.96 11.705
## base_A  1.750 0.693 0.845  53.71  4.570
```

```
## base_B 1.775 0.640 0.285  94.75 18.015
## base_C 3.356 0.155 0.865  10.31  1.439
```

```
round(rbind(std_resid = sapply(20:22, function(i) rstandard(lm(Y ~ X, ol[c(1:19, i),
 ↪  ]))[20]),
            leverage  = sapply(20:22, function(i) hatvalues(lm(Y ~ X, ol[c(1:19, i),
             ↪  ]))[20]),
            flag_lev  = 2 * 2 / 20), 3)
```

```
##              20    21     22
## std_resid 4.003 0.771 -4.014
## leverage  0.067 0.550  0.550
## flag_lev  0.200 0.200  0.200
```

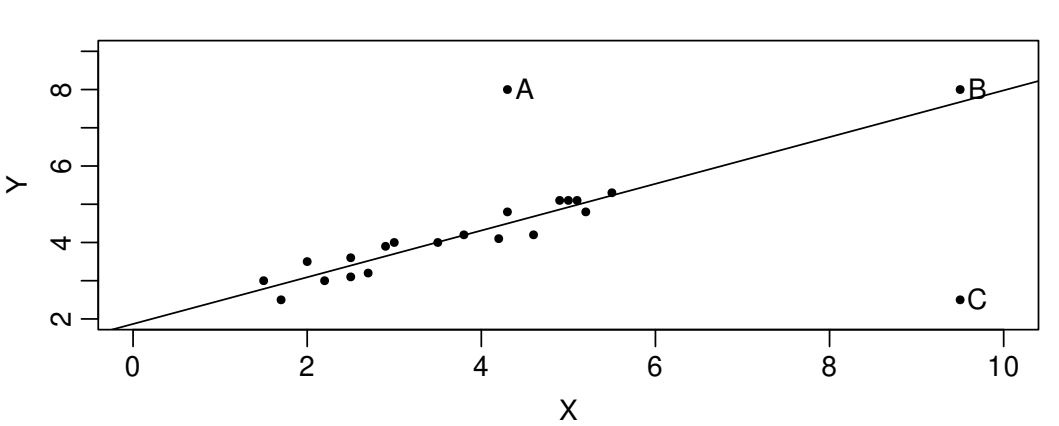


Figure 2.8: Nineteen base points with the line fitted to them, and the three unusual points (FREES Figure 2.7).

The base points are well fitted: $R^2 = 89\%$, $s = 0.29$, $t(b_1) = 11.7$. Adding A, an outlier at an ordinary x, leaves the line much where it was but drops R^2 to 54% and triples s. B has A's y-value at a large x and lies near the base line: it is a high leverage point but not an outlier, and it **raises** R^2 to 95% by enlarging SST while leaving SSE nearly unchanged. C has B's x and a low y-value,

so it is both: the slope falls from 0.61 to 0.16, R^2 to 10%, and the t ratio to 1.4. One point in twenty can do this, because the coefficients are weighted sums of the responses (Proposition 2.17) with weights that grow with $|x_i - \bar{x}|$. Here removing the outlier A would restore the fit, while removing the high leverage point B would lower R^2.

Example 2.24 (Lottery sales, continued: Kenosha; FREES §2.6)**.** The point at the upper left of the scatter plot is the ZIP code containing Kenosha, on the Illinois border, whose sales are far above what its population predicts.

```
i_ken <- unname(which.max(residuals(fit_lot)))
c(row = i_ken, POP = lottery$POP[i_ken], SALES = lottery$SALES[i_ken],
  std_resid = rstandard(fit_lot)[[i_ken]], leverage = hatvalues(fit_lot)[[i_ken]])
```

```
##         row         POP       SALES   std_resid    leverage
##     9.00000  4464.00000 21587.80000     4.86614     0.02389
```

```
fit_ken <- lm(SALES ~ POP, data = lottery[-i_ken, ])
round(rbind(with_Kenosha = one_fit_lot <- c(coef(fit_lot), s = s_lot,
                  R2_pct = 100 * summary(fit_lot)$r.squared,
                  t_b1 = coef(summary(fit_lot))["POP", "t value"]),
```

```
    without = c(coef(fit_ken), s = summary(fit_ken)$sigma,
        R2_pct = 100 * summary(fit_ken)$r.squared,
        t_b1 = coef(summary(fit_ken))["POP", "t value"])), 3)
```

```
##                (Intercept)   POP    s R2_pct  t_b1
## with_Kenosha         469.70 0.647 3792  78.55 13.26
## without              -43.46 0.662 2728  88.28 18.82
```

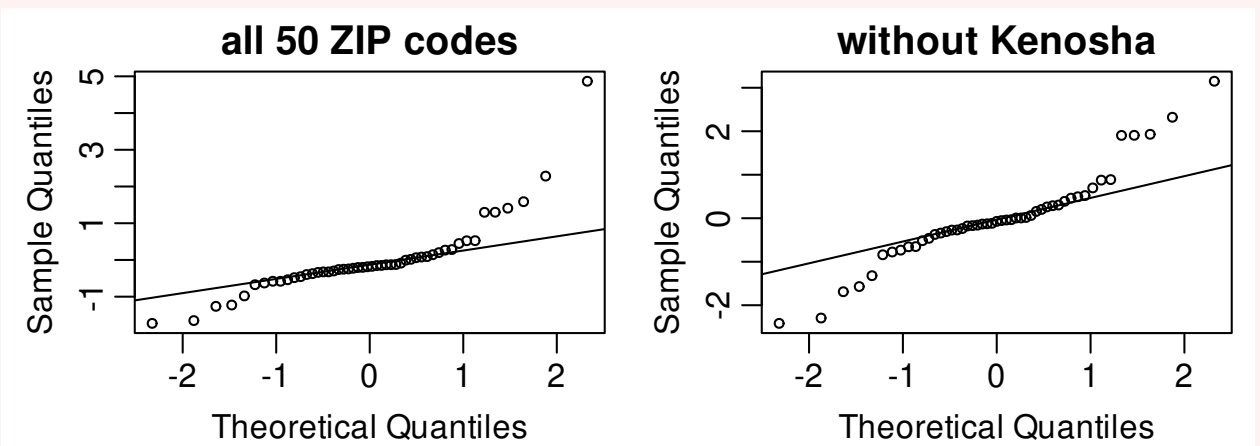


Figure 2.9: Normal quantile plots of the standardised residuals with and without Kenosha (FREES Figure 2.9).

Kenosha is an outlier with a standardised residual of 4.9 and ordinary leverage. The slope barely changes when it is removed, from 0.647 to 0.662, but its t ratio rises from 13.3 to 18.8 because s falls by more than \$1,000 (from 3792 to 2728), and R^2 rises from 78.5% to 88.3%; the smaller s matters for prediction intervals. The quantile plot with Kenosha climbs away from the line at the upper end; without it the points follow the line much more closely, with mild departures in both tails, so the apparent non-normality was one observation, not skewness shared by all.

7 Computation in R

FREES §2.8 ·ISLR §3.6.1–3.6.3.

7.1 Reading regression output

FREES §2.8 annotates the output of Excel, SAS and R for the lottery regression; every printed quantity is a formula of this unit.

```r
summary(fit_lot)
```

```
##
## Call:
## lm(formula = SALES ~ POP, data = lottery)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -6047  -1461   -670    486  18229
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 469.7036   702.9062    0.67     0.51
## POP           0.6471     0.0488   13.26   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3790 on 48 degrees of freedom
## Multiple R-squared:  0.785,        Adjusted R-squared:  0.781
## F-statistic:  176 on 1 and 48 DF,  p-value: <2e-16
```

```r
anova(fit_lot)
```

```
## Analysis of Variance Table
##
## Response: SALES
##           Df     Sum Sq    Mean Sq F value Pr(>F)
## POP        1 2527165015 2527165015     176 <2e-16 ***
## Residuals 48  690116755   14377432
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
X_l <- model.matrix(fit_lot); y_l <- lottery$SALES; n_l <- nrow(X_l)
b_l <- solve(crossprod(X_l), crossprod(X_l, y_l))           # (X'X)^{-1} X'y
e_l <- y_l - X_l %*% b_l
SSE_l <- sum(e_l^2); SST_l <- sum((y_l - mean(y_l))^2); SSR_l <- SST_l - SSE_l
s2_l <- SSE_l / (n_l - 2)
se_l <- sqrt(s2_l * diag(solve(crossprod(X_l))))
t_l <- drop(b_l) / se_l
c(n = n_l, b0 = b_l[1], b1 = b_l[2], se_b0 = se_l[1], se_b1 = se_l[2], t_b0 = t_l[1],
  t_b1 = t_l[2])
```

```
##                  n                 b0                 b1
##           50.00000          469.70360            0.64709
## se_b0.(Intercept)          se_b1.POP  t_b0.(Intercept)
##          702.90619            0.04881            0.66823
##          t_b1.POP
##          13.25794
```

```
c(p_b0 = 2 * pt(-abs(t_l[1]), n_l - 2), p_b1 = 2 * pt(-abs(t_l[2]), n_l - 2),
  s = sqrt(s2_l), R2 = SSR_l / SST_l, R2_adj = 1 - s2_l / (SST_l / (n_l - 1)),
  multiple_R = sqrt(SSR_l / SST_l))
```

```
## p_b0.(Intercept)          p_b1.POP                 s
##        5.072e-01         1.158e-17         3.792e+03
##                R2            R2_adj        multiple_R
##        7.855e-01         7.810e-01         8.863e-01
```

```
c(SSR = SSR_l, SSE = SSE_l, SST = SST_l, MSR = SSR_l / 1, MSE = s2_l,
  F = (SSR_l / 1) / s2_l, p_F = pf((SSR_l / 1) / s2_l, 1, n_l - 2, lower.tail = FALSE))
```

```
##       SSR       SSE       SST       MSR       MSE         F
## 2.527e+09 6.901e+08 3.217e+09 2.527e+09 1.438e+07 1.758e+02
```

```
##       p_F
## 1.158e-17
```

The coefficient table holds $\hat{\beta}_j$, $\mathrm{SE}(\hat{\beta}_j)$, the t ratio for $H_0\colon \beta_j = 0$ and its two-sided p-value (Definition 2.36); the residual standard error is s on $n-k-1$ degrees of freedom; `Multiple R-squared` and `Adjusted R-squared` are (18); the F statistic is (23) on $(k, n-k-1)$ degrees of freedom; and `anova` prints the regression and error rows of the ANOVA table, the total row being their sum.

7.2 The ISLR lab

The `Boston` data of `ISLR2`: the median house value `medv` of 506 census tracts with 12 predictors, among them `lstat`, the percentage of households of low socioeconomic status.

```
1 lm.fit <- lm(medv ~ lstat, data = Boston)
```

```r
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ lstat, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -15.168  -3.990  -1.318   2.034  24.500
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  34.5538     0.5626    61.4   <2e-16 ***
## lstat        -0.9500     0.0387   -24.5   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.22 on 504 degrees of freedom
## Multiple R-squared:  0.544,        Adjusted R-squared:  0.543
## F-statistic:  602 on 1 and 504 DF,  p-value: <2e-16
```

```
names(lm.fit)
```

```
##  [1] "coefficients"  "residuals"     "effects"
##  [4] "rank"          "fitted.values" "assign"
##  [7] "qr"            "df.residual"   "xlevels"
## [10] "call"          "terms"         "model"
```

```
coef(lm.fit)
```

```
## (Intercept)       lstat
##       34.55       -0.95
```

```
confint(lm.fit)
```

```
##               2.5 % 97.5 %
## (Intercept) 33.448 35.659
## lstat       -1.026 -0.874
```

`predict` gives the confidence interval for the mean response and the prediction interval for a new response at chosen values of the predictor (Theorem 2.41): at `lstat` $= 10$ the fitted value is 25.05, the confidence interval $(24.47, 25.63)$ and the prediction interval $(12.83, 37.28)$.

```r
predict(lm.fit, data.frame(lstat = c(5, 10, 15)), interval = "confidence")
```

```
##     fit   lwr   upr
## 1 29.80 29.01 30.60
## 2 25.05 24.47 25.63
## 3 20.30 19.73 20.87
```

```r
predict(lm.fit, data.frame(lstat = c(5, 10, 15)), interval = "prediction")
```

```
##     fit    lwr   upr
## 1 29.80 17.566 42.04
## 2 25.05 12.828 37.28
## 3 20.30  8.078 32.53
```

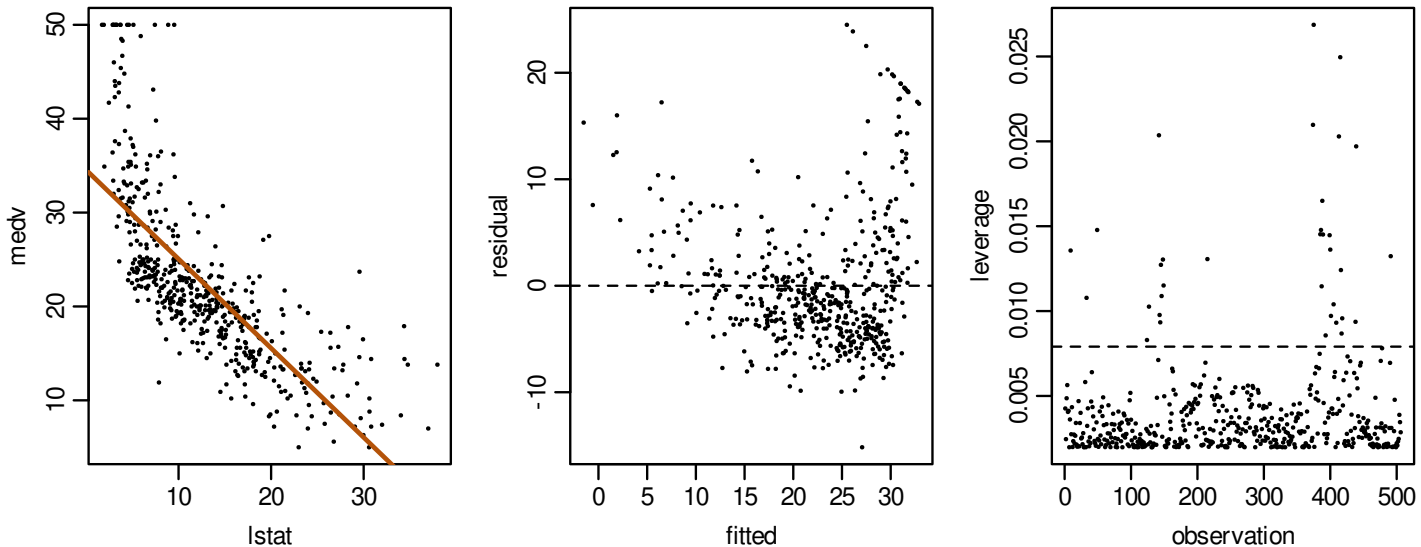


Figure 2.10: `medv` against `lstat` with the least squares line; residuals against fitted values; leverages with the flag $2(k+1)/n$.

The residuals bend: negative in the middle, positive at both ends, so the relation is not linear. Leverage is computed by `hatvalues`, and `which.max` names the observation with the largest.

```
c(max_leverage = max(hatvalues(lm.fit)), at = unname(which.max(hatvalues(lm.fit))),
  mean_leverage = 2 / 506, lstat_there = Boston$lstat[which.max(hatvalues(lm.fit))],
  max_lstat = max(Boston$lstat))
```

```
##  max_leverage            at mean_leverage   lstat_there
##      0.026865    375.000000      0.003953     37.970000
##     max_lstat
##     37.970000
```

Several predictors are joined by `+`, all of them by `.`, and all but one by `- age`; `update` refits with a changed formula.

```
lm.fit2 <- lm(medv ~ lstat + age, data = Boston)
round(coef(summary(lm.fit2)), 5)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.22276    0.73085  45.458  0.00000
## lstat       -1.03207    0.04819 -21.416  0.00000
## age          0.03454    0.01223   2.826  0.00491
```

```r
lm.fit3 <- lm(medv ~ ., data = Boston)
summary(lm.fit3)
```

```
##
## Call:
## lm(formula = medv ~ ., data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -15.130  -2.767  -0.581   1.941  26.253
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.61727    4.93604    8.43  3.8e-16 ***
## crim         -0.12139    0.03300   -3.68  0.00026 ***
## zn            0.04696    0.01388    3.38  0.00077 ***
## indus         0.01347    0.06214    0.22  0.82852
## chas          2.83999    0.87001    3.26  0.00117 **
## nox         -18.75802    3.85135   -4.87  1.5e-06 ***
## rm            3.65812    0.42025    8.70  < 2e-16 ***
```

```
## age           0.00361    0.01333    0.27  0.78659
## ... [output truncated]
```

```
c(R2 = summary(lm.fit3)$r.squared, RSE = summary(lm.fit3)$sigma)
```

```
##      R2     RSE
## 0.7343 4.7980
```

```
lm.fit4 <- update(lm.fit3, ~ . - age)
c(R2_without_age = summary(lm.fit4)$r.squared, RSE_without_age =
 ↪  summary(lm.fit4)$sigma)
```

```
##  R2_without_age RSE_without_age
##          0.7343           4.7935
```

Dropping **age**, whose p-value is 0.79, leaves R^2 unchanged to three decimals and lowers the residual standard error, as Theorem 2.31 requires of a variable with $|t| < 1$.

7.3 Formulas against package output

The fit is **medv** on **lstat**, **age** and **rm**, $k = 3$.

```
fit <- lm(medv ~ lstat + age + rm, data = Boston)
X <- model.matrix(fit); y <- Boston$medv; e <- residuals(fit)
```

```r
n <- nrow(X); k <- ncol(X) - 1L
c(n = n, k = k, df = n - k - 1L)
```

```
##   n   k  df
## 506   3 502
```

```r
XtX   <- t(X) %*% X
b_man <- drop(solve(XtX, t(X) %*% y))          # (X'X)^{-1} X'Y without forming the inverse
cbind(manual = b_man, lm = coef(fit), diff = b_man - coef(fit))
```

```
##                manual        lm       diff
## (Intercept) -1.175311 -1.175311  1.722e-12
## lstat       -0.668513 -0.668513 -1.699e-14
## age          0.009091  0.009091  1.568e-15
## rm           5.019133  5.019133 -2.585e-13
```

`lm` does not form $\mathbf{X}'\mathbf{X}$ at all; it factors $\mathbf{X} = \mathbf{QR}$, which avoids squaring the condition number. Equation (5) is how least squares is reasoned about, not how it is computed.

```r
s2    <- sum(e^2) / (n - k - 1)                    # MSE
V_man <- s2 * solve(XtX)                           # s^2 (X'X)^{-1}
c(max_abs_diff = max(abs(V_man - vcov(fit))), s2_manual = s2, s2_lm =
  ↪ summary(fit)$sigma^2)
```

```
## max_abs_diff    s2_manual        s2_lm
##    5.240e-13    3.072e+01    3.072e+01
```

```r
H <- X %*% solve(XtX) %*% t(X); M <- diag(n) - H
c(symmetric = max(abs(H - t(H))), idempotent = max(abs(H %*% H - H)),
  HM_zero = max(abs(H %*% M)), MX_zero = max(abs(M %*% X)),
  tr_H = sum(diag(H)), tr_M = sum(diag(M)))
```

```
##  symmetric idempotent    HM_zero    MX_zero       tr_H
##  2.082e-16  3.803e-15  3.803e-15  4.476e-12  4.000e+00
##       tr_M
##  5.020e+02
```

```r
c(fitted = max(abs(drop(H %*% y) - fitted(fit))), resid = max(abs(drop(M %*% y) - e)),
  leverage = max(abs(diag(H) - hatvalues(fit))),
  Xt_e = max(abs(t(X) %*% e)), sum_e = sum(e), yhat_t_e = sum(fitted(fit) * e))
```

```
##    fitted     resid  leverage      Xt_e     sum_e  yhat_t_e
## 1.428e-12 1.450e-12 3.775e-15 9.095e-13 4.233e-14 2.248e-12
```

```r
r_man <- e / (sqrt(s2) * sqrt(1 - hatvalues(fit)))
c(max_abs_diff = max(abs(r_man - rstandard(fit))), h_min = min(hatvalues(fit)),
  h_max = max(hatvalues(fit)))
```

```
## max_abs_diff        h_min        h_max
##     0.000000     0.002052     0.077074
```

```r
se_man <- sqrt(diag(V_man)); t_man <- coef(fit) / se_man
p_man  <- 2 * pt(abs(t_man), df = n - k - 1, lower.tail = FALSE)
tab    <- coef(summary(fit))
round(cbind(se_man, se_lm = tab[, 2], t_man, t_lm = tab[, 3], p_man, p_lm = tab[, 4]),
 ↪  8)
```

```
##              se_man   se_lm    t_man     t_lm p_man  p_lm
## (Intercept) 3.18192 3.18192  -0.3694  -0.3694 0.712 0.712
## lstat       0.05436 0.05436 -12.2985 -12.2985 0.000 0.000
## age         0.01122 0.01122   0.8106   0.8106 0.418 0.418
## rm          0.45431 0.45431  11.0479  11.0479 0.000 0.000
```

```r
SSE <- sum(e^2); SST <- sum((y - mean(y))^2); SSR <- SST - SSE
seq_ss <- anova(fit)[["Sum Sq"]]                    # sequential sums of squares
c(SSR = SSR, sum_sequential = sum(seq_ss[1:k]), SSE = SSE, anova_residual = seq_ss[k + 1L],
  SSR_projection = sum(((H - matrix(1/n, n, n)) %*% y)^2))
```

```
##            SSR sum_sequential            SSE anova_residual
##          27297          27297          15419          15419
## SSR_projection
##          27297
```

```r
R2 <- 1 - SSE / SST; adjR2 <- 1 - (n - 1) / (n - k - 1) * (1 - R2)
F_man <- (SSR / k) / (SSE / (n - k - 1))
c(R2 = R2, R2_lm = summary(fit)$r.squared, R2_cor2 = cor(y, fitted(fit))^2,
  adj = adjR2, adj_lm = summary(fit)$adj.r.squared,
  F = F_man, F_lm = summary(fit)$fstatistic[["value"]],
  F_anova_null = anova(lm(medv ~ 1, data = Boston), fit)[["F"]][2],
  R2_from_F = k * F_man / (k * F_man + n - k - 1))
```

```
##           R2        R2_lm      R2_cor2          adj
##       0.6390       0.6390       0.6390       0.6369
```

```
##       adj_lm            F         F_lm F_anova_null
##       0.6369     296.2377     296.2377     296.2377
##    R2_from_F
##       0.6390
```

```r
x0 <- c(1, lstat = 10, age = 60, rm = 6.5); nd0 <- data.frame(lstat = 10, age = 60, rm = 6.5)
lev0 <- drop(t(x0) %*% solve(XtX) %*% x0); tq <- qt(0.975, n - k - 1)
pr <- predict(fit, nd0, se.fit = TRUE)
c(yhat_manual = sum(x0 * coef(fit)), yhat_R = unname(pr$fit),
  se_fit_manual = sqrt(s2 * lev0), se_fit_R = pr$se.fit, width_ratio = sqrt((1 + lev0) / lev0))
```

```
##   yhat_manual        yhat_R se_fit_manual      se_fit_R
##       25.3094       25.3094        0.2658        0.2658
##   width_ratio
##       20.8763
```

```
rbind(manual = c(pr$fit + c(-1, 1) * tq * sqrt(s2 * lev0), pr$fit + c(-1, 1) * tq *
 ↪  sqrt(s2 * (1 + lev0))),
      R      = c(predict(fit, nd0, interval = "confidence")[2:3],
                 predict(fit, nd0, interval = "prediction")[2:3]))
```

```
##          [,1]  [,2]  [,3]  [,4]
## manual 24.79 25.83 14.41 36.21
## R      24.79 25.83 14.41 36.21
```

At this point the prediction interval is 21 times as wide as the confidence interval, the ratio being $\sqrt{(1+h_0)/h_0}$ by Theorem 2.41.

```
m_no <- lm(medv ~ lstat - 1, data = Boston)
c(R2_with = summary(lm.fit)$r.squared, R2_without = summary(m_no)$r.squared,
  SSE_with = sum(residuals(lm.fit)^2), SSE_without = sum(residuals(m_no)^2),
  sum_resid_without = sum(residuals(m_no)))
```

```
##           R2_with        R2_without          SSE_with
##            0.5441            0.4486       19472.3814
##       SSE_without sum_resid_without
##       165199.2060         4217.3843
```

Without an intercept the residuals do not sum to zero, Theorem 2.26 does not apply, and R switches to an uncentred R^2, $1-\text{SSE}/\sum_i Y_i^2$, which is not comparable with the centred one (here 0.449 against 0.544, with an SSE 8.5 times larger).

7.4 Maximum likelihood in R (beyond the syllabus)

```
s2_mle <- SSE / n
ll_man <- -n/2 * log(2*pi) - n/2 * log(s2_mle) - SSE / (2 * s2_mle)
c(s2_unbiased = s2, s2_mle = s2_mle, ratio = s2_mle / s2, factor = (n - k - 1) / n,
  loglik_manual = ll_man, loglik_R = as.numeric(logLik(fit)), npar_R =
   ↪  attr(logLik(fit), "df"))
```

```
##   s2_unbiased        s2_mle         ratio        factor
##       30.7154       30.4726        0.9921        0.9921
## loglik_manual      loglik_R        npar_R
##    -1582.4402    -1582.4402        5.0000
```

`logLik` uses $\hat{\sigma}^2_{\text{MLE}}$ of Proposition 2.42 and counts $k+2$ parameters, the coefficients and σ.

7.5 The sampling theory by simulation

The design is held fixed, as A1–A5 require, and only the errors are redrawn: the means of the estimates, their variances, the moments of SSE/σ^2 and its independence from $\hat{\boldsymbol{\beta}}$ are Theorems 2.18, 2.19, 2.34.

```r
set.seed(1)
Xs <- X[1:60, ]; ns <- nrow(Xs); ks <- ncol(Xs) - 1L
beta_true <- c(10, -0.5, 0.02, 3); sigma <- 4; B <- 5000
bhat <- matrix(NA_real_, B, ks + 1L); sse <- numeric(B)
for (bb in seq_len(B)) {
  ysim <- drop(Xs %*% beta_true) + rnorm(ns, sd = sigma)
  f <- lm.fit(Xs, ysim)
  bhat[bb, ] <- f$coefficients; sse[bb] <- sum(f$residuals^2)
}
rbind(true = beta_true, mean_bhat = colMeans(bhat))
```

```
##              [,1]    [,2]    [,3]  [,4]
## true       10.000 -0.5000 0.02000 3.000
## mean_bhat   9.895 -0.4986 0.01977 3.016
```

```r
round(cbind(empirical = apply(bhat, 2, var), theory = sigma^2 * diag(solve(t(Xs) %*%
  Xs))), 5)
```

```
##             empirical    theory
## (Intercept) 110.78282 110.66690
## lstat         0.02010   0.02031
## age           0.00069   0.00071
## rm            2.53789   2.54066
```

```
c(mean_SSE_over_s2 = mean(sse) / sigma^2, df = ns - ks - 1L,
  var_SSE_over_s2 = var(sse / sigma^2), two_df = 2 * (ns - ks - 1L))
```

```
## mean_SSE_over_s2             df  var_SSE_over_s2
##             56.1           56.0             108.7
##           two_df
##            112.0
```

```
round(cor(bhat, sse), 4)
```

```
##         [,1]
## [1,]  0.0183
## [2,] -0.0113
## [3,]  0.0079
## [4,] -0.0185
```

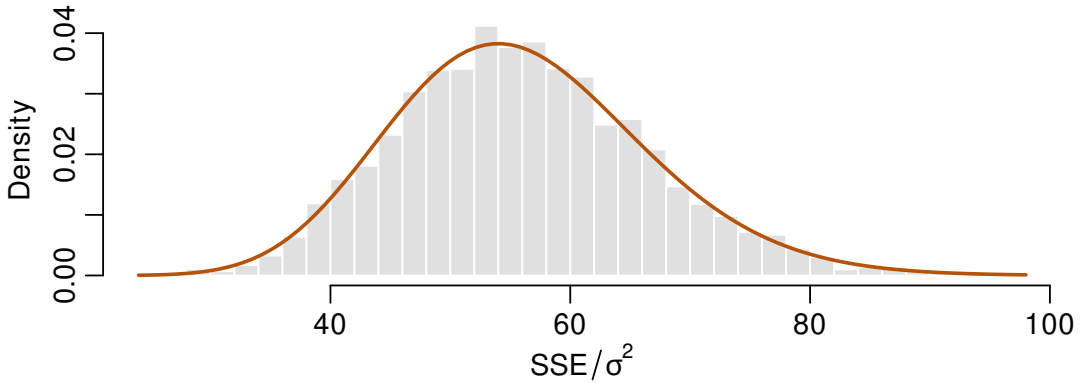


Figure 2.11: Simulated SSE/σ^2 over 5000 replications on a fixed design of 60 rows against the χ^2_{56} density.

The mean and variance of SSE/σ^2 are those of χ^2_{56}, 56 and 112, and its correlations with the four estimates are zero to Monte-Carlo error.

8 Exercises

Exercise 2.1 (FREES exercise 2.1)**.** For the data $x = (2, -6, 7)$, $y = (3, 4, 6)$, fit a line by least squares: determine r, b_1 and b_0.

Solution. $\bar{x} = 1$ and $\bar{y} = 13/3$; the deviations are $(1, -7, 6)$ and $(-4/3, -1/3, 5/3)$, so $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = -4/3 + 7/3 + 10 = 11$, $\sum_i (x_i - \bar{x})^2 = 1 + 49 + 36 = 86$ and $\sum_i (y_i - \bar{y})^2 = (16 + 1 + 25)/9 = 14/3$. By (1) and Corollary 2.14,

$$r = \frac{11}{\sqrt{86 \cdot 14/3}} = 0.549, \qquad b_1 = \frac{11}{86} = 0.128, \qquad b_0 = \frac{13}{3} - \frac{11}{86} = 4.205.$$

```
xa <- c(2, -6, 7); ya <- c(3, 4, 6)
c(r = cor(xa, ya), b1 = 11/86, b0 = 13/3 - 11/86); coef(lm(ya ~ xa))
```

```
##      r     b1     b0
## 0.5491 0.1279 4.2054
## (Intercept)          xa
##      4.2054      0.1279
```

■

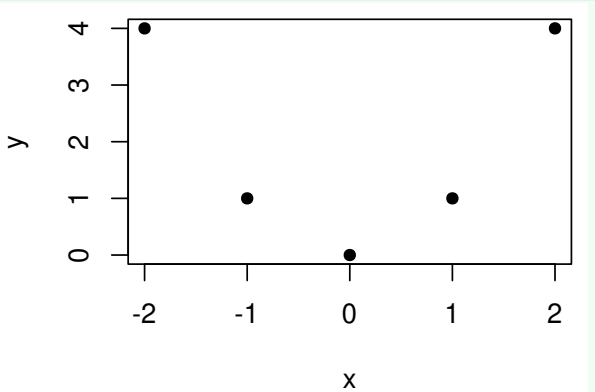
Exercise 2.2 (FREES exercise 2.2)**.** For $x = (-2, -1, 0, 1, 2)$ and $y = x^2 = (4, 1, 0, 1, 4)$: (a) graph the data; (b) check that $r = 0$.

Solution.

```
xq <- -2:2; yq <- xq^2
c(r = cor(xq, yq), cross_products = sum((xq - mean(xq)) * (yq - mean(yq))))
```

```
##              r cross_products
##              0              0
```

```
plot(xq, yq, pch = 16, xlab = "x", ylab = "y")
```



(b) $\bar{x} = 0$ and $\bar{y} = 2$, so $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i(y_i - 2) = (-2)(2) + (-1)(-1) + 0 + (1)(-1) + (2)(2) = 0$ and $r = 0$: a perfect relation that is not linear has zero correlation, since by Proposition 2.2 $|r| = 1$ needs a straight line. ■

Exercise 2.3 (FREES exercise 2.3)**.** With a and c constants and standardised values $u_i = (x_i - \bar{x})/s_x$, $v_i = (y_i - \bar{y})/s_y$: (a) verify $0 \leqslant (n-1)^{-1} \sum_i (au_i - cv_i)^2 = a^2 + c^2 - 2acr$; (b) deduce $2ac(r-1) \leqslant (a-c)^2$; (c) take $a = c$ to show $r \leqslant 1$; (d) take $a = -c$ to show $r \geqslant -1$; (e) when is $r = -1$, and when $r = 1$?

Solution. (a) The left side is a mean of squares, hence non-negative. Expanding, $(n-1)^{-1} \sum_i (au_i - cv_i)^2 = a^2 (n-1)^{-1} \sum_i u_i^2 + c^2 (n-1)^{-1} \sum_i v_i^2 - 2ac(n-1)^{-1} \sum_i u_i v_i$, and $(n-1)^{-1} \sum_i u_i^2 = s_x^2 / s_x^2 = 1$, likewise for v, while $(n-1)^{-1} \sum_i u_i v_i = r$ by (1). (b) $a^2 + c^2 - 2acr \geqslant 0$ is $2acr \leqslant a^2 + c^2$; subtracting $2ac$ from both sides, $2ac(r-1) \leqslant a^2 - 2ac + c^2 = (a-c)^2$. (c) With $a = c \neq 0$, (b) reads $2a^2(r-1) \leqslant 0$, so $r \leqslant 1$. (d) With $a = -c \neq 0$, (b) reads $-2a^2(r-1) \leqslant 4a^2$, that is $r - 1 \geqslant -2$, so $r \geqslant -1$. (e) $r = 1$ iff the sum in (a) vanishes at $a = c$, that is $u_i = v_i$ for every i: each y_i is the same number of standard deviations from its mean as x_i, the points lying on a line of positive slope s_y/s_x. $r = -1$ iff $u_i = -v_i$ for every i, a line of negative

slope. This is Proposition 2.2 by another route. ■

Exercise 2.4 (FREES exercise 2.4)**.** Show that $b_0 = \sum_i w_{i,0} y_i$ is a weighted sum of the responses and express $w_{i,0}$ through the slope weights w_i of Proposition 2.17.

Solution. $b_0 = \bar{y} - b_1 \bar{x} = \sum_i \frac{1}{n} y_i - \bar{x} \sum_i w_i y_i = \sum_i (\frac{1}{n} - \bar{x} w_i) y_i$, so $w_{i,0} = 1/n - \bar{x} w_i$. The weights sum to 1, since $\sum_i w_i = 0$, and b_0 is the mean of the responses when $\bar{x} = 0$. ■

Exercise 2.5 (FREES exercise 2.5)**.** (a) With $slope_i = (y_i - \bar{y})/(x_i - \bar{x})$ the slope from $(\bar{x}, \bar{y})$ to (x_i, y_i), show $b_1 = \sum_i weight_i \, slope_i / \sum_i weight_i$ and identify $weight_i$. (b) With $\bar{x} = 4$, $\bar{y} = 3$, $x_1 = 2$, $y_1 = 6$, find $slope_1$ and $weight_1$.

Solution. (a) With $weight_i = (x_i - \bar{x})^2$, $\sum_i weight_i \, slope_i = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ and $\sum_i weight_i = \sum_i (x_i - \bar{x})^2$, whose ratio is b_1 by Corollary 2.14: the least squares slope is the average of the point slopes weighted by squared distance from $\bar{x}$, which is why distant points dominate. (b) $slope_1 = (6 - 3)/(2 - 4) = -1.5$ and $weight_1 = (2 - 4)^2 = 4$. ■

Exercise 2.6 (FREES exercise 2.6)**.** Let $b_{y|x}$ be the slope from regressing y on x and $b_{x|y}$ that from regressing x on y, both with intercepts. Show $|r| = \sqrt{b_{y|x}\, b_{x|y}}$.

Solution. By Corollary 2.14, $b_{y|x} = r\, s_y/s_x$ and, exchanging the roles of the variables, $b_{x|y} = r\, s_x/s_y$. Their product is r^2, so $|r| = \sqrt{b_{y|x} b_{x|y}}$; the two slopes share the sign of r. For $r \neq 0$, drawn on the same axes the second line has slope $1/b_{x|y} = s_y/(r s_x)$, and the ratio of the first slope to it is r^2: the lines coincide iff $r^2 = 1$, and otherwise the regression of y on x is the flatter. In standard units the fitted line is $\hat{v} = ru$, so a value one standard deviation above average in x is predicted r standard deviations from the average in y, above it when $r > 0$ and below it when $r < 0$, and $|r| \leqslant 1$: the regression toward the mean of Unit 1. ■

Exercise 2.7 (FREES exercise 2.7)**.** For regression through the origin, $y_i = \beta_1 x_i + \varepsilon_i$: (a) verify $b_1 = \sum_i x_i y_i / \sum_i x_i^2$; (b) find the least squares estimate in the model $y_i = \beta_1 z_i^2 + \varepsilon_i$.

Solution. (a) $\mathrm{d} \sum_i (y_i - b x_i)^2 / \mathrm{d}b = -2 \sum_i x_i (y_i - b x_i) = 0$ gives $b \sum_i x_i^2 = \sum_i x_i y_i$, and the second derivative $2 \sum_i x_i^2 > 0$ makes it the minimum; this is Corollary 2.15. (b) The model is linear in β_1 with explanatory variable $x_i = z_i^2$, so $b_1 = \sum_i z_i^2 y_i / \sum_i z_i^4$. ■

Exercise 2.8 (FREES exercise 2.8)**.** Show (a) $\sum_i (y_i - \bar{y})^2 = \sum_i y_i^2 - n\bar{y}^2$; (b) $\sum_i (y_i - \bar{y})(x_i - \bar{x}) = \sum_i x_i y_i - n\bar{x}\bar{y}$; (c) $b_1 = \sum_i (y_i - \bar{y})(x_i - \bar{x}) / \sum_i (x_i - \bar{x})^2$; and (d) $b_1 = (\sum_i x_i y_i - n\bar{x}\bar{y}) / (\sum_i x_i^2 - n\bar{x}^2)$.

Solution. (a) $\sum_i (y_i - \bar{y})^2 = \sum_i y_i^2 - 2\bar{y} \sum_i y_i + n\bar{y}^2 = \sum_i y_i^2 - 2n\bar{y}^2 + n\bar{y}^2$. (b) $\sum_i (y_i - \bar{y})(x_i - \bar{x}) = \sum_i x_i y_i - \bar{x} \sum_i y_i - \bar{y} \sum_i x_i + n\bar{x}\bar{y} = \sum_i x_i y_i - n\bar{x}\bar{y}$. (c) is the first form of (11), from the normal equations. (d) Substitute (a), applied to x, and (b) into (c). ■

Exercise 2.9 (FREES exercise 2.9)**.** Suppose x_i takes only the values 0 and 1: n_1 observations have $x = 0$ with mean response $\bar{y}_1$, and $n_2 = n - n_1$ have $x = 1$ with mean $\bar{y}_2$. Show $b_1 = \bar{y}_2 - \bar{y}_1$ and $b_0 = \bar{y}_1$, and give $\mathrm{SE}(b_1)$ in terms of n_1, n_2 and s.

Solution. $\bar{x} = n_2/n$ and $x_i^2 = x_i$, so $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2 = n_2 - n_2^2/n = n_1 n_2/n$, and $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i x_i y_i - n\bar{x}\bar{y} = n_2\bar{y}_2 - \frac{n_2}{n}(n_1\bar{y}_1 + n_2\bar{y}_2) = \frac{n_1 n_2}{n}(\bar{y}_2 - \bar{y}_1)$. By Corollary 2.14 the ratio is $b_1 = \bar{y}_2 - \bar{y}_1$, and $b_0 = \bar{y} - b_1\bar{x} = \frac{1}{n}(n_1\bar{y}_1 + n_2\bar{y}_2) - \frac{n_2}{n}(\bar{y}_2 - \bar{y}_1) = \bar{y}_1$. The fitted values are the two group means, so SSE is the within-group sum of squares and s^2 the pooled variance

on $n-2$ degrees of freedom; by Corollary 2.20,

$$\mathrm{SE}(b_1) = \frac{s}{\sqrt{n_1 n_2/n}} = s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}},$$

so $t(b_1)$ is the pooled two-sample t statistic: regression on one indicator is the two-sample comparison of means. ■

Exercise 2.10 (FREES exercise 2.10)**.** Nursing home data of 2000 (`WiscNursingHome`, Unit 1). Part 1: (a) the correlations (i) between `TPY` and `LOGTPY` $=\ln$ `TPY`, (ii) among `TPY`, `NUMBED` and `SQRFOOT`, (iii) between `TPY` and `NUMBED`/10; (b) scatter plots of `TPY` against `NUMBED` and against `SQRFOOT`; (c) regressions of `TPY` on `NUMBED`, of `TPY` on `SQRFOOT`, of `LOGTPY` on $\ln$ `NUMBED` and of `LOGTPY` on $\ln$ `SQRFOOT`, each summarised by R^2 and the t ratio of the slope. Part 2: refit `TPY` on `NUMBED` for 2001.

Solution.

```
nh <- read.csv("data/frees/csv/WiscNursingHome.csv")
nh0 <- subset(nh, CRYEAR == 2000); nh1 <- subset(nh, CRYEAR == 2001)
c(cor_TPY_logTPY = cor(nh0$TPY, log(nh0$TPY)), cor_TPY_NUMBED_over_10 = cor(nh0$TPY,
 ↪  nh0$NUMBED / 10))
```

```
##          cor_TPY_logTPY cor_TPY_NUMBED_over_10
##                  0.9372                 0.9791
```

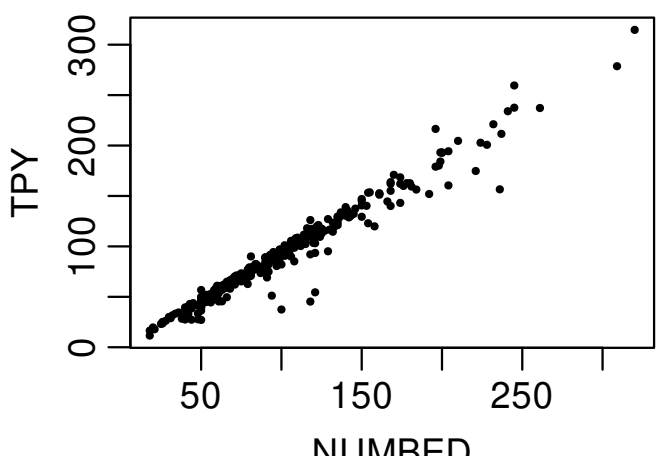
```
round(cor(nh0[, c("TPY", "NUMBED", "SQRFOOT")], use = "complete.obs"), 3)
```

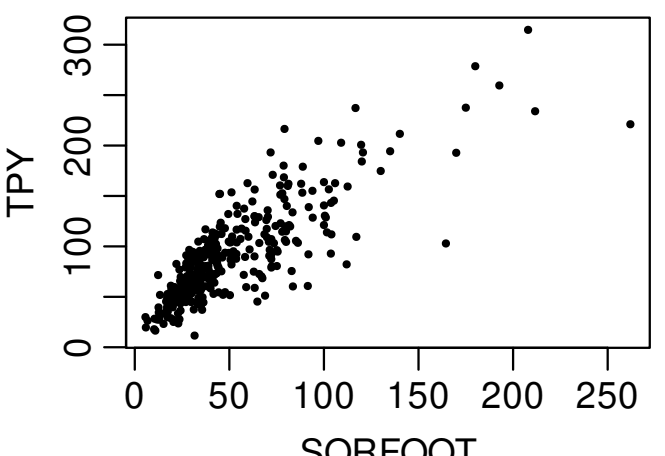
```
##           TPY NUMBED SQRFOOT
## TPY     1.000  0.979   0.824
## NUMBED  0.979  1.000   0.819
## SQRFOOT 0.824  0.819   1.000
```

```
one_slr <- function(f, d) { m <- lm(f, d)
  c(n = nobs(m), b1 = coef(m)[[2]], R2 = summary(m)$r.squared, t_b1 =
   ↪  coef(summary(m))[2, 3]) }
round(rbind(TPY_NUMBED = one_slr(TPY ~ NUMBED, nh0), TPY_SQRFOOT = one_slr(TPY ~
 ↪  SQRFOOT, nh0),
            logTPY_logNUMBED = one_slr(log(TPY) ~ log(NUMBED), nh0),
```

```
    logTPY_logSQRFOOT = one_slr(log(TPY) ~ log(SQRFOOT), nh0),
    TPY_NUMBED_2001 = one_slr(TPY ~ NUMBED, nh1)), 3)
```

```
##                        n     b1    R2   t_b1
## TPY_NUMBED           362 0.921 0.959  91.35
## TPY_SQRFOOT          357 1.103 0.680  27.45
## logTPY_logNUMBED     362 1.012 0.948  81.23
## logTPY_logSQRFOOT    357 0.687 0.677  27.25
## TPY_NUMBED_2001      355 0.932 0.976 120.39
```





(a)(i) The correlation between `TPY` and its logarithm is 0.937: the logarithm is increasing, so the association is strong, but it is not linear, so $r < 1$. (ii) `TPY` and `NUMBED` are correlated at 0.979, the square footage with either at about 0.82. (iii) Dividing `NUMBED` by 10 leaves the correlation at 0.979, by the scale invariance of Proposition 2.2. (b) Patient years rise linearly with beds along a tight band; against square footage the band is wider. (c) On `NUMBED`, $R^2 = 0.959$ with $t = 91$; on `SQRFOOT`, $R^2 = 0.680$ with $t = 27$ on the 357 facilities that report it, so beds fit far better. In logarithms the slope on $\ln$ `NUMBED` is 1.01, an elasticity of one, with $R^2 = 0.948$; on $\ln$ `SQRFOOT`, $R^2 = 0.677$. Part 2: in 2001 the slope on `NUMBED` is 0.932 against 0.921 in 2000, with $R^2 = 0.976$: the relation is stable. ■

Exercise 2.11 (FREES exercise 2.11)**.** For $n = 3$, $e_2 = 24$ and $e_3 = -1$; determine e_1.

Solution. By Theorem 2.11(ii) the residuals of a fit with an intercept sum to zero, so $e_1 = -(24 - 1) = -23$. ■

Exercise 2.12 (FREES exercise 2.12)**.** With $r = 0$, $n = 15$ and $s_y = 10$, determine s.

Solution. By Corollary 2.32, $R^2 = r^2 = 0$, so SSR $= 0$ and SSE $=$ SST $= (n-1)s_y^2 = 14(100) = 1400$; then $s^2 = \text{SSE}/(n-2) = 1400/13 = 107.7$ and $s = 10.38$. The line explains nothing, and s exceeds s_y because a degree of freedom was spent on the useless slope. ■

Exercise 2.13 (FREES exercise 2.13)**.** (a) Show $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$; (b) deduce $\text{SSR} = b_1^2 s_x^2 (n-1)$; (c) establish $R^2 = r^2$.

Solution. (a) $\hat{y}_i - \bar{y} = b_0 + b_1 x_i - \bar{y} = (\bar{y} - b_1\bar{x}) + b_1 x_i - \bar{y} = b_1(x_i - \bar{x})$. (b) Squaring and summing, $\text{SSR} = b_1^2 \sum_i (x_i - \bar{x})^2 = b_1^2 (n-1) s_x^2$. (c) With $b_1 = r\, s_y/s_x$, $\text{SSR} = r^2(n-1)s_y^2 = r^2 \text{SST}$, so $R^2 = \text{SSR}/\text{SST} = r^2$; this is Corollary 2.32. ■

Exercise 2.14 (FREES exercise 2.14)**.** Show that the average residual is zero.

Solution. The first normal equation, $\sum_i (y_i - b_0 - b_1 x_i) = 0$, is $\sum_i e_i = 0$; in general it is the first row of $\mathbf{X}'\mathbf{e} = \mathbf{0}$, Theorem 2.11(ii), and holds because the intercept column $\mathbf{1}$ is in the design. ■

Exercise 2.15 (FREES exercise 2.15)**.** (a) If $\bar{x} = 0$ or $\bar{y} = 0$, show that $r(y, x) = 0$ iff $\sum_i y_i x_i = 0$. (b) Show that the residuals are uncorrelated with the explanatory variable, and (c) with the fitted values.

Solution. (a) $\sum_i (y_i - \bar{y})(x_i - \bar{x}) = \sum_i y_i x_i - n\bar{x}\bar{y}$ by Exercise 2.8(b), and the second term vanishes when either mean is zero; so the numerator of r is $\sum_i y_i x_i$ and, the denominator being positive, $r = 0$ iff the sum is zero. (b) $\bar{e} = 0$ by Exercise 2.14, and $\sum_i x_i e_i = 0$ is the second normal equation (Theorem 2.11(i)), so by (a) with y replaced by e, $r(e, x) = 0$. (c) $\sum_i \hat{y}_i e_i = 0$ is Theorem 2.11(iii), and $\bar{e} = 0$ again gives $r(e, \hat{y}) = 0$. ■

Exercise 2.16 (FREES exercise 2.16)**.** Show (a) $R^2 = 1 - \frac{n-2}{n-1}\frac{s^2}{s_y^2}$; (b) $s = s_y\sqrt{(1 - r^2)\frac{n-1}{n-2}}$; (c) $t(b_1) = \sqrt{n-2}\,\dfrac{r}{\sqrt{1 - r^2}}$.

Solution. (a) $R^2 = 1 - \text{SSE/SST}$ with $\text{SSE} = (n-2)s^2$ and $\text{SST} = (n-1)s_y^2$. (b) Solve (a) for

s^2 using $R^2 = r^2$: $s^2 = s_y^2(1-r^2)(n-1)/(n-2)$. (c) By Definition 2.24 and Corollary 2.14,

$$t(b_1) = \frac{b_1}{\mathrm{SE}(b_1)} = \frac{r\, s_y/s_x}{s/(s_x\sqrt{n-1})} = \frac{r\, s_y\sqrt{n-1}}{s}$$
$$= \frac{r\, s_y\sqrt{n-1}}{s_y\sqrt{(1-r^2)(n-1)/(n-2)}} = \sqrt{n-2}\,\frac{r}{\sqrt{1-r^2}}.$$

The t statistic is a function of r and n alone, symmetric in x and y, so regressing x on y gives the same t; for the lottery data, $\sqrt{48}\,(0.8863)/\sqrt{1-0.7855} = 13.3$. ■

Exercise 2.17 (FREES exercise 2.17)**.** In a regression with one predictor on $n = 100$ observations the tenth residual is large. What percentage of SSE is due to it if (a) $e_{10} = 8s$; (b) $e_{10} = 4s$; (c) $e_{10} = 4s$ with $n = 20$?

Solution. SSE $= (n-2)s^2$, so $e_{10}^2/\mathrm{SSE} = c^2/(n-2)$ when $e_{10} = cs$: (a) $64/98 = 65.3\%$; (b) $16/98 = 16.3\%$; (c) $16/18 = 88.9\%$. A residual of $8s$ at $n = 100$ owns two thirds of the error sum of squares, and $c^2 \leqslant n-2$ is forced, since one squared residual cannot exceed their sum. ■

Exercise 2.18 (FREES exercise 2.18)**.** With $n = 20$, $\bar{x} = 0$, $\bar{y} = 9$, $s_x = 1$, $s_y = 10$ and $s = 7$, determine the standard error of a prediction at $x_* = 1$.

Solution. By Theorem 2.41(iii),

$$\text{SE(pred)} = 7\sqrt{1 + \frac{1}{20} + \frac{(1-0)^2}{19(1)}} = 7\sqrt{1.1026} = 7.35,$$

almost all of it the σ of the new observation; $\bar{y}$ and s_y are not needed. ■

Exercise 2.19 (FREES exercise 2.19)**.** Anscombe's data (`AnscombesData`): (a) compute the means and standard deviations of each column; (b) regress y_1, y_2, y_3 on x_1 and y_4 on x_2, and verify $b_0 \approx 3.0$, $b_1 \approx 0.5$, $s \approx 1.237$, $R^2 \approx 0.677$ in each case; (c) draw the four scatter plots; (d) discuss.

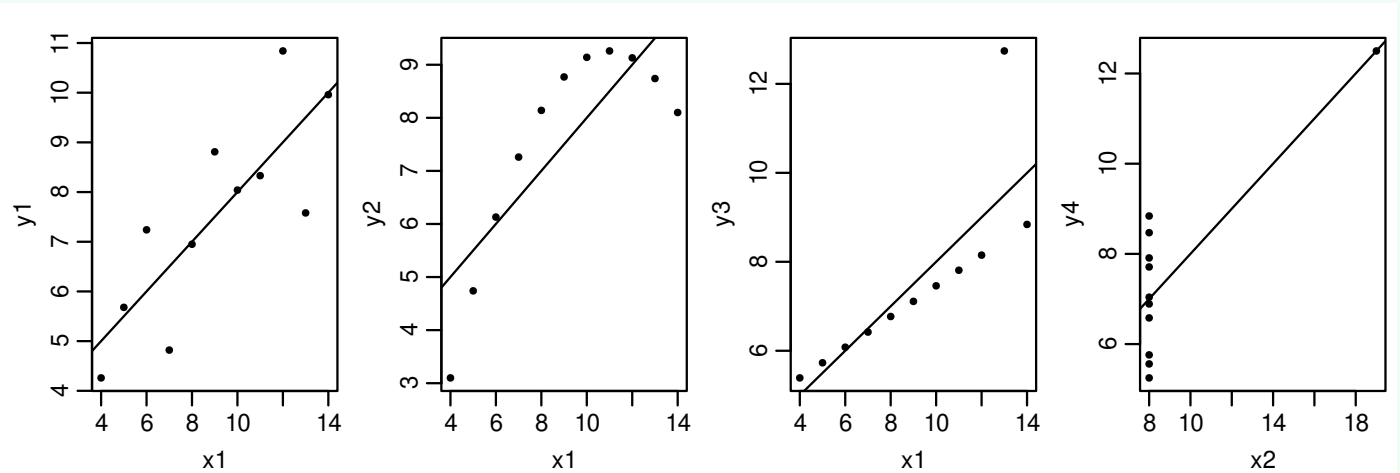
Solution.

```
an <- read.csv("data/frees/csv/AnscombesData.csv")
round(sapply(an[, -1], function(v) c(mean = mean(v), sd = sd(v))), 3)
```

```
##         x1    y1    y2   y3    x2    y4
## mean 9.000 7.501 7.501 7.50 9.000 7.501
## sd   3.317 2.032 2.032 2.03 3.317 2.031
```

```
one_fit_a <- function(f) { m <- lm(f, an)
  c(b0 = coef(m)[[1]], b1 = coef(m)[[2]], s = summary(m)$sigma, R2 =
   ↪  summary(m)$r.squared) }
round(rbind(y1_x1 = one_fit_a(y1 ~ x1), y2_x1 = one_fit_a(y2 ~ x1),
            y3_x1 = one_fit_a(y3 ~ x1), y4_x2 = one_fit_a(y4 ~ x2)), 3)
```

```
##          b0  b1     s    R2
## y1_x1 3.000 0.5 1.237 0.667
## y2_x1 3.001 0.5 1.237 0.666
## y3_x1 3.002 0.5 1.236 0.666
## y4_x2 3.002 0.5 1.236 0.667
```



(a)–(b) The two x columns share mean 9.00 and standard deviation 3.32, the four y columns mean 7.50 and standard deviation 2.03, and the four fits agree in b_0, b_1, s and R^2 to two decimals, with $R^2 = 0.666$ or 0.667 in each case: the exercise's 0.677 is a misprint in FREES. (c)–(d) The plots differ entirely: a linear relation with scatter; a smooth curve that a line misrepresents; a linear relation with one outlier that tilts the line; and a single high leverage point that alone determines the slope, the other ten x values being equal. Summary statistics and the fitted line cannot distinguish the four, so a regression is read with its scatter plot. ■

Exercise 2.20 (FREES exercise 2.20)**.** Nursing home data of 2001, `LOGTPY` $=$ ln `TPY` on `LOGNUMBED` $=$ ln `NUMBED`: (a) summary statistics, the correlation and a scatter plot; (b) the fit, with R^2, the slope and its t ratio; (c) at the 5% level, with p-values, test $H_0\colon \beta_1 = 0$ against $\beta_1 \neq 0$ and $H_0\colon \beta_1 = 1$ against $\beta_1 \neq 1$, $\beta_1 > 1$ and $\beta_1 < 1$; (d) a point estimate and 95% and 99% confidence intervals for the expected change in `LOGTPY` when `LOGNUMBED` changes by 2; (e) at 100 beds, the predicted `LOGTPY`, its standard error of prediction, a 95% prediction interval, the same converted to patient years, and a 90% interval.

Solution.

```
nh1$LOGTPY <- log(nh1$TPY); nh1$LOGNUMBED <- log(nh1$NUMBED)
round(rbind(LOGTPY = descr(nh1$LOGTPY), LOGNUMBED = descr(nh1$LOGNUMBED)), 3)
```

```
##             mean median    sd   min   max
## LOGTPY     4.368  4.396 0.510 2.511 6.088
## LOGNUMBED 4.457  4.500 0.492 2.890 6.125
```

```
fit_nh <- lm(LOGTPY ~ LOGNUMBED, data = nh1)
c(cor = cor(nh1$LOGTPY, nh1$LOGNUMBED), R2 = summary(fit_nh)$r.squared, n =
 ↪  nobs(fit_nh))
```

```
##      cor       R2        n
##   0.9830   0.9664 355.0000
```

```
round(coef(summary(fit_nh)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.1747     0.0454   -3.85   0.0001
## LOGNUMBED     1.0192     0.0101  100.73   0.0000
```

```
b1n <- coef(fit_nh)[["LOGNUMBED"]]; se1n <- coef(summary(fit_nh))["LOGNUMBED", 2];
 ↪  t1 <- (b1n - 1) / se1n
c(t_beta1_is_1 = t1, crit_two = qt(0.975, 353), crit_one = qt(0.95, 353),
  p_two = 2 * pt(-abs(t1), 353), p_greater = pt(t1, 353, lower.tail = FALSE),
   ↪  p_less = pt(t1, 353))
```

```
## t_beta1_is_1     crit_two     crit_one        p_two
##      1.90057      1.96671      1.64918      0.05817
##    p_greater       p_less
##      0.02909      0.97091
```

```
rbind(estimate = 2 * b1n, CI95 = 2 * (b1n + c(-1, 1) * qt(0.975, 353) * se1n),
      CI99 = 2 * (b1n + c(-1, 1) * qt(0.995, 353) * se1n))
```

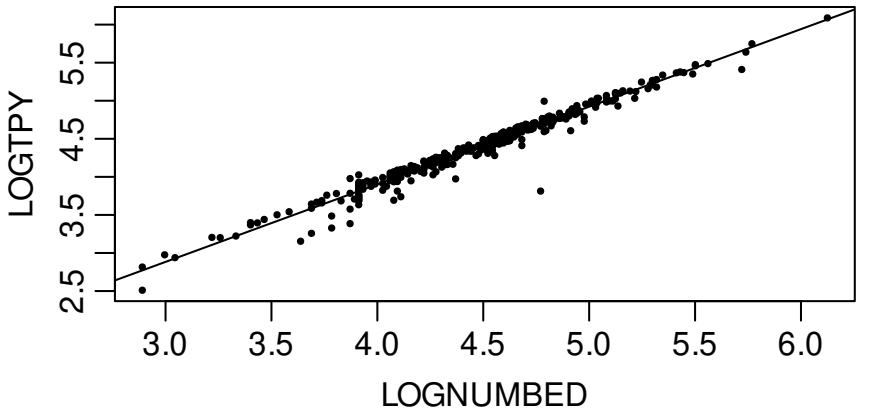
```
##           [,1]  [,2]
## estimate 2.038 2.038
## CI95     1.999 2.078
## CI99     1.986 2.091
```

```r
nd_nh <- data.frame(LOGNUMBED = log(100))
pr_nh <- predict(fit_nh, nd_nh, se.fit = TRUE)
c(fit = pr_nh$fit, se_pred = sqrt(summary(fit_nh)$sigma^2 + pr_nh$se.fit^2))
```

```
##   fit.1 se_pred
## 4.51904 0.09388
```

```r
rbind(log_scale_95 = predict(fit_nh, nd_nh, interval = "prediction"),
      patient_years_95 = exp(predict(fit_nh, nd_nh, interval = "prediction")),
      patient_years_90 = exp(predict(fit_nh, nd_nh, interval = "prediction", level =
      ↪  0.90)))
```

```
##      fit    lwr     upr
## 1  4.519  4.334   4.704
## 1 91.747 76.280 110.351
## 1 91.747 78.588 107.110
```



(a)–(b) The two logarithms are correlated at 0.983; the slope is 1.019 with standard error 0.010, $t = 101$ and $R^2 = 0.966$. (c) H_0: $\beta_1 = 0$ is rejected at any level. For H_0: $\beta_1 = 1$ the t ratio is $(1.019 - 1)/0.010 = 1.90$: against $\beta_1 \neq 1$ it is below $t_{353,0.975} = 1.97$, so H_0 is not rejected, $p = 0.058$; against $\beta_1 > 1$ it exceeds $t_{353,0.95} = 1.65$, so H_0 is rejected, $p = 0.029$; against $\beta_1 < 1$ it is not rejected, $p = 0.971$. (d) The expected change is $2b_1 = 2.04$, with intervals $(2.00, 2.08)$ and $(1.99, 2.09)$, twice those for β_1 by Definition 2.36 with $\mathbf{c} = (0, 2)'$. (e) At $\ln 100 = 4.605$ the prediction is 4.52 with standard error of prediction 0.094, nearly all of it s; the 95% interval

$(4.33, 4.70)$ becomes $(76.3, 110.4)$ patient years around 91.7 after exponentiation, and the 90% interval is $(78.6, 107.1)$. ■

Exercise 2.21 (FREES exercise 2.21)**.** Initial public offerings (`IPO`): one-year `RETURN` and 1997 revenue `REV` in millions of dollars for the 110 of 116 firms reporting both, with `LnREV` $= \ln$ `REV`. (a) With $r($`RETURN`$,$ `REV`$) = -0.0175$: (i) fit `RETURN` on `REV` from the summary statistics; (ii) the fitted return of Hyperion Telecommunications, `REV` $= 95.55$. (b) With the fit on `LnREV`: (i) the fitted return of Hyperion, compared with a(ii); (ii) test at 10% whether logarithmic revenue affects returns; (iii) test at 10% the null of no relation against a positive relation. (c) The correlation between `LnREV` and `RETURN`. (d) At `LnREV` $= 2$, the point prediction and a 95% prediction interval. (e) Inktomi, `LnREV` $= 1.76$ and `RETURN` $= 4.333$: (i) its residual; (ii) its share of SSE; (iii) the definition of a high leverage point; (iv) whether Inktomi is one.

Solution.

```
ipo <- read.csv("data/frees/csv/IPO.csv"); ipo <- ipo[complete.cases(ipo[,
 ↪  c("RETURN", "REV")]), ]
round(t(sapply(ipo[, c("RETURN", "REV", "LnREV")], descr)), 3)
```

```
##           mean median      sd    min      max
## RETURN   0.106 -0.130   0.824 -0.938    4.333
## REV    134.487 39.971 261.881  0.099 1455.761
## LnREV    3.686  3.688   1.698 -2.316    7.283
```

```r
b1_rev <- cor(ipo$RETURN, ipo$REV) * sd(ipo$RETURN) / sd(ipo$REV)
b0_rev <- mean(ipo$RETURN) - b1_rev * mean(ipo$REV)
c(r = cor(ipo$RETURN, ipo$REV), b1 = b1_rev, b0 = b0_rev, hyperion_REV = b0_rev +
 ↪  b1_rev * 95.55)
```

```
##           r          b1          b0 hyperion_REV
##  -0.01752069  -0.00005514   0.11311036   0.10784127
```

```r
fit_ipo <- lm(RETURN ~ LnREV, data = ipo)
round(coef(summary(fit_ipo)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.4381     0.1861   2.354   0.0204
## LnREV        -0.0902     0.0459  -1.965   0.0520
```

```r
c(s = summary(fit_ipo)$sigma, R2 = summary(fit_ipo)$r.squared, r_LnREV =
  cor(ipo$LnREV, ipo$RETURN),
  hyperion_LnREV = unname(predict(fit_ipo, data.frame(LnREV = log(95.55)))),
  crit_two_10 = qt(0.95, 108), crit_one_10 = qt(0.90, 108))
```

```
##             s            R2      r_LnREV hyperion_LnREV
##       0.81364       0.03452      -0.18579       0.02685
##   crit_two_10   crit_one_10
##       1.65909       1.28944
```

```r
predict(fit_ipo, data.frame(LnREV = 2), interval = "prediction")
```

```
##      fit   lwr   upr
## 1 0.2577 -1.37 1.885
```

```r
i_ink <- which.max(ipo$RETURN)
c(company = ipo$COMPANY[i_ink], LnREV = ipo$LnREV[i_ink], RETURN =
  ipo$RETURN[i_ink])
```

```
##         company           LnREV          RETURN
## "Inktomi Corp."   "1.755268361"   "4.333333333"
```

```
c(residual = residuals(fit_ipo)[[i_ink]], share_of_SSE =
  residuals(fit_ipo)[[i_ink]]^2 / sum(residuals(fit_ipo)^2),
  leverage = hatvalues(fit_ipo)[[i_ink]], mean_leverage = 2 / 110, flag = 4 / 110,
  std_resid = rstandard(fit_ipo)[[i_ink]])
```

```
##      residual  share_of_SSE      leverage mean_leverage
##       4.05353       0.22982       0.02095       0.01818
##          flag     std_resid
##       0.03636       5.03500
```

(a) $b_1 = r\,s_y/s_x = -0.0175(0.824)/261.9 = -5.5 \times 10^{-5}$ and $b_0 = 0.106 - b_1(134.5) = 0.113$; Hyperion's fitted return is $0.113 - 5.5 \times 10^{-5}(95.55) = 0.108$. (b)(i) On the logarithmic scale the fit is $0.4381 - 0.0902\,$`LnREV` and Hyperion's fitted return is $0.4381 - 0.0902 \ln 95.55 = 0.027$: two models, two fitted values. (ii) H_0: $\beta_1 = 0$ against $\beta_1 \neq 0$ at 10%: $|t| = 1.97$ exceeds $t_{108,0.95} = 1.66$, so H_0 is rejected, with $p = 0.052$. (iii) Against $\beta_1 > 0$ the rejection region is $t > t_{108,0.90} = 1.29$, and $t = -1.97$ is not in it: H_0 is not rejected, the evidence pointing the other way. (c) $r = -0.186$, negative like the slope, and $r^2 = 0.0345 = R^2$. (d) $0.438 - 0.090(2) = 0.258$, with prediction interval $(-1.37, 1.89)$: the interval is 3.3 wide because $s = 0.81$ is large relative to the fitted values. (e)(i) Inktomi's residual is $4.333 - \{0.438 - 0.090(1.76)\} = 4.05$; (ii) its square

is 23% of SSE; (iii) a high leverage point is unusual in the explanatory variable, with h_{ii} above $2(k+1)/n$ (Definition 2.43); (iv) its leverage 0.021 is barely above the mean 0.018 and below the flag 0.036, since 1.76 is one standard deviation below the mean of `LnREV`: Inktomi is an outlier, with standardised residual 5.0, and not a high leverage point. ■

Exercise 2.22 (FREES exercise 2.22)**.** Regress `LIFEEXP` on `FERTILITY` (`UNLifeExpectancy`, Unit 1). (a) The fitted life expectancy of the United States, `FERTILITY` $= 2.0$; (b) a 95% prediction interval for Dominica, not in the regression, at 2.0; (c) China, `FERTILITY` $= 1.7$ and `LIFEEXP` $=$ 72.5: its residual and the residual as a multiple of s; (d) test at 5% the prior hypothesis that the slope is -6.0 against the alternative that it is greater, with an approximate p-value.

Solution.

```
un <- read.csv("data/frees/csv/UNLifeExpectancy.csv")
fit_un <- lm(LIFEEXP ~ FERTILITY, data = un)
round(coef(summary(fit_un)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   83.738     1.0439   80.22        0
## FERTILITY     -5.274     0.2887  -18.27        0
```

```
c(n = nobs(fit_un), s = summary(fit_un)$sigma, R2 = summary(fit_un)$r.squared)
```

```
##        n        s       R2
## 181.0000   6.6149   0.6508
```

```
rbind(US = predict(fit_un, data.frame(FERTILITY = 2.0)),
      Dominica = predict(fit_un, data.frame(FERTILITY = 2.0), interval =
       ↪  "prediction"))
```

```
##      fit   lwr   upr
## US 73.19 73.19 73.19
## 1  73.19 60.08 86.30
```

```
res_china <- 72.5 - predict(fit_un, data.frame(FERTILITY = 1.7))
c(residual = unname(res_china), multiple_of_s = unname(res_china) /
 ↪  summary(fit_un)$sigma)
```

```
##      residual multiple_of_s
##       -2.2731       -0.3436
```

```r
b1u <- coef(fit_un)[["FERTILITY"]]; se1u <- coef(summary(fit_un))["FERTILITY", 2]
c(b1 = b1u, t = (b1u + 6) / se1u, crit = qt(0.95, nobs(fit_un) - 2), p = pt((b1u +
  6) / se1u, nobs(fit_un) - 2, lower.tail = FALSE))
```

```
##         b1          t       crit          p
## -5.273501  2.516447  1.653411  0.006367
```

The regression uses the 181 countries reporting fertility: $\widehat{\texttt{LIFEEXP}} = 83.7 - 5.27\,\texttt{FERTILITY}$, $s = 6.6$, $R^2 = 0.65$. (a) At 2.0 births per woman the fitted life expectancy is 73.2 years. (b) Dominica's prediction interval is $(60.1, 86.3)$, about $\pm 2s$. (c) China's fitted value is 74.8, so its residual is -2.3 years, 0.34 of s: unremarkable. (d) H_0: $\beta_1 = -6$ against $\beta_1 > -6$: $t = (-5.2735 + 6)/0.2887 = 2.52$ exceeds $t_{179,0.95} = 1.65$, so H_0 is rejected with $p = 0.006$; the slope is less steep than -6. ■

Exercise 2.23 (FREES exercise 3.1)**.** A data set of $n = 100$ observations has $s_y = 80$; a regression on three explanatory variables gives $s = 50$. (a) Calculate R^2_a; (b) complete the ANOVA table; (c) calculate R^2.

Solution. Two mean squares determine everything: $\text{SST} = (n-1)s_y^2$ and $\text{SSE} = (n-k-1)s^2$, and $\text{SSR} = \text{SST} - \text{SSE}$ by Theorem 2.26.

```
nA <- 100; kA <- 3; syA <- 80; sA <- 50
SSTa <- (nA - 1) * syA^2; SSEa <- (nA - kA - 1) * sA^2; SSRa <- SSTa - SSEa
c(SST = SSTa, SSR = SSRa, SSE = SSEa, MSR = SSRa / kA, MSE = sA^2, F = (SSRa / kA) /
  sA^2,
  R2 = SSRa / SSTa, R2_adj = 1 - sA^2 / syA^2)
```

```
##          SST          SSR          SSE          MSR          MSE
## 633600.0000 393600.0000 240000.0000 131200.0000   2500.0000
##            F           R2      R2_adj
##      52.4800       0.6212       0.6094
```

(a) $R_a^2 = 1 - s^2/s_y^2 = 1 - 2500/6400 = 0.609$, from (18) without any sum of squares. (b)

Source	Sum of squares	df	Mean square
Regression	393,600	3	131,200
Error	240,000	96	2,500
Total	633,600	99	

(c) $R^2 = 393{,}600/633{,}600 = 0.621$, above R^2_a as always when $k \geqslant 1$ and $R^2 < 1$. ■

Exercise 2.24 (FREES exercise 3.2)**.** With $n = 100$, $k = 3$, $s = 50$ and

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 100 & 20 & 20 & 20 \\ 20 & 90 & 30 & 40 \\ 20 & 30 & 80 & 50 \\ 20 & 40 & 50 & 70 \end{pmatrix},$$

determine (a) $\text{SE}(b_3)$; (b) the estimated $\text{cov}(b_2, b_3)$; (c) the estimated $\text{cor}(b_2, b_3)$; (d) the estimated variance of $4b_2 + 3b_3$.

Solution. By Theorem 2.19 the estimated covariance matrix is $s^2(\mathbf{X}'\mathbf{X})^{-1}$, so $\widehat{\mathrm{SE}}(b_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$ and $\widehat{\mathrm{cov}}(b_i, b_j) = s^2[(\mathbf{X}'\mathbf{X})^{-1}]_{ij}$; the correlation cancels s^2 and can be read off $(\mathbf{X}'\mathbf{X})^{-1}$ alone. For the linear combination, **variance under a linear map** with $\mathbf{c} = (0, 0, 4, 3)'$ gives $\widehat{\mathrm{var}}(\mathbf{c}'\hat{\boldsymbol{\beta}}) = s^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c} = s^2\{16(80) + 9(70) + 2(4)(3)(50)\}$.

```
XtXi <- rbind(c(100, 20, 20, 20), c(20, 90, 30, 40), c(20, 30, 80, 50), c(20, 40,
 ↪  50, 70))
sF <- 50; cF <- c(0, 0, 4, 3)
c(se_b3 = sF * sqrt(XtXi[4, 4]), cov_b23 = sF^2 * XtXi[3, 4],
  cor_b23 = XtXi[3, 4] / sqrt(XtXi[3, 3] * XtXi[4, 4]),
  var_lin = sF^2 * drop(t(cF) %*% XtXi %*% cF), se_lin = sF * sqrt(drop(t(cF) %*%
   ↪  XtXi %*% cF)))
```

```
##        se_b3      cov_b23      cor_b23      var_lin
##     418.3300  125000.0000       0.6682 7775000.0000
##       se_lin
##    2788.3687
```

(a) $\widehat{\mathrm{SE}}(b_3) = 50\sqrt{70} = 418$; (b) $2500(50) = 125{,}000$; (c) $50/\sqrt{80 \cdot 70} = 0.668$; (d) $2500(1280 + 630 + 1200) = 7{,}775{,}000$, of which the covariance term contributes $3{,}000{,}000$: ignoring it would

understate the standard error of $4b_2 + 3b_3$ by the factor $\sqrt{7775/4775} = 1.28$. ■

Exercise 2.25 (FREES exercise 3.3)**.** For the data

$$\mathbf{x}_1 = (-1, 2, 4, 6)', \quad \mathbf{x}_2 = (0, 0, 1, 1)', \quad \mathbf{y} = (0, 1, 5, 8)',$$

the fit of $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ has $s = 1.373$, $\mathbf{b} = (0.15, 0.692, 2.88)'$ and

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 0.53846 & -0.07692 & -0.15385 \\ -0.07692 & 0.15385 & -0.69231 \\ -0.15385 & -0.69231 & 4.11538 \end{pmatrix}.$$

(a) Write $\mathbf{y}$ and $\mathbf{X}$; (b) determine $\hat{y}_3$; (c) SE(b_2); (d) $t(b_1)$.

Solution. (a) $\mathbf{X}$ has the intercept column first: it is 4×3 with rows $(1, -1, 0)$, $(1, 2, 0)$, $(1, 4, 1)$, $(1, 6, 1)$, and $n - k - 1 = 1$.

```
Xf <- cbind(1, x1 = c(-1, 2, 4, 6), x2 = c(0, 0, 1, 1)); yf <- c(0, 1, 5, 8)
Af <- solve(crossprod(Xf)); bf <- drop(Af %*% crossprod(Xf, yf))
```

```
ef <- yf - Xf %*% bf; sf <- sqrt(sum(ef^2) / (4 - 3))
round(Af, 5)
```

```
##                    x1      x2
##     0.53846 -0.07692 -0.1538
## x1 -0.07692  0.15385 -0.6923
## x2 -0.15385 -0.69231  4.1154
```

```
sprintf("b%d = %.5f", 0:2, bf)
```

```
## [1] "b0 = 0.15385" "b1 = 0.69231" "b2 = 2.88462"
```

```
c(b0 = bf[1], b1 = bf[2], b2 = bf[3], s = sf, yhat3 = drop(Xf[3, ] %*% bf),
  se_b2 = sf * sqrt(Af[3, 3]), t_b1 = bf[2] / (sf * sqrt(Af[2, 2])),
  p_b1 = 2 * pt(-abs(bf[2] / (sf * sqrt(Af[2, 2]))), df = 1))
```

```
##      b0   b1.x1   b2.x2       s   yhat3   se_b2 t_b1.x1
##  0.1538  0.6923  2.8846  1.3728  5.8077  2.7849  1.2857
## p_b1.x1
##  0.4208
```

(b) $\hat{y}_3 = 0.154 + 0.692(4) + 2.885(1) = 5.81$; (c) with $s = 1.3728$, $\mathrm{SE}(b_2) = 1.3728\sqrt{4.11538} = 2.785$; (d) $t(b_1) = 0.69231/(1.3728\sqrt{0.15385}) = 1.29$ on one degree of freedom, whose two-sided

p-value is 0.42 and 5% critical value 12.7: four observations and three parameters leave almost nothing with which to estimate σ^2. ■

Exercise 2.26 (FREES exercise 3.4, parts (a)–(f), (i), (j))**.** The lottery data with the eight explanatory variables of `WiscLottery` (`PERPERHH`, `MEDSCHYR`, `MEDHVL`, `PRCRENT`, `PRC55P`, `HHMEDAGE`, `MEDINC`, `POP`). (a) Summary statistics; how many standard deviations above the mean are `MEDSCHYR` and `MEDHVL` for observation 11 (Shorewood)? (b) Which three variables are most correlated with `SALES`? (c) A scatterplot matrix; where is observation 11 in the plot of `SALES` against `MEDSCHYR`? (d) Fit `SALES` on all eight variables and report s, R^2, R^2_a. (e) Is `MEDSCHYR` significant? State the hypotheses, the criterion and the decision. (f) Fit `SALES` on `MEDSCHYR`, `MEDHVL` and `POP`; compare s, R^2, R^2_a with (d). (i) Refit (f) without observation 11: is `MEDSCHYR` significant? (j) Refit (f) without observation 9.

Solution.

```
1 round(t(sapply(lottery[, -1], descr)), 2)
```

```
##             mean median       sd   min     max
## PERPERHH    2.71    2.7     0.21   2.2     3.2
```

```
## MEDSCHYR   12.70   12.6     0.55  12.2    15.9
## MEDHVL     57.09   53.9    18.37  34.5   120.0
## PRCRENT    24.68   24.0     9.34   6.0    62.0
## PRC55P     39.70   40.0     7.51  25.0    56.0
## HHMEDAGE   48.76   48.0     4.14  41.0    59.0
## MEDINC     45.12   43.1     9.78  27.9    70.7
## SALES    6494.83 2426.4  8103.01 189.0 33181.4
## POP      9311.04 4405.5 11098.16 280.0 39098.0
```

```
c(z_MEDSCHYR_11 = (lottery$MEDSCHYR[11] - mean(lottery$MEDSCHYR)) /
  sd(lottery$MEDSCHYR),
  z_MEDHVL_11 = (lottery$MEDHVL[11] - mean(lottery$MEDHVL)) / sd(lottery$MEDHVL))
```

```
## z_MEDSCHYR_11   z_MEDHVL_11
##         5.810         3.424
```

```
round(sort(cor(lottery[, -1])[, "SALES"], decreasing = TRUE), 3)
```

```
##    SALES      POP   MEDHVL  PRCRENT   MEDINC MEDSCHYR
##    1.000    0.886    0.572    0.496    0.469    0.467
## PERPERHH HHMEDAGE   PRC55P
##   -0.136   -0.365   -0.386
```

```
fit_stats3 <- function(m) c(n = nobs(m), s = summary(m)$sigma, R2 =
 ↪  summary(m)$r.squared,
                            R2_adj = summary(m)$adj.r.squared)
fit_l8 <- lm(SALES ~ . - ZIP, data = lottery)
fit_l3 <- lm(SALES ~ MEDSCHYR + MEDHVL + POP, data = lottery)
fit_l3_no11 <- update(fit_l3, data = lottery[-11, ]); fit_l3_no9 <- update(fit_l3,
 ↪  data = lottery[-9, ])
round(rbind(eight = fit_stats3(fit_l8), three = fit_stats3(fit_l3),
            three_no_11 = fit_stats3(fit_l3_no11), three_no_9 =
             ↪  fit_stats3(fit_l3_no9)), 3)
```

```
##              n    s    R2 R2_adj
## eight       50 3580 0.837  0.805
## three       50 3445 0.830  0.819
## three_no_11 49 3440 0.827  0.815
## three_no_9  49 2585 0.899  0.893
```

```
round(coef(summary(fit_l8))["MEDSCHYR", ], 4); c(crit = qt(0.975, 41))
```

```
##   Estimate Std. Error    t value   Pr(>|t|)
## -3761.7376  1660.9387    -2.2648     0.0289
```

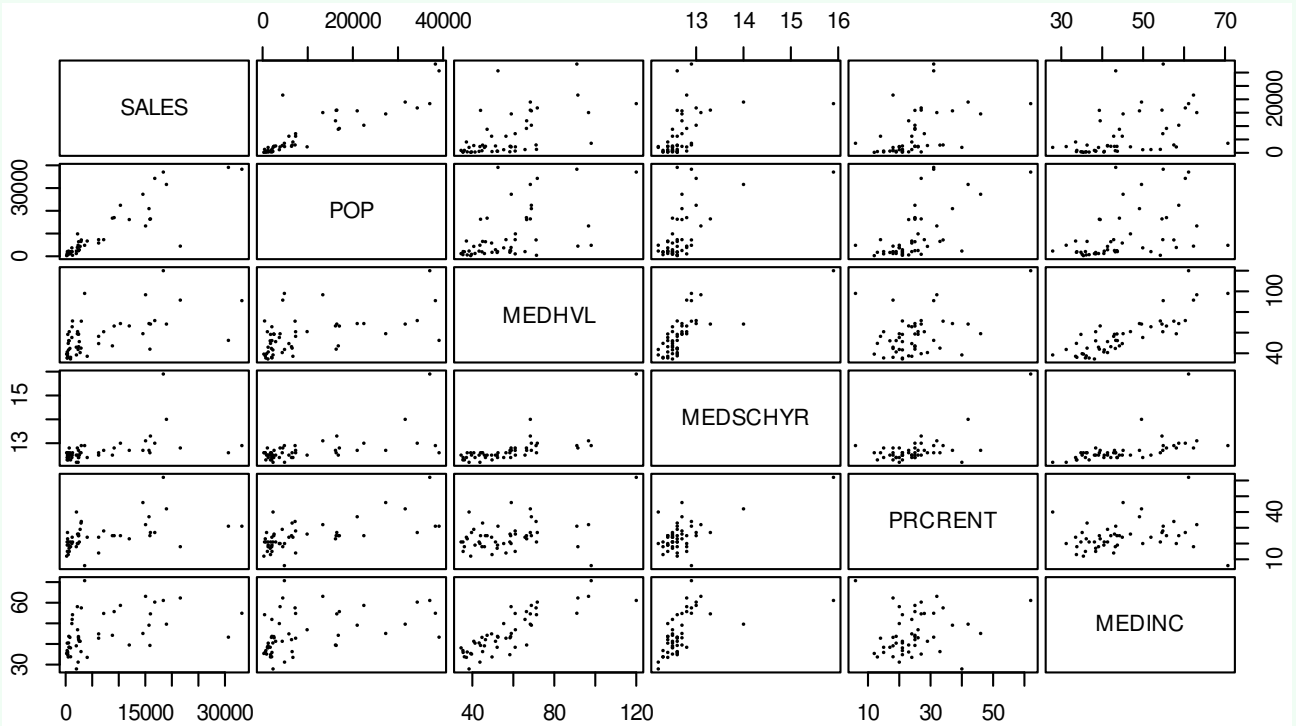
```
## crit
## 2.02
```

```
round(coef(summary(fit_l3)), 4)
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 41804.5804 15827.9288   2.641   0.0112
## MEDSCHYR    -3830.0366  1355.1256  -2.826   0.0069
## MEDHVL        127.5152    38.6194   3.302   0.0019
## POP             0.6483     0.0559  11.594   0.0000
```

```
round(coef(summary(fit_l3_no11))["MEDSCHYR", ], 4); c(crit = qt(0.975, 45))
```

```
##   Estimate Std. Error    t value   Pr(>|t|)
## -1860.5060  2298.7858    -0.8093     0.4226
##  crit
## 2.014
```



(a) Observation 11 is 5.8 standard deviations above the mean in `MEDSCHYR` and 3.4 in `MEDHVL`. (b) `POP` (0.886), `MEDHVL` (0.572) and `PRCRENT` (0.496). (c) In the panel of `SALES` against `MEDSCHYR` observation 11 sits alone at the far right with moderate sales. (d) $s = 3580$, $R^2 = 0.837$, $R^2_a = 0.805$

on 41 residual degrees of freedom. (e) H_0: $\beta_{\texttt{MEDSCHYR}} = 0$ against $\neq 0$; reject at 5% if $|t| > t_{41,0.975} = 2.02$; $t = -2.26$ with $p = 0.029$, so H_0 is rejected. (f) The three-variable model has $s = 3445$, $R^2 = 0.830$ and $R^2_a = 0.819$: a lower R^2, as Proposition 2.28 requires of a submodel, but a smaller s, hence by (18) a larger R^2_a. (i) Without Shorewood the `MEDSCHYR` t ratio is -0.81 with $p = 0.42$, so H_0 is no longer rejected: the significance in (e) rested on one high leverage observation. (j) Without Kenosha, s falls to 2585 and R^2 rises to 0.899, the outlier of Example 2.24 again. ■

Exercise 2.27 (FREES exercise 3.5, parts (a)–(f))**.** Insurer expenses (`NAICExpense`, Unit 1), with the outputs `LONGLOSS`, `SHORTLOSS`, `GPWPERSONAL`, `GPWCOMM`, the size measures `ASSETS`, `CASH`, the indicators `GROUP`, `STOCK`, `MUTUAL`, and the wages `STAFFWAGE`, `AGENTWAGE`. (a) Summary statistics of the response and the non-binary explanatory variables, with their skewness; note the negative values. (b) Transform each non-binary variable by $\ln(1 + x)$, with `LNEXPENSES` $= \ln(1 +$ `EXPENSES`$)$. (c) Which three variables are most correlated with `LNEXPENSES`? (d) A box plot of `LNEXPENSES` by `GROUP`: which level has higher expenses? (e) Fit `LNEXPENSES` on all eleven explanatory variables and report s, R^2, R^2_a. (f) Fit the eight-variable model dropping `CASH`, `STOCK` and `MUTUAL`: (i) s, R^2, R^2_a; (ii) interpret the coefficient of `GPWCOMM`; (iii) the expected increase in

EXPENSES when GPWCOMM increases by \$1, at the medians.

Solution.

```
naic <- read.csv("data/frees/csv/NAICExpense.csv")
skew <- function(v) mean((v - mean(v))^3) / mean((v - mean(v))^2)^1.5
vars <- c("EXPENSES", "LONGLOSS", "SHORTLOSS", "GPWPERSONAL", "GPWCOMM", "ASSETS",
 ↪  "CASH",
          "STAFFWAGE", "AGENTWAGE")
round(t(sapply(naic[, vars], function(v) { v <- v[!is.na(v)]
  c(n = length(v), median = median(v), mean = mean(v), sd = sd(v), min = min(v),
   ↪  skew = skew(v),
    negative = sum(v < 0)) })), 3)
```

```
##                n median   mean     sd    min  skew negative
## EXPENSES     384  0.009  0.043  0.121 -0.002 6.513        4
## LONGLOSS     384  0.002  0.025  0.083 -0.071 6.429       14
## SHORTLOSS    384  0.004  0.037  0.120 -0.003 6.556        9
## GPWPERSONAL 384  0.000  0.053  0.174 -0.004 6.269        8
## GPWCOMM      384  0.024  0.123  0.318 -0.001 7.208        1
## ASSETS       384  0.057  0.357  1.015  0.000 5.593        0
```

```
## CASH         384  0.050  0.333  0.969  0.000 5.858        0
## STAFFWAGE    384 84.382 87.182 11.803 51.727 0.802        0
## AGENTWAGE    365 78.766 80.154  9.104 47.471 0.223        0
```

```
L <- as.data.frame(lapply(naic[, vars], function(v) log(1 + v))); names(L) <-
 ↪  paste0("LN", vars)
L$GROUP <- naic$GROUP; L$STOCK <- naic$STOCK; L$MUTUAL <- naic$MUTUAL
round(sort(cor(L[, paste0("LN", vars)], use = "complete.obs")[, "LNEXPENSES"],
 ↪  decreasing = TRUE), 3)
```

```
##    LNEXPENSES   LNSHORTLOSS    LNLONGLOSS     LNGPWCOMM
##         1.000         0.930         0.918         0.864
##      LNASSETS       LNCASH LNGPWPERSONAL   LNSTAFFWAGE
##         0.863         0.860         0.809         0.101
##   LNAGENTWAGE
##         0.087
```

```
tapply(L$LNEXPENSES, L$GROUP, median)
```

```
##        0        1
## 0.002267 0.015806
```

```r
fit_n11 <- lm(LNEXPENSES ~ ., data = L)
fit_n8 <- lm(LNEXPENSES ~ LNASSETS + GROUP + LNLONGLOSS + LNSHORTLOSS +
 ↪  LNGPWPERSONAL + LNGPWCOMM +
                LNSTAFFWAGE + LNAGENTWAGE, data = L)
round(rbind(eleven = fit_stats3(fit_n11), eight = fit_stats3(fit_n8)), 4)
```

```
##           n      s     R2 R2_adj
## eleven 365 0.0224 0.9422 0.9404
## eight  365 0.0224 0.9414 0.9400
```

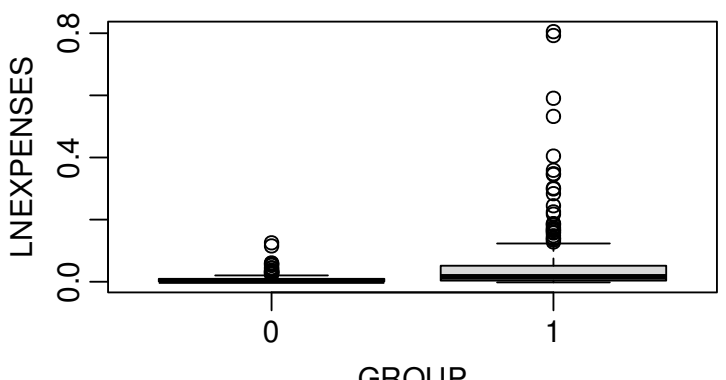
```r
round(coef(summary(fit_n8)), 4)
```

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.0124     0.0457 -0.2705   0.7869
## LNASSETS        0.0442     0.0074  5.9955   0.0000
## GROUP          -0.0014     0.0026 -0.5513   0.5818
## LNLONGLOSS      0.4582     0.0450 10.1836   0.0000
## LNSHORTLOSS     0.3132     0.0325  9.6313   0.0000
## LNGPWPERSONAL   0.0632     0.0191  3.3037   0.0011
## LNGPWCOMM       0.0689     0.0161  4.2874   0.0000
```

```
## LNSTAFFWAGE     0.0027     0.0143  0.1918   0.8480
## LNAGENTWAGE     0.0002     0.0166  0.0116   0.9907
```

```r
mG <- median(naic$GPWCOMM); mE <- median(naic$EXPENSES)
c(median_GPWCOMM = mG, median_EXPENSES = mE,
  dEXPENSES_per_dollar = coef(fit_n8)[["LNGPWCOMM"]] * (1 + mE) / (1 + mG))
```

```
##       median_GPWCOMM      median_EXPENSES
##             0.023807             0.008504
## dEXPENSES_per_dollar
##             0.067828
```



(a) Expenses, the losses, the premiums, assets and cash are heavily skewed to the right, with skewness between 5.6 and 7.2 and means several times the medians, while the two wage variables are nearly symmetric, with skewness 0.80 and 0.22; expenses, both losses and both premium variables take negative values for a few companies, so the plain logarithm is undefined for them, and `AGENTWAGE` is missing for 19 companies. (b) The shifted logarithm $\ln(1+x)$ is defined for every value here, the smallest being -0.071. (c) `LNSHORTLOSS` (0.930), `LNLONGLOSS` (0.918) and `LNGPWCOMM` (0.864). (d) Affiliated companies, `GROUP` $= 1$, have the higher expenses, a median of 0.016 against 0.002. (e) On the 365 companies with complete data, $s = 0.0224$, $R^2 = 0.942$, $R^2_a = 0.940$. (f)(i) The eight-variable model has $s = 0.0224$, $R^2 = 0.941$ and $R^2_a = 0.940$: dropping three variables costs a thousandth of R^2 and nothing in R^2_a. (ii) The coefficient 0.069 of `LNGPWCOMM` is an elasticity between the shifted variables: a 1% increase in $1 +$ `GPWCOMM` is associated with a 0.069% increase in $1 +$ `EXPENSES`, other variables fixed. (iii) Differentiating $\ln(1+E) = \cdots + 0.069 \ln(1+G)$ gives $\mathrm{d}E = 0.069\,(1+E)/(1+G)\,\mathrm{d}G$; at the medians, $E = 0.0085$ and $G = 0.0238$ million, a dollar more of commercial premium goes with 0.068 dollars more expense. ■

Exercise 2.28 (FREES exercise 3.6, parts (a)–(c))**.** Regress life expectancy, `LIFEEXP`, on the three variables `FERTILITY`, `PUBLICEDUCATION` and `lnHEALTH` = ln `PRIVATEHEALTH`. (a) Interpret the coefficient of public education. (b) Interpret the coefficient of health expenditure without the logarithmic scale. (c) Is `PUBLICEDUCATION` significant? State the hypotheses, the criterion and the decision.

Solution.

```
un$lnHEALTH <- log(un$PRIVATEHEALTH)
fit_un3 <- lm(LIFEEXP ~ FERTILITY + PUBLICEDUCATION + lnHEALTH, data = un)
round(coef(summary(fit_un3)), 4)
```

```
##                 Estimate Std. Error  t value Pr(>|t|)
## (Intercept)      85.6264     2.0033  42.7422   0.0000
## FERTILITY        -5.3993     0.3308 -16.3242   0.0000
## PUBLICEDUCATION  -0.1846     0.2685  -0.6875   0.4928
## lnHEALTH         -1.0296     0.9431  -1.0917   0.2768
```

```
c(n = nobs(fit_un3), R2 = summary(fit_un3)$r.squared, crit = qt(0.975, nobs(fit_un3)
 ↪  - 4),
  cor_PUBED_LIFEEXP = cor(un$PUBLICEDUCATION, un$LIFEEXP, use = "complete.obs"))
```

```
##                   n                  R2                crit
##           152.0000              0.6450              1.9761
## cor_PUBED_LIFEEXP
##              0.0733
```

The fit uses the 152 countries with all four variables. (a) Holding fertility and private health expenditure fixed, a country spending one more percentage point of GDP on public education has an expected life expectancy 0.18 years **lower**, although the simple correlation of the two variables is positive (0.07). (b) The coefficient -1.03 of $\ln$ `PRIVATEHEALTH` means that a 1% increase in private health expenditure, as a share of GDP, is associated with a change of $-1.03/100 = -0.010$ years in expected life expectancy, the other variables fixed; doubling it, a change of $\ln 2$ in the logarithm, with -0.71 years. (c) H_0: $\beta_{\texttt{PUBED}} = 0$ against $\neq 0$; reject at 5% if $|t| > t_{148,0.975} = 1.98$; $t = -0.69$ with $p = 0.49$, so H_0 is not rejected. ■

Exercise 2.29 (ISLR §3.7, exercise 1)**.** Describe the null hypotheses of the p-values in ISLR Table 3.4, the regression of **sales** on **TV**, **radio** and **newspaper** (Example 2.18), and the conclusions they support, in terms of the variables rather than the coefficients.

Solution. Each p-value tests, by Definition 2.36, $H_0\colon \beta_j = 0$ against $\beta_j \neq 0$ with the other two media held fixed: for `TV`, that television advertising has no association with sales in markets with given radio and newspaper budgets; likewise for `radio` and for `newspaper`; and for the intercept, that sales are zero when nothing is spent. The p-values below 10^{-4} for `TV` and `radio` reject their nulls: each medium is associated with sales after adjusting for the other two. The p-value 0.86 for `newspaper` does not: given the television and radio budgets, newspaper spending carries no evidence of an association, its simple association in Example 2.10 being borrowed from radio. ■

Exercise 2.30 (ISLR §3.7, exercise 4)**.** With $n = 100$ observations on one predictor, a linear and a cubic regression $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$ are fitted. (a) If the true relation is linear, which has the lower training RSS? (b) The lower test RSS? (c)–(d) The same when the truth is non-linear to an unknown degree.

Solution. (a) The cubic, or a tie: the linear design's column space is contained in the cubic's, so by Proposition 2.28 the cubic's training RSS cannot exceed the linear one's, whatever the truth. (b) The linear model is expected to have the lower test RSS: both are unbiased for a linear truth, since the cubic contains it, and the cubic estimates two more coefficients, which adds variance and

no reduction in bias (the decomposition of Unit 1). (c) The cubic again, by the same containment; the training RSS says nothing about the truth. (d) Not enough information: if the truth is far from linear the cubic's lower bias outweighs its extra variance and it wins on test data; if the truth is nearly linear the linear model wins as in (b). ■

Exercise 2.31 (ISLR §3.7, exercise 5)**.** For regression without an intercept, $\hat{y}_i = x_i\hat{\beta}$ with $\hat{\beta} = \sum_i x_i y_i / \sum_j x_j^2$. Write $\hat{y}_i = \sum_j a_j y_j$ and identify a_j.

Solution. Substituting Corollary 2.15,

$$\hat{y}_i = x_i \frac{\sum_j x_j y_j}{\sum_l x_l^2} = \sum_j a_j y_j, \qquad a_j = \frac{x_i x_j}{\sum_l x_l^2} :$$

the fitted values are linear combinations of the responses with weights fixed by the design. In general $\hat{\mathbf{Y}} = \mathbf{HY}$, so the weight of Y_j in $\hat{Y}_i$ is h_{ij}; here $\mathbf{H} = \mathbf{x}\mathbf{x}'/(\mathbf{x}'\mathbf{x})$, whose (i,j) entry is a_j. With an intercept, $\mathbf{H1} = \mathbf{1}$ makes $\sum_j h_{ij} = 1$ for every i, and the weight an observation gives itself is its leverage. ■

Exercise 2.32 (ISLR §3.7, exercise 6)**.** Show that the least squares line always passes through $(\bar{x}, \bar{y})$.

Solution. By Corollary 2.14, $b_0 = \bar{y} - b_1\bar{x}$, so the line's height at $\bar{x}$ is $b_0 + b_1\bar{x} = \bar{y}$. It is the first normal equation, $\sum_i e_i = 0$, and needs the intercept: regression through the origin passes through $(0, 0)$ instead. ■

Exercise 2.33 (ISLR §3.7, exercise 7)**.** Prove $R^2 = r^2$ for simple linear regression, assuming $\bar{x} = \bar{y} = 0$.

Solution. With zero means, $b_1 = \sum_i x_i y_i / \sum_i x_i^2$ and $b_0 = 0$, so $\mathrm{TSS} = \sum_i y_i^2$ and

$$\mathrm{RSS} = \sum_i (y_i - b_1 x_i)^2 = \sum_i y_i^2 - 2b_1 \sum_i x_i y_i + b_1^2 \sum_i x_i^2 = \sum_i y_i^2 - \frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2},$$

hence

$$R^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = \frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2 \sum_i y_i^2} = r^2$$

by (1) with the means zero. Without the assumption, replace x_i and y_i by their deviations, which changes neither R^2 (Corollary 2.32) nor r (Proposition 2.2). ■

Exercise 2.34 (ISLR §3.7, exercise 8)**.** Regress `mpg` on `horsepower` in `Auto`. (a) Is there a relation; how strong; positive or negative; the predicted `mpg` at 98 horsepower with 95% confidence and prediction intervals. (b) Plot the data with the line. (c) Diagnostic plots.

Solution.

```
fit_hp <- lm(mpg ~ horsepower, data = Auto)
round(coef(summary(fit_hp)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  39.9359     0.7175   55.66        0
## horsepower   -0.1578     0.0064  -24.49        0
```

```
c(p_slope = coef(summary(fit_hp))["horsepower", "Pr(>|t|)"])
```

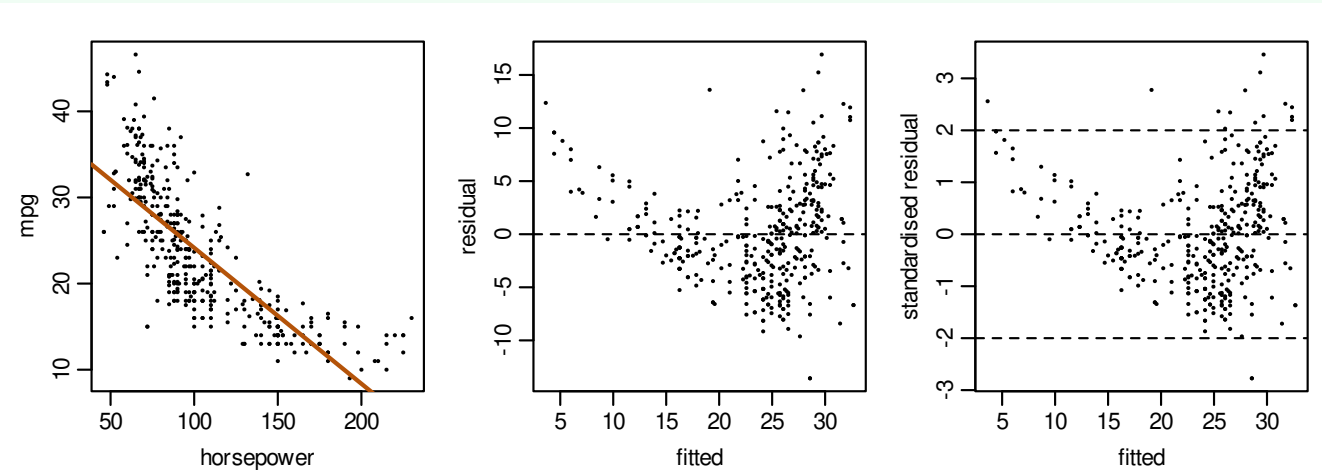
```
##   p_slope
## 7.032e-81
```

```r
c(R2 = summary(fit_hp)$r.squared, RSE = summary(fit_hp)$sigma, mean_mpg =
  mean(Auto$mpg),
  RSE_over_mean = summary(fit_hp)$sigma / mean(Auto$mpg))
```

```
##           R2          RSE      mean_mpg RSE_over_mean
##       0.6059       4.9058        23.4459        0.2092
```

```r
rbind(confidence = predict(fit_hp, data.frame(horsepower = 98), interval =
  "confidence"),
      prediction = predict(fit_hp, data.frame(horsepower = 98), interval =
        "prediction"))
```

```
##     fit   lwr   upr
## 1 24.47 23.97 24.96
## 1 24.47 14.81 34.12
```



(a) The slope -0.158 has $t = -24.5$ and a p-value of 7×10^{-81}: there is a relation, and it is negative, each additional unit of horsepower lowering the expected mileage by 0.16 miles per gallon. $R^2 = 0.61$, and the residual standard error 4.9 is 21% of the mean mileage 23.4. At 98 horsepower the fitted value is 24.5, the confidence interval $(24.0, 25.0)$ and the prediction interval $(14.8, 34.1)$. (b)–(c) The line underestimates mileage at low and high horsepower and overestimates it in the middle: the residuals against the fitted values form a U, the mark of a relation that is not linear; the standardised residuals also spread more at high fitted values. ■

Exercise 2.35 (ISLR §3.7, exercise 9, parts (a)–(d))**.** For `Auto`: (a) a scatterplot matrix; (b) the correlation matrix excluding `name`; (c) regress `mpg` on all other variables: is there a relation, which predictors are significant, what does the `year` coefficient say; (d) diagnostic plots: large residuals, high leverage points.

Solution.

```
A <- Auto[, names(Auto) != "name"]
round(cor(A), 2)
```

```
##                mpg cylinders displacement horsepower weight
## mpg           1.00     -0.78        -0.81      -0.78  -0.83
## cylinders    -0.78      1.00         0.95       0.84   0.90
## displacement -0.81      0.95         1.00       0.90   0.93
## horsepower   -0.78      0.84         0.90       1.00   0.86
## weight       -0.83      0.90         0.93       0.86   1.00
## acceleration  0.42     -0.50        -0.54      -0.69  -0.42
## year          0.58     -0.35        -0.37      -0.42  -0.31
## origin        0.57     -0.57        -0.61      -0.46  -0.59
##              acceleration  year origin
## mpg                  0.42  0.58   0.57
```

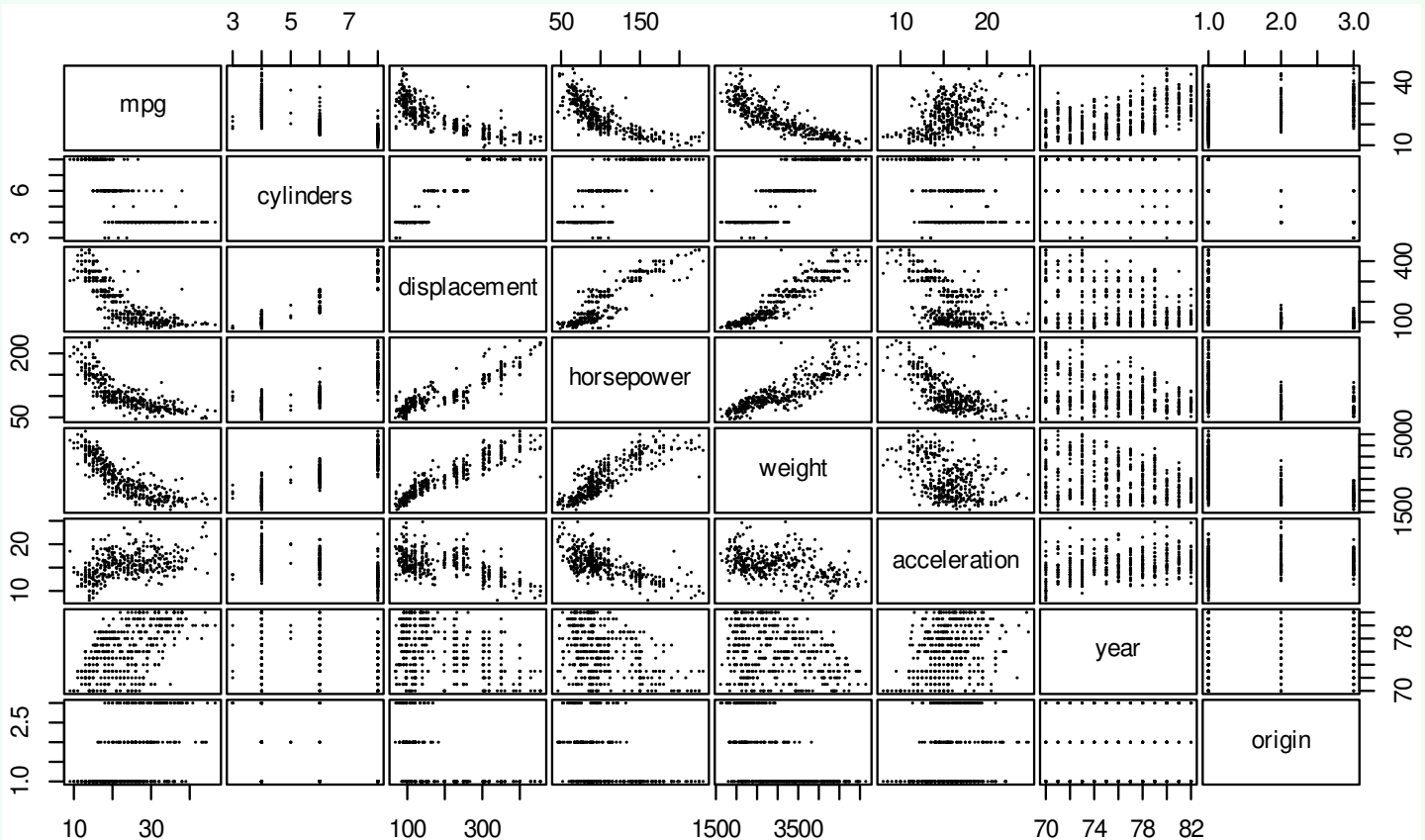
```
## cylinders           -0.50 -0.35  -0.57
## displacement        -0.54 -0.37  -0.61
## horsepower          -0.69 -0.42  -0.46
## weight              -0.42 -0.31  -0.59
## acceleration         1.00  0.29   0.21
## year                 0.29  1.00   0.18
## origin               0.21  0.18   1.00
```

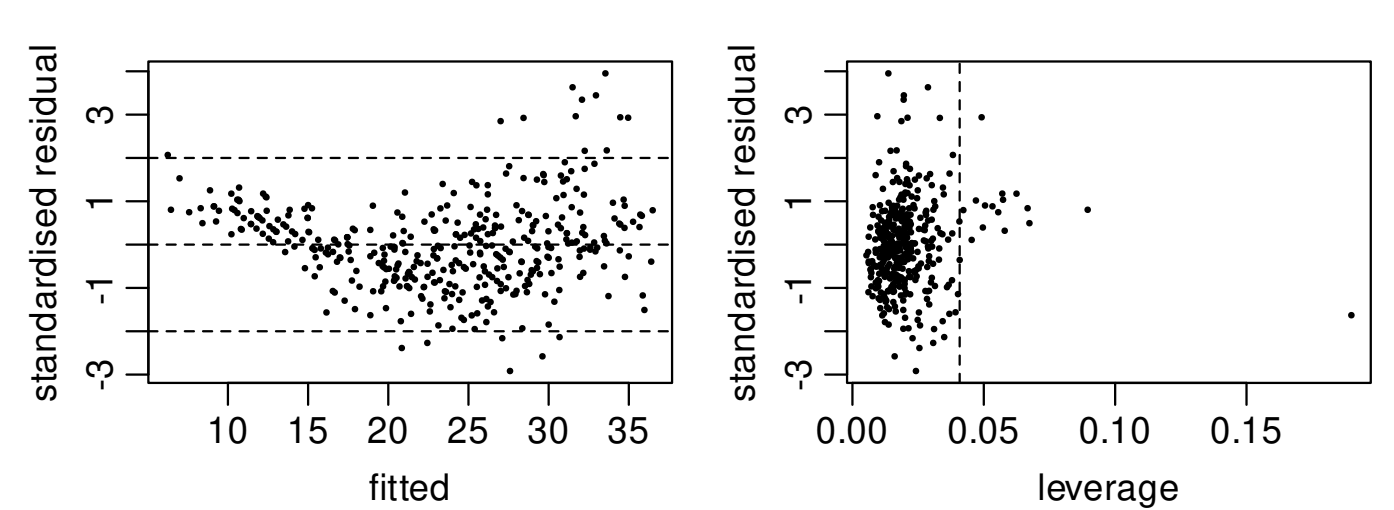
```
fit_auto <- lm(mpg ~ ., data = A)
round(coef(summary(fit_auto)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -17.2184     4.6443 -3.7074   0.0002
## cylinders    -0.4934     0.3233 -1.5261   0.1278
## displacement  0.0199     0.0075  2.6474   0.0084
## horsepower   -0.0170     0.0138 -1.2295   0.2196
## weight       -0.0065     0.0007 -9.9288   0.0000
## acceleration  0.0806     0.0988  0.8152   0.4155
## year          0.7508     0.0510 14.7288   0.0000
## origin        1.4261     0.2781  5.1275   0.0000
```

```r
c(R2 = summary(fit_auto)$r.squared, F = summary(fit_auto)$fstatistic[["value"]],
  max_std_resid = max(abs(rstandard(fit_auto))),
  at = unname(which.max(abs(rstandard(fit_auto)))),
  max_leverage = max(hatvalues(fit_auto)),
  at_lev = unname(which.max(hatvalues(fit_auto))), flag = 2 * 8 / 392)
```

```
##            R2             F max_std_resid            at
##       0.82148     252.42805       3.95189     321.00000
## max_leverage        at_lev          flag
##       0.18991      14.00000       0.04082
```


mpg
cylinders
displacement
horsepower
weight
acceleration
year
origin



(a)–(b) Mileage falls with cylinders, displacement, horsepower and weight, whose mutual correlations are 0.84 to 0.95, and rises with year and origin. (c) $F = 252$ on $(7, 384)$ degrees of freedom: a relation exists. At 5% the significant predictors are `displacement`, `weight`, `year` and `origin`; `cylinders`, `horsepower` and `acceleration` are not, their simple associations being absorbed by the correlated size variables. The `year` coefficient 0.75 says that, with the other variables fixed, a car one model year newer is expected to travel 0.75 more miles per gallon. (d) The standardised

residuals curve upward at high fitted values, so the relation is again not linear; observation 321 has the largest standardised residual, 4.0, and observation 14 the largest leverage, 0.19, nearly five times the flag $2(k+1)/n = 0.04$. ■

Exercise 2.36 (ISLR §3.7, exercise 11)**.** Generate x_i standard normal and $y_i = 2x_i + \varepsilon_i$, $n = 100$, with `set.seed(1)`. (a) Regress y on x without an intercept and report the coefficient, its standard error, t and p-value. (b) The same for x on y. (c) The relation between the two. (d) Show that for regression through the origin

$$t = \frac{\sqrt{n-1}\,\sum_i x_i y_i}{\sqrt{(\sum_i x_i^2)(\sum_i y_i^2) - (\sum_i x_i y_i)^2}},$$

and confirm it numerically. (e) Deduce that the t statistics in (a) and (b) are equal. (f) Show numerically that the same holds with an intercept.

Solution. (d) With $\hat\beta = \sum_i x_i y_i / \sum_i x_i^2$ and, for one parameter and no intercept, $\mathrm{SE}(\hat\beta) = \sqrt{\sum_i (y_i - x_i\hat\beta)^2 / \{(n-1)\sum_i x_i^2\}}$ (the proofs of Theorems 2.9(iv) and 2.22 and Definition 2.24

hold for any $\mathbf{X}$ of full column rank p, with $n-p$ in place of $n-k-1$; here $\mathbf{X}=\mathbf{x}$ has one column, so the divisor is $n-1$),

$$t=\frac{\hat\beta}{\mathrm{SE}(\hat\beta)}=\frac{\sum_i x_i y_i}{\sum_i x_i^2}\cdot\frac{\sqrt{n-1}\sqrt{\sum_i x_i^2}}{\sqrt{\sum_i (y_i-x_i\hat\beta)^2}},\qquad \sum_i (y_i-x_i\hat\beta)^2=\sum_i y_i^2-\frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2},$$

the second by expanding the square and substituting $\hat\beta$; multiplying numerator and denominator by $\sqrt{\sum_i x_i^2}$ gives the display. (e) The expression is symmetric in x and y, so exchanging their roles cannot change it.

```
set.seed(1); xa <- rnorm(100); ya <- 2 * xa + rnorm(100)
tf <- sqrt(99) * sum(xa * ya) / sqrt(sum(xa^2) * sum(ya^2) - sum(xa * ya)^2)
round(rbind(y_on_x = coef(summary(lm(ya ~ xa + 0)))[1, ], x_on_y =
 ↪  coef(summary(lm(xa ~ ya + 0)))[1, ]), 5)
```

```
##         Estimate Std. Error t value Pr(>|t|)
## y_on_x    1.9939    0.10648   18.73        0
## x_on_y    0.3911    0.02089   18.73        0
```

```r
c(closed_form = tf, with_intercept_y_on_x = coef(summary(lm(ya ~ xa)))[2, 3],
  with_intercept_x_on_y = coef(summary(lm(xa ~ ya)))[2, 3])
```

```
##           closed_form with_intercept_y_on_x
##                 18.73                 18.56
## with_intercept_x_on_y
##                 18.56
```

(a)–(c) The coefficient of y on x is 1.99, that of x on y is 0.39, and the two t statistics are the same, 18.7, as (d)–(e) require; the coefficients differ because $\sum_i x_i^2 \neq \sum_i y_i^2$. (f) With an intercept the two t statistics again coincide, by Exercise 2.16(c): $t = \sqrt{n-2}\, r/\sqrt{1-r^2}$ depends on the symmetric r alone, so testing $\beta_1 = 0$ is testing $r = 0$, a statement about the pair and not about a direction. ■

Exercise 2.37 (ISLR §3.7, exercise 12)**.** For regression through the origin: (a) when is the coefficient of X on Y equal to that of Y on X? (b) Generate $n = 100$ observations for which they differ; (c) for which they coincide.

Solution. (a) By Corollary 2.15 the two are $\sum_i x_i y_i / \sum_i x_i^2$ and $\sum_i x_i y_i / \sum_i y_i^2$, equal iff $\sum_i x_i^2 = \sum_i y_i^2$ (or the numerator is zero).

```
set.seed(4); xb <- rnorm(100); yb <- 3 * xb + rnorm(100)
c(y_on_x = coef(lm(yb ~ xb + 0))[[1]], x_on_y = coef(lm(xb ~ yb + 0))[[1]], sum_x2 =
 ↪  sum(xb^2), sum_y2 = sum(yb^2))
```

```
##   y_on_x   x_on_y   sum_x2   sum_y2
##   2.8445   0.3062  83.6129 776.7771
```

```
yc <- sample(xb)                                  # a permutation: the same squares
c(y_on_x = coef(lm(yc ~ xb + 0))[[1]], x_on_y = coef(lm(xb ~ yc + 0))[[1]], sum_x2 =
 ↪  sum(xb^2), sum_y2 = sum(yc^2))
```

```
##   y_on_x   x_on_y   sum_x2   sum_y2
## -0.01329 -0.01329 83.61295 83.61295
```

(b) With $y = 3x + \varepsilon$ the sums of squares differ and so do the coefficients. (c) A permutation of x has the same sum of squares, and the two coefficients agree. ■

Exercise 2.38 (ISLR §3.7, exercise 13). With `set.seed(1)`: (a) x, 100 draws from $N(0,1)$; (b) `eps`, 100 draws from $N(0, 0.25)$; (c) $y = -1 + 0.5x + \varepsilon$: the length of y, and β_0, β_1; (d) a scatter plot; (e) the least squares fit against the truth; (f) the fitted and the population lines on the plot, with a legend; (g) a quadratic term: does it improve the fit? (h)–(i) repeat with less noise (variance 0.01) and more noise (variance 1); (j) the confidence intervals for β_0, β_1 in the three cases.

Solution.

```r
sim_fit <- function(sd_eps) {
  set.seed(1); x <- rnorm(100); eps <- rnorm(100, sd = sd_eps); y <- -1 + 0.5 * x +
   ↪  eps
  m <- lm(y ~ x)
  list(m = m, x = x, y = y, sd = sd_eps, ci = confint(m),
       quad = coef(summary(lm(y ~ x + I(x^2))))["I(x^2)", ])
}
s_mid <- sim_fit(0.5); s_low <- sim_fit(0.1); s_high <- sim_fit(1)
c(length_y = length(s_mid$y), beta0 = -1, beta1 = 0.5)
```

```
## length_y    beta0    beta1
##    100.0     -1.0      0.5
```

```r
round(coef(summary(s_mid$m)), 4)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.0188     0.0485 -21.010        0
## x             0.4995     0.0539   9.273        0
```

```r
round(s_mid$quad, 4)
```

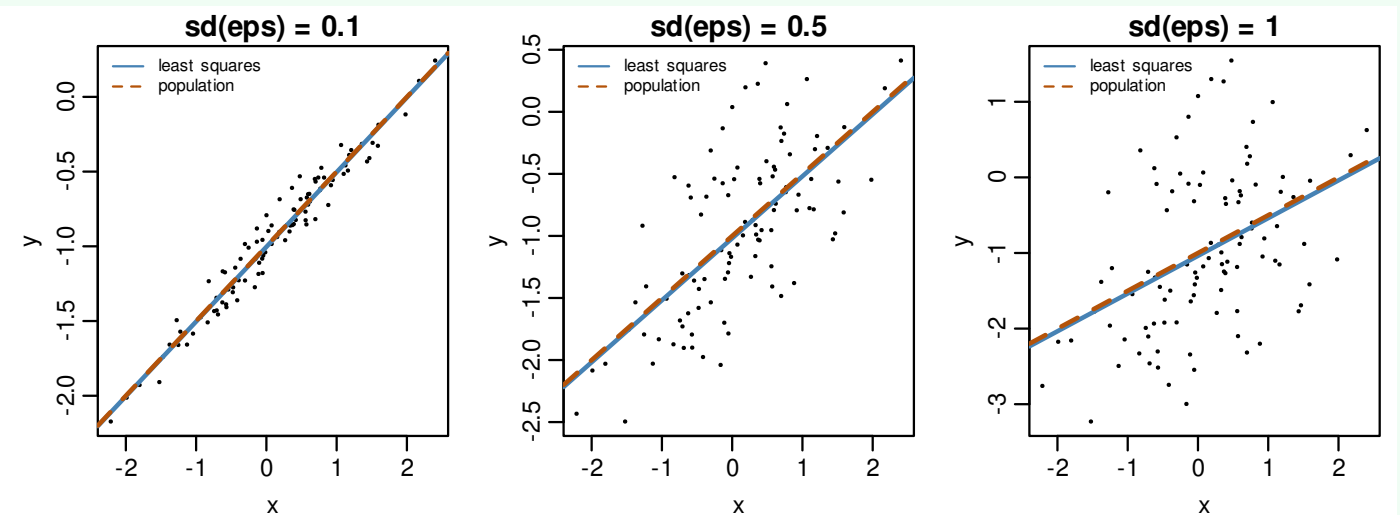
```
##   Estimate Std. Error    t value   Pr(>|t|)
##    -0.0595     0.0424    -1.4029     0.1638
```

```r
round(rbind(noise_0.25 = c(s_mid$ci[1, ], s_mid$ci[2, ]), noise_0.01 = c(s_low$ci[1, ], s_low$ci[2, ]),
            noise_1 = c(s_high$ci[1, ], s_high$ci[2, ])), 4)
```

```
##             2.5 %  97.5 %  2.5 % 97.5 %
## noise_0.25 -1.115 -0.9226 0.3926 0.6064
## noise_0.01 -1.023 -0.9845 0.4785 0.5213
## noise_1    -1.230 -0.8452 0.2852 0.7127
```

```r
round(rbind(noise_0.25 = coef(s_mid$m), noise_0.01 = coef(s_low$m), noise_1 = coef(s_high$m)), 4)
```

```
##             (Intercept)      x
## noise_0.25       -1.019 0.4995
## noise_0.01       -1.004 0.4999
## noise_1          -1.038 0.4989
```



(c) y has length 100, $\beta_0 = -1$ and $\beta_1 = 0.5$. (d) A linear relation with scatter. (e) $\hat{\beta}_0 = -1.02$ and $\hat{\beta}_1 = 0.50$, each within one standard error of the truth. (g) The quadratic term has $t = -1.40$ and $p = 0.16$: no evidence of curvature, as expected of a linear truth. (h)–(j) With error variance 0.01

the fit is $(-1.004, 0.500)$ and the slope interval $(0.479, 0.521)$; with variance 1 the fit is $(-1.04, 0.50)$ and the slope interval $(0.29, 0.71)$: the interval width scales with σ exactly as (14) says, the design being the same in all three cases, and all three intervals cover the truth. ■

Exercise 2.39 (ISLR §3.7, exercise 15, parts (a)–(c))**.** Predict the per-capita crime rate `crim` in `Boston`. (a) Regress it on each predictor alone: which associations are significant? (b) Regress it on all predictors: for which can $H_0\colon \beta_j = 0$ be rejected? (c) Plot the simple coefficients against the multiple ones.

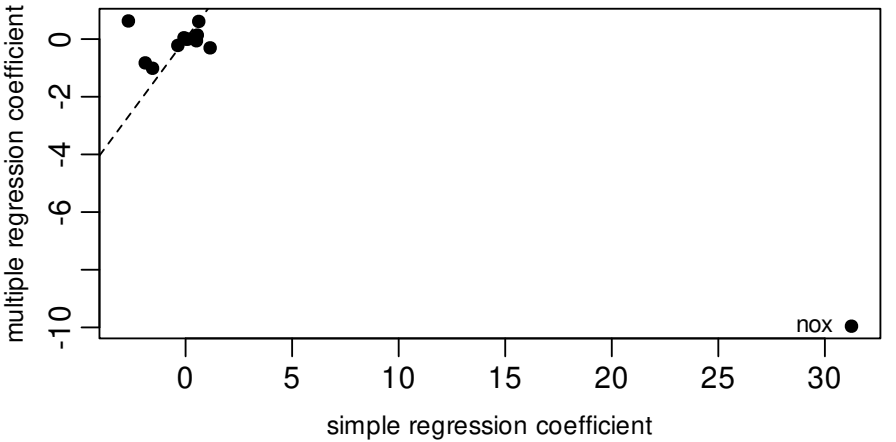
Solution.

```
preds <- setdiff(names(Boston), "crim")
uni <- t(sapply(preds, function(v) { m <- lm(as.formula(paste("crim ~", v)), data =
 ↪  Boston)
  c(coef = coef(m)[[2]], p = coef(summary(m))[2, 4]) }))
fit_crim <- lm(crim ~ ., data = Boston)
multi <- coef(summary(fit_crim))[-1, c(1, 4)]
round(cbind(simple = uni[, "coef"], p_simple = uni[, "p"], multiple = multi[, 1],
 ↪  p_multiple = multi[, 2]), 4)
```

```
##          simple p_simple multiple p_multiple
## zn      -0.0739   0.0000   0.0457     0.0153
## indus    0.5098   0.0000  -0.0584     0.4857
## chas    -1.8928   0.2094  -0.8254     0.4858
## nox     31.2485   0.0000  -9.9576     0.0604
## rm      -2.6841   0.0000   0.6289     0.3007
## age      0.1078   0.0000  -0.0008     0.9623
## dis     -1.5509   0.0000  -1.0122     0.0004
## rad      0.6179   0.0000   0.6125     0.0000
## tax      0.0297   0.0000  -0.0038     0.4658
## ptratio  1.1520   0.0000  -0.3041     0.1034
## lstat    0.5488   0.0000   0.1388     0.0674
## medv    -0.3632   0.0000  -0.2201     0.0003
```

```
1 c(R2_multiple = summary(fit_crim)$r.squared)
```

```
## R2_multiple
##      0.4493
```



(a) Alone, every predictor but `chas` ($p = 0.21$) is significantly associated with the crime rate. (b) Together, H_0 is rejected at 5% for `zn`, `dis`, `rad` and `medv` only, with `nox` and `lstat` near the boundary; the model explains 45% of the variation. (c) The two sets of coefficients are unrelated, and `nox` moves from 31 alone to -10 in the joint model: its simple association is carried by the predictors it is correlated with, Example 2.10 on a larger scale. ■

Exercise 2.40 (SOA sample question 11)**.** A regression model gives, for five observations, $y = (2, 5, 6, 8, 4)$ and $\hat{f}(x) = (4, 3, 9, 3, 6)$. Calculate the sum of squared errors: (A) -35; (B) -5; (C) 5; (D) 35; (E) 46.

Solution. By Definition 2.16, $\text{SSE} = \sum_i (y_i - \hat{y}_i)^2 = (-2)^2 + 2^2 + (-3)^2 + 5^2 + (-2)^2 = 4 + 4 + 9 + 25 + 4 = 46$, which is (E). (A) and (B) are impossible, a sum of squares being non-negative; the residuals themselves sum to $-2 + 2 - 3 + 5 - 2 = 0$, and neither 5, the number of observations, nor 35 is a sum of squares of these residuals. ■

Exercise 2.41 (SOA sample question 18)**.** For a simple linear regression on 25 observations, $\sum_i e_i^2 = 230$ and $R^2 = 0.64$. Calculate the total sum of squares: (A) 605.94; (B) 638.89; (C) 690.77; (D) 701.59; (E) 750.87.

Solution. By (18), $R^2 = 1 - \text{SSE}/\text{SST}$, so $\text{SST} = \text{SSE}/(1 - R^2) = 230/0.36 = 638.89$, which is (B); $n = 25$ is not needed, and $\text{SSR} = \text{SST} - \text{SSE} = 408.89$ checks as 0.64×638.89. None of the other options equals $230/(1 - R^2)$; they would require $1 - R^2$ of 0.380, 0.333, 0.328 or 0.306. ■

Exercise 2.42 (SOA sample question 23)**.** A cafeteria sells coffee at 1.00 for 12 ounces, 1.20 for 16 and 1.40 for 20, and will price any size at the least squares regression of price on size. Toby and Karen each buy a 24-ounce coffee; calculate what they save by sharing one 48-ounce coffee instead: (A) it costs 0.40 more; (B) the same; (C) they save 0.40; (D) 0.80; (E) 1.20.

Solution. The three points are collinear, so by Proposition 2.2 $r = 1$ and the least squares line passes through all three with SSE $= 0$: the slope is $(1.40 - 1.00)/(20 - 12) = 0.05$ per ounce and the intercept $1.00 - 0.05(12) = 0.40$; the same comes from (11), with $\bar{x} = 16$, $\bar{y} = 1.20$, $b_1 = \{(-4)(-0.2) + 0 + (4)(0.2)\}/\{16 + 0 + 16\} = 1.6/32 = 0.05$. Two 24-ounce cups cost $2(0.40 + 1.20) = 3.20$; one 48-ounce cup costs $0.40 + 2.40 = 2.80$. They save 0.40, the fixed charge of the second cup, which is (C). (A) would follow from adding, not subtracting, the intercept; (B) from a line through the origin, which the data reject; (D) and (E) from doubling or tripling the intercept. ■

Exercise 2.43 (SOA sample question 24)**.** A regression of a mutual fund's return on four predictors plus an intercept over 105 months reports $F = 20.0$; the software failed before computing R^2. Which is true: (A) insufficient information, but R^2 could be calculated given RSS; (B) insufficient

information, but it could be calculated given TSS and RSS; (C) $R^2 = 0.44$; (D) $R^2 = 0.56$; (E) $R^2 = 0.80$.

Solution. By Corollary 2.39, R^2 depends on SSE and SST only through their ratio, and F fixes that ratio: with $k = 4$ and $n - k - 1 = 100$,

$$F = \frac{(\text{SST} - \text{SSE})/4}{\text{SSE}/100} = 20 \iff \frac{\text{SST} - \text{SSE}}{\text{SSE}} = 0.80 \iff \frac{\text{SST}}{\text{SSE}} = 1.80,$$

so $R^2 = 1 - 1/1.80 = 0.444$, which is (C); the corollary gives it directly as $kF/(kF + n - k - 1) = 80/180$. (A) and (B) are false because nothing beyond F, n and k is needed. (D) is $1 - R^2 = \text{SSE}/\text{SST} = 0.556$, the unexplained share, and (E) is $kF/(n-k-1) = \text{SSR}/\text{SSE} = 0.80$, the ratio of explained to unexplained variation. ■

Exercise 2.44 (SOA sample question 44)**.** Dental claims of $n = 100$ participants with the predictor sex, coded 0 and 1. Actuary 1 fits $Y = \beta + \varepsilon$ and Actuary 2 fits $Y = \beta_0 + \beta_1 \,\text{Sex} + \varepsilon$, with RSS = 250,000 and TSS = 490,000. Calculate the F statistic testing whether Actuary 2's model is a significant improvement: (A) 92; (B) 93; (C) 94; (D) 95; (E) 96.

Solution. Actuary 1's model is the intercept-only model, whose residual sum of squares is TSS, so nothing further about it is needed. By (23) with $k = 1$,

$$F = \frac{(\text{TSS} - \text{RSS})/1}{\text{RSS}/(n-2)} = \frac{240{,}000}{250{,}000/98} = 94.08 \quad \text{on } F_{1,98},$$

which is (C). The other options are the same ratio with the wrong residual degrees of freedom, 96, 97, 99 and 100 in place of $n - 2 = 98$. Two checks: by Proposition 2.40, $|t(b_1)| = \sqrt{94.08} = 9.70$, and by Exercise 2.9 this is the pooled two-sample t statistic comparing the mean claims of the two sexes; and $R^2 = 1 - 250/490 = 0.4898$ reproduces F through (23). ■

Exercise 2.45 (SOA sample question 53)**.** Which statement about $Y = \beta_0 + \beta_1 X + \varepsilon$ is NOT true: (A) β_0 is the expected value of Y; (B) β_1 is the average increase in Y associated with a one-unit increase in X; (C) ε is typically assumed independent of X; (D) the equation defines the population regression line; (E) least squares is commonly used to estimate β_0 and β_1.

Solution. (A) is false, hence the answer: by A1, $\mathsf{E}[Y \mid X] = \beta_0 + \beta_1 X$, so $\beta_0 = \mathsf{E}[Y \mid X = 0]$, the expected response when $X = 0$, and the unconditional expectation is $\beta_0 + \beta_1 \mathsf{E}[X]$. (B) is the interpretation of a coefficient in §1.2, $\beta_1 = \partial \mathsf{E}[Y \mid X]/\partial X$; (C) is a stronger form of A2: independence of ε and X together with $\mathsf{E}[\varepsilon] = 0$ implies A2; (D) is the population regression line of the same remark; (E) is Definition 2.5. ■